A Model-Free Assessment of the Importance of Subjective Beliefs for Asset Pricing*

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Abstract

Are belief dynamics or risks and risk attitudes more important for asset pricing? Allowing both, I use survey data combined with subjective-belief versions of stochastic discount factor (SDF) volatility bounds to shed new light on this question. I estimate lower bounds for the volatility of the SDF attributable to (i) risks relevant for investor marginal utility, versus (ii) subjective belief distortions. The estimates suggest that risks, particularly long-term risks, make up at least 50% of SDF volatility, with many estimates near 100%. Conversely, I estimate that belief distortions contribute much less to SDF volatility. Two example models—one with growth extrapolation and one with disagreement—can account for my estimates, provided they feature a large-enough *subjective risk* mechanism, which is the indirect impact beliefs have through induced marginal utility volatility.

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1 Introduction

Are risks or beliefs more important for asset pricing? Undoubtedly, both matter to some degree. Risk-based stories identify important factors such as long-term uncertainty, cyclical risk tolerance, bank balance sheets, and so on. On the other hand, violations of full-information rational expectations in macro-finance surveys suggest that subjective belief dynamics also play a role. In the theoretical literature, frameworks alternatively emphasizing risks or beliefs can account for a variety of facts in asset markets.

Against this backdrop, I seek to *quantify* the role of risk versus beliefs. To tackle this question at the appropriate level of generality, I take a "model-free" approach similar to Hansen and Jagannathan (1991) and Alvarez and Jermann (2005) but in a context where subjective beliefs may differ from rational expectations. More specifically, by combining financial survey data with historical returns, I estimate the fraction of stochastic discount factor (SDF) volatility due to belief distortions versus marginal utility. Marginal utility volatility arises from uncertainty and preferences, so I label this component "risk." I find that risk accounts for at least 50% of SDF volatility, with some lower-bound estimates close to 100%. Given the large belief distortions implied by surveys, which I take at face value, risk can dominate SDF volatility only if belief distortions induce risk for marginal investors—a mechanism I formalize and term *subjective risk*.

There are two broad benefits of my model-free SDF approach. First, since a theoretical model is often studied through the lens of its SDF, my estimates can serve as useful targets for researchers seeking to align their models with data. The paper provides a proof-of-concept in two example models. Second, since the SDF provides a representation for *all* asset prices, my results can help unify disparate results from various specific markets. That is, rather than ask how much beliefs matter for, say, stock market volatility, this paper asks how much beliefs matter for asset pricing in general.

The basic identification problem. To account for large expected excess returns, we know that the SDF must be highly volatile (Hansen and Jagannathan, 1991). What is not known is whether this volatility stems primarily from beliefs or risk. The problem shows up most clearly in the price of an Arrow-Debreu security for state x:

$$\text{AD-Price}(x) = \text{Probability}(x) \times \underbrace{\text{Beliefs}(x) \times \text{MU}(x)}_{\text{SDF}(x)},$$

where Probability(x) is the objective state probability, Beliefs(x) denotes investors' belief distortion (ratio of subjective-to-objective probability), and MU(x) is the representative investor's marginal utility in state x. It is this latter piece, $\mathrm{MU}(x)$, that captures risk and risk attitudes. With enough data, perhaps the econometrician can figure out the state frequency $\mathrm{Probability}(x)$. But beyond that, asset prices only identify the product $\mathrm{Beliefs}(x) \times \mathrm{MU}(x)$, a challenge originally articulated by Harrison and Kreps (1979). While the rational expectations paradigm imposes $\mathrm{Beliefs}(x) = 1$, I follow the behavioral finance tradition and use surveys to learn about $\mathrm{Beliefs}(x)$.

Solution and results. Clearly, surveys provide *some information* about beliefs, but precisely what information and how should it be used? My core idea, different from the literature, is that survey data on expected returns are connected to risk by optimization. If investors are optimistic about an asset's future returns, and they have traded until they are marginal, then they cannot simply think the asset is a "good deal." Instead, they *must perceive the asset to be risky*. Survey expected returns thus reveal something about investor marginal utility (MU), which captures risk sensitivities. On the other hand, realized returns contain information about the total volatility of the SDF (Hansen and Jagannathan, 1991), which is affected by both risks and beliefs. By comparing survey expected returns to objective expected returns, one thus learns how much of SDF volatility comes from MU volatility, i.e., how much stems from risk. I formalize this idea and explore variations.

More specifically, with few assumptions, I obtain various bounds of the form

$$\frac{\text{Volatility of MU}}{\text{Volatility of SDF}} \geq \frac{\text{Subjective risk premium}}{\text{Objective risk premium}}. \tag{1}$$

Take two extreme examples to build intuition. If investors held rational expectations, survey and objective expected returns would coincide. In that case, (1) implies that over 100% of SDF volatility is risk-based. Of course, this is the right answer: rational expectations leave no room for belief distortions, and so the SDF is entirely risk-based. By contrast, if investors were risk-neutral, survey expected returns would equal risk-free rates. The right-hand-side of (1) would be 0%, consistent with the reality under risk-neutrality that all SDF volatility is due to beliefs, and none due to MU.

My main evidence, building on the logic above, suggests that at least half of total SDF volatility stems from risk. The key reason is that subjective equity premia are large, almost as large as their true unconditional counterparts. Relatedly, Adam, Matveev and Nagel (2021) show that survey respondents do not report risk-neutral expectations (meaning survey expected returns are significantly above riskless rates). I take their idea a step further and use (1) to *quantify* the fraction of SDF volatility due to risk.

I examine variations on the asset returns used and the bound (1) itself. For example, I investigate not only the aggregate market risk premium, but also premia on leveraged portfolios, market-timing portfolios, and proxies for the growth-optimal portfolio. I also investigate volatility bounds pertaining to the permanent components of MU and the SDF, which then requires bringing in survey data on long-term bonds. Finally, I estimate the contributions of belief distortions to SDF volatility, in two different ways: one method directly estimates belief distortions, while the second method reformulates the bound (1) as an upper-bound on the role of belief distortions to make an indirect inference. In some of these variations, I find that nearly 100% of SDF volatility must be risk-based and 0% belief-based. That is, I do what behavioral finance does—take surveys at face value—and still find a dominant role for risk.

What type of risk matters? I provide some direction by estimating a subjective version of the Alvarez and Jermann (2005) bounds. In particular, I estimate that the permanent component of MU is responsible for at least 90% of total MU volatility. This says that investors perceive substantial *long-term risks*.

Models and interpretation. Finally, I analyze several example models to illustrate and interpret the results. A key question is how can beliefs display large systematic variation, even though most SDF volatility is risk-based? The answer: belief distortions must manifest primarily as a risk factor—that is, belief fluctuations induce risk. Isolating this nuanced role for beliefs provides a way forward for the behavioral finance literature.

First, I compare a long-run risks model (Bansal and Yaron, 2004) against a growth extrapolation model with "perceived long-run risks" (related to Bidder and Dew-Becker, 2016; Collin-Dufresne, Johannes and Lochstoer, 2017; Nagel and Xu, 2022). The latter is interesting because both beliefs and risks are operational: investors fear, and price, a hypothetical long-run risk, even though they are wrong. As a proof-of-concept, I calibrate the growth extrapolation model to my estimated volatility ratios, showing how these ratios are very informative targets. I also investigate what happens when the direct role for beliefs is magnified, by introducing sentiment shocks. In that case, the transitory component of the SDF is given too large a role, standing against existing evidence. This turns out to be a more general property: within this class of models, the direct role of beliefs cannot be too large. Instead, belief distortions *must* matter mostly by creating marginal utility volatility, which I refer to as *subjective risks*.

I also study a model where investors "agree to disagree" (similar to Dumas, Kurshev and Uppal, 2009; Ehling, Graniero and Heyerdahl-Larsen, 2018; Panageas, 2020). This exercise serves two purposes. First, I demonstrate how to aggregate beliefs and marginal utilities in a way that is both consistent with the "consensus belief" in surveys as well as my own volatility bounds. This allows me to reconcile, in a calibrated model, the large amount of disagreement in surveys with the minor role for belief distortions in SDF volatility. Second, the disagreement model reinforces my earlier conclusion about *subjective risks*. The SDF cannot inherit too much volatility directly from belief distortions, because that would magnify transitory SDF fluctuations. Instead, belief distortions *must* matter primarily through induced marginal utility volatility. Essentially, the disagreement must create trade that the marginal trader views as sufficiently risky.

Related literature and other approaches. My approach relies on volatility bounds, similar to Alvarez and Jermann (2005), Bakshi and Chabi-Yo (2012), and Augenblick and Lazarus (2026). While such approaches are relatively model-free, they can only provide partial information about SDFs, risks, and beliefs. If one wants more information, one needs more assumptions. For instance, some papers put structure on marginal utility and treat beliefs as a "residual" needed to match asset market data (Ghosh and Roussellet, 2023; Chen, Hansen and Hansen, 2024b). Other approaches put structure on both preferences and beliefs and estimate beliefs jointly with the rest of the model (Szőke, 2022; Maenhout, Vedolin and Xing, 2025; Bhandari, Borovička and Ho, 2025).

While these approaches are useful, my goal is to see how much can be said without such assumptions. I do not take a stand on investor preferences, the underlying objective dynamics, or the types of risks. As long as we trust the survey data to proxy for investor beliefs, my procedure imposes minimal additional assumptions. The cost is that I can obtain only bounds on the volatilities of SDFs and marginal utilities, rather than their actual values.

A separate stream of literature combines present-value relations with survey data to help resolve "excess volatility" puzzles (Chen, Da and Zhao, 2013; Delao and Myers, 2021, 2024; Delao, Han and Myers, 2023; Bordalo, Gennaioli, Porta and Shleifer, 2019, 2024a; Bordalo, Gennaioli, La Porta and Shleifer, 2024b; Bretscher, Malkhozov, Tamoni and Yang, 2024; Cassella, Chen, Gulen and Liu, 2025). My paper differs by focusing on excess returns rather than excess volatility, and examining the SDF rather than a particular asset's present-value relation. However, there is a connection between my work and this literature. If the market return is an important factor for the SDF, then its volatility is a key contributor to SDF volatility. In that case, subjective beliefs' role in SDF volatility and stock market volatility are linked, further reinforcing my conclusion that belief distortions must matter primarily via induced risk. Bretscher et al. (2024) make a related point, that belief distortions show up in asset-price data largely as a risk factor.

2 Theory: Subjective belief bounds

2.1 General environment

Let $\mathbb{P}$ denote the objective probability measure and $\mathbb{E}$ the associated expectation operator. I allow the marginal investor to possess *subjective beliefs* and use a distorted probability and expectation, which I denote by $\tilde{\mathbb{P}}$ and $\tilde{\mathbb{E}}$. To maintain a minimal amount of consistency for beliefs, I assume that $\mathbb{P}$ and $\tilde{\mathbb{P}}$ agree on null sets, so that their discrepancy can be modeled via the positive martingale B_t (likelihood ratio):

$$B_t = \left(\frac{d\tilde{\mathbb{P}}}{d\mathbb{P}}\right)_t \tag{2}$$

The subjective probability is defined by $\tilde{\mathbb{P}}(\mathcal{E}) := \mathbb{E}[\frac{B_t}{B_0}\mathbf{1}_{\mathcal{E}}]$ for any event $\mathcal{E}$ that resides in the time-t information set. This assumption on subjective beliefs is quite general and captures almost all belief-based models in the literature.[1] Because of its pervasiveness, Cui, Delao and Myers (2024) takes (2) as a starting point to extract interpretable common variation out of a variety of different surveys.

I assume absence of arbitrage opportunities. (Because $\mathbb{P}$ and $\tilde{\mathbb{P}}$ are mutually absolutely continuous, there is no ambiguity to saying "arbitrage" without reference to either probability.) In that case, let $(S_t)_{t\geq 0}$ denote the *stochastic discount factor* (SDF) process that prices all assets, i.e., S is positive and for any gross return R_{t+1} we have

$$1 = \mathbb{E}_t\Big[\frac{S_{t+1}}{S_t}R_{t+1}\Big], \tag{3}$$

where $\mathbb{E}_t$ denotes the conditional expectation operator.

I also assume an investor Euler equation. That is, there is a subjective SDF $\tilde{S}_t$ that prices assets under investor beliefs:

$$1 = \tilde{\mathbb{E}}_t\Big[\frac{\tilde{S}_{t+1}}{\tilde{S}_t}R_{t+1}\Big]. \tag{4}$$

The interpretation of equation (4) is that $\tilde{S}_t$ represents the *investors' marginal utility of*

[1] Assumption (2) even permits frameworks, such as Molavi, Tahbaz-Salehi and Vedolin (2024), in which agents can only entertain misspecified models based on a strict subset of relevant statistical factors. The models that are ruled out by (2) are those where the law of iterated expectations (LIE) is violated—e.g., the model-averaging belief specifications in Fuster, Laibson and Mendel (2010) and the distant-memory diagnostic belief model of Bianchi, Ilut and Saijo (2024). Other models, like the fading memory model of Nagel and Xu (2022), may violate LIE when interpreted literally but can be written in an observationally equivalent form where LIE holds. Therefore, that model would be included in the general setup here.

wealth—I refer to $\tilde{S}$ as marginal utility, in contrast to the objective SDF S. Whereas (2)-(3) are weak conditions, (4) contains two substantive assumptions. First, I am assuming that some investors are marginal in all the relevant markets considered in this paper. Second, because I identify the survey results with the subjective probability $\tilde{\mathbb{P}}$, I am assuming that these marginal investors hold beliefs identical to those in the surveys.

Without loss of any generality, impose the normalization $S_0 = \tilde{S}_0 = B_0 = 1$. Then, comparing (3)-(4) and using (2), we have

$$S_t = \tilde{S}_t B_t. \tag{5}$$

The SDF is the product of investor marginal utility and investor beliefs. Roughly speaking, the idea of this paper is to use survey data to learn about the volatilities of S and $\tilde{S}$ separately, thereby providing information about the belief distortion B.

In addition to the volatilities of S and $\tilde{S}$, information is contained in the volatilities of their permanent components. Under fairly general conditions, which I assume, the SDF obeys a unique factorization into *permanent and transitory components* (Alvarez and Jermann, 2005; Hansen and Scheinkman, 2009a; Qin and Linetsky, 2016, 2017):

$$S_t = G_t H_t, \tag{6}$$

where G is trend-stationary (transitory component) and H is a $\mathbb{P}$-martingale (permanent component). An analogous factorization holds for investors' marginal utility:

$$\tilde{S}_t = \tilde{G}_t \tilde{H}_t, \tag{7}$$

where $\tilde{G}$ is trend-stationary and $\tilde{H}$ is a $\tilde{\mathbb{P}}$-martingale.

Luckily, equation (5) then implies that

$$H_t = \tilde{H}_t B_t \tag{8}$$

and hence $G_t = \tilde{G}_t$. To see this, notice that $\mathbb{E}_0[H_t] = H_0 \mathbb{E}_0[\frac{B_t}{B_0}\frac{\tilde{H}_t}{\tilde{H}_0}] = H_0 \tilde{\mathbb{E}}_0[\frac{\tilde{H}_t}{\tilde{H}_0}] = H_0$, so that H and $\tilde{H}$ are indeed $\mathbb{P}$- and $\tilde{\mathbb{P}}$-martingales, respectively, when they satisfy (8). The interpretations of $\tilde{H}$ and H are different: $\tilde{H}$ captures the impact of permanent risks on investor marginal utility, while $H = \tilde{H}B$ captures the joint impact of permanent risks and belief distortions. The permanent components H and $\tilde{H}$ deserve attention, in light of evidence from Alvarez and Jermann (2005) and Bakshi and Chabi-Yo (2012) that G carries a relatively minor contribution to overall SDF volatility. What is not yet agreed is whether the prominence of H is due to $\tilde{H}$ or B.

This entire theoretical setup has referenced a single investor belief and marginal utility, when in fact there is clearly heterogeneity in the real world and in the surveys that I use empirically. Appendix D shows how the presence of heterogeneity can be accommodated in this setup. In particular, the common practice of averaging across survey respondents—the so-called consensus forecast—allows us to interpret B, $\tilde{S}$, and $\tilde{H}$ as the average belief, the belief-weighted average marginal utility, and its permanent component. The example disagreement models in Section 7 allow me to provide more interpretation for these weighted-average aggregates.

Finally, I make frequent reference to a few special returns in this theory section. First, I denote the *one-period risk-free rate* R_t^f, which I assume is traded.

Second, I occasionally consider the *growth-optimal return* R_{t+1}^*. This portfolio maximizes the growth rate of wealth, i.e., it solves $\max_R \mathbb{E}_t[\log(R)]$ subject to the pricing constraint (3). The solution is $R_{t+1}^* = \frac{S_t}{S_{t+1}}$ (Roll, 1973; Bansal and Lehmann, 1997). Thus, there is a well-known equivalence between extracting the SDF from the data and finding the growth-optimal portfolio. Note that there is also a *perceived growth-optimal return* $\tilde{R}_{t+1}^*$, which is defined analogously by $\arg\max_R \tilde{\mathbb{E}}_t[\log(R)]$. I also refer to $\tilde{R}_{t+1}^*$.

Third, I make use of the *long bond return* R_{t+1}^∞, defined as the one-period holding return on an arbitrarily long-maturity discount bond:

$$R_{t+1}^\infty := \lim_{T\to\infty} \frac{\mathbb{E}_{t+1}[\frac{S_T}{S_{t+1}}]}{\mathbb{E}_t[\frac{S_T}{S_t}]}.$$

The long bond is studied in Kazemi (1992), Martin and Ross (2013), and Ross (2015). In particular, it turns out that R_{t+1}^∞ identifies the transitory component of the SDF:

$$R_{t+1}^\infty = \lim_{T\to\infty} \frac{\mathbb{E}_{t+1}[\frac{S_T}{S_{t+1}}]}{\mathbb{E}_t[\frac{S_T}{S_t}]} = \lim_{T\to\infty} \frac{G_t}{G_{t+1}} \frac{\mathbb{E}_{t+1}[\frac{H_T}{H_{t+1}} G_T]}{\mathbb{E}_t[\frac{H_T}{H_t} G_T]} = \frac{G_t}{G_{t+1}}, \tag{9}$$

because the ratio of conditional expectations converges to 1 under mild conditions.

2.2 Volatility bounds

I consider *entropy measures of volatility*. For any positive random variable X, define

$$L_t(X) := \log\left(\mathbb{E}_t[X]\right) - \mathbb{E}_t\left[\log(X)\right] \tag{10}$$

$$\tilde{L}_t(X) := \log\left(\tilde{\mathbb{E}}_t[X]\right) - \tilde{\mathbb{E}}_t\left[\log(X)\right]. \tag{11}$$

Entropy measures played an important role in Alvarez and Jermann (2005) and have also been studied in other asset-pricing contexts (Backus, Chernov and Martin, 2011; Backus, Chernov and Zin, 2014; Martin, 2017). By Jensen's inequality, $L_t(X) \geq 0$ and $\tilde{L}_t(X) \geq 0$. In the special case that X is conditionally log-normal, $L_t(X_{t+1}) = \frac{1}{2}\text{Var}_t[\log X_{t+1}]$. For other distributions, entropy overweights the left tail of a distribution, relative to variance.

Whereas $L_t(X)$ represents objective volatility, $\tilde{L}_t(X)$ represents subjective, or perceived, volatility. To make interpretable comparisons, I sometimes invoke the following condition relating objective and subjective entropies. This condition allows me to bound time-series averages of conditional entropies.

Definition 1. X satisfies the *unconditional entropy inequality* (UEI) if $L(X_{t+1}) \geq \mathbb{E}[\tilde{L}_t(X_{t+1})]$.

When does a variable X to satisfy UEI? To understand this condition, first note the following property, analogous to the "law of total variance" but for entropies:

$$L(X) = \mathbb{E}[L_t(X)] + L(\mathbb{E}_t[X]).$$

Total entropy is the average of conditional entropy plus the entropy of the conditional mean. With this law in mind, and noting that $L(\mathbb{E}_t[X]) \geq 0$, one obvious situation where UEI holds is if $L_t(X) \geq \tilde{L}_t(X)$ on average, meaning that the subjective measure understates risk to X on average. This case is relevant because, empirically, survey measures of risk tend to be lower than actual risk—see Ben-David, Graham and Harvey (2013) as well as Figure 1 below. This case is also theoretically appealing, because when the underlying joint dynamics are conditionally log-normal over sufficiently short time periods (i.e., no jumps), then $L_t(X) = \tilde{L}_t(X)$ and so UEI holds (see Appendix A.8). Most macro-finance models have this property. A second situation where UEI is appropriate is if $L(\mathbb{E}_t[X])$ is large, meaning that X has significant predictability. This case is relevant to my specific application because actual returns do tend to have some predictability.

A first result establishes volatility bounds on the SDF and its various components. All proofs are contained in Appendix A. Going forward, I make frequent use of the notation $\Delta X_{t+1} := \frac{X_{t+1}}{X_t}$ for any variable X.

Lemma 1. *The following conditional volatility bounds hold for all returns R:*

$$L_t(\Delta S_{t+1}) \geq \mathbb{E}_t\big[\log R_{t+1}\big] - \log R_t^f \tag{12}$$

$$\tilde{L}_t(\Delta \tilde{S}_{t+1}) \geq \tilde{\mathbb{E}}_t\big[\log R_{t+1}\big] - \log R_t^f \tag{13}$$

$$L_t(\Delta H_{t+1}) \geq \mathbb{E}_t\big[\log R_{t+1} - \log R_{t+1}^\infty\big] \tag{14}$$

$$\tilde{L}_t(\Delta \tilde{H}_{t+1}) \geq \tilde{\mathbb{E}}_t\big[\log R_{t+1} - \log R_{t+1}^\infty\big] \tag{15}$$

The SDF bound (12), which arises purely due to absence of arbitrage, says that SDF volatility is at least as large as any risk-adjusted expected return. The risk-adjustment comes from the concavity of log. This is the "growth-optimal bound" uncovered by Bansal and Lehmann (1997). It also resembles the familiar bound in Hansen and Jagannathan (1991); there, risk-adjusted returns are measured by Sharpe ratios, which lower-bound the standard deviation of the SDF.

What is new is the marginal utility bound (13). Whereas $L_t(\Delta S_{t+1})$ measures the true conditional volatility of $S = B\tilde{S}$, which captures contributions from beliefs and risks, the object $\tilde{L}_t(\Delta \tilde{S}_{t+1})$ isolates the role of risks. If conditional log-normality holds and time-periods are short, then $\tilde{L}_t(\Delta \tilde{S}_{t+1}) \approx L_t(\Delta \tilde{S}_{t+1})$, so that the perceived marginal utility volatility equals true marginal utility volatility. In that case, comparing the volatilities from (12)-(13) provides information about the importance of the belief distortion B, which is the wedge between S and $\tilde{S}$. This is the basic idea behind the entire paper.

A similar discussion holds for the permanent components H and $\tilde{H}$. Whereas (14) measures the total volatility coming from beliefs (B) and long-run risks ($\tilde{H}$) together, the subjective bound (15) measures the perceived volatility of long-run risks ($\tilde{H}$) alone.

The most important economic insight is that elicitations of subjective excess returns, such as $\tilde{\mathbb{E}}_t[\log R_{t+1}] - \log R_t^f$ and $\tilde{\mathbb{E}}_t[\log R_{t+1} - \log R_t^{\infty}]$, can be informative about risks. Is it not possible that investors are just optimistic about an asset return R, which has nothing to do with their marginal utility $\tilde{S}$? The Euler equation (4) rules this out: if investors are optimistic about an asset, but they do not wish to buy more of it, they *must perceive it to be sufficiently risky*. This implies that their marginal utility $\tilde{S}$ covaries with the asset in question. Thus, when investors report large subjective risk premia, we learn that they must have large marginal utility volatility.

Building on the conditional results of Lemma 1, I also present unconditional bounds. For the objective probability, this is straightforward. On the other hand, the subjective bound does not generalize immediately to a useful unconditional version. Indeed, the *unconditional* subjective expectation $\tilde{\mathbb{E}}$ is not directly observable; taking time-series averages of *conditional* subjective expectations $\tilde{\mathbb{E}}_t$ will not yield an estimate of $\tilde{\mathbb{E}}$ unless beliefs are rational. We can make progress if we invoke UEI.[2]

[2]Note that the unconditional bounds are presented in terms of $R_t^f \Delta S_{t+1}$ and $R_t^f \Delta \tilde{S}_{t+1}$, rather than ΔS_{t+1} and $\Delta \tilde{S}_{t+1}$. This is for mathematical convenience, since the first two objects are martingales under $\mathbb{P}$ and $\tilde{\mathbb{P}}$, respectively. That said, this is quantitatively unimportant, since the risk-free rate is quite smooth. Indeed, note that

$$L(\Delta S) = L(1/R^f) + L(R^f \Delta S).$$

In the data, $L(1/R^f) \approx 0.0005$ (see Table G.1), which is only about $1/100$ of the log equity premium.

Proposition 1. *The following unconditional volatility bounds hold for all returns R:*

$$L\big(R_t^f \Delta S_{t+1}\big) \geq \mathbb{E}\big[\log R_{t+1} - \log R_t^f\big] \tag{16}$$
$$L\big(\Delta H_{t+1}\big) \geq \mathbb{E}\big[\log R_{t+1} - \log R_{t+1}^{\infty}\big] \tag{17}$$

Additionally, if UEI holds for $R_t^f \Delta\tilde{S}_{t+1}$ *and* $\Delta\tilde{H}_{t+1}$*, then*

$$L\big(R_t^f \Delta\tilde{S}_{t+1}\big) \geq \mathbb{E}\Big[\tilde{\mathbb{E}}_t\big[\log R_{t+1} - \log R_t^f\big]\Big] \tag{18}$$
$$L\big(\Delta\tilde{H}_{t+1}\big) \geq \mathbb{E}\Big[\tilde{\mathbb{E}}_t\big[\log R_{t+1} - \log R_{t+1}^{\infty}\big]\Big] \tag{19}$$

While the results so far hold in levels, we may want a ratio version in order to lower-bound the fraction of long-run SDF volatility that is due to marginal utility rather than beliefs. This approach requires an assumption on the assets we examine—namely, the asset studied in historical returns should be "closer" to objective growth-optimal than the surveyed asset is to subjective growth-optimal. This holds, for example, if the chosen asset is actually the objective growth-optimal return, but in other cases too. The benefit is this approach permits a more intuitive scale as a ratio of volatilities.

Definition 2. For any return R, define its *distance to growth-optimal* $\delta_t(R)$ and *distance to perceived growth-optimal* $\tilde{\delta}_t(R)$, respectively, by

$$\delta_t(R) := \mathbb{E}_t[\log R_{t+1}^*] - \mathbb{E}_t[\log R_{t+1}]$$
$$\tilde{\delta}_t(R) := \tilde{\mathbb{E}}_t[\log \tilde{R}_{t+1}^*] - \tilde{\mathbb{E}}_t[\log R_{t+1}].$$

Proposition 2. *Suppose UEI holds for* $\Delta\tilde{S}_{t+1}$ *and* $\Delta\tilde{H}_{t+1}$*. Let* R *and* $\tilde{R}$ *be any two returns such that, on average,* R *is closer to growth-optimal than* $\tilde{R}$ *is to perceived growth-optimal, in the sense that* $\mathbb{E}[\delta_t(R)] \leq \mathbb{E}[\tilde{\delta}_t(\tilde{R})]$*. Then, the following volatility bounds hold:*

$$\frac{L(R_t^f \Delta\tilde{S}_{t+1})}{L(R_t^f \Delta S_{t+1})} \geq \min\left\{1,\ \frac{\mathbb{E}\big[\tilde{\mathbb{E}}_t[\log \tilde{R}_{t+1} - \log R_t^f]\big]}{\mathbb{E}[\log R_{t+1} - \log R_t^f]}\right\} \tag{20}$$
$$\frac{L(\Delta\tilde{H}_{t+1})}{L(\Delta H_{t+1})} \geq \min\left\{1,\ \frac{\mathbb{E}\big[\tilde{\mathbb{E}}_t[\log \tilde{R}_{t+1} - \log R_{t+1}^{\infty}]\big]}{\mathbb{E}[\log R_{t+1} - \log R_{t+1}^{\infty}]}\right\} \tag{21}$$

What type of risk matters? I split marginal utility shocks into permanent risks ($\tilde{H}$) and transitory risks (G). The next result quantifies $\tilde{H}$. This is a subjective (and conditional) version of Alvarez and Jermann (2005) bound, which instead quantifies H.

Corollary 1.

$$\frac{\tilde{L}_t(\Delta \tilde{H}_{t+1})}{\tilde{L}_t(\Delta \tilde{S}_{t+1})} \geq \min\left\{1, \max\left\{0, \frac{\tilde{\mathbb{E}}_t[\log R_{t+1} - \log R^{\infty}_{t+1}]}{\tilde{\mathbb{E}}_t[\log R_{t+1}] - \log R^f_t}\right\}\right\} \tag{22}$$

for any return R such that $\tilde{\mathbb{E}}_t \log R_{t+1} > \log R^f_t$.

Finally, whereas the bounds so far estimate the size of marginal utility and the SDF, I also want to directly estimate the size of belief distortions. This can be done by measuring the deviations between subjective and objective expectations of the same return.

Proposition 3. *For any return $\tilde{R}$ for which $\mathbb{E}[\tilde{L}_t(\tilde{R}_{t+1})] \geq \mathbb{E}[\tilde{L}_t(\tilde{R}^{-1}_{t+1})]$,*

$$L(\Delta B_{t+1}) \geq \left|\mathbb{E}[\log \tilde{R}_{t+1}] - \mathbb{E}\left[\tilde{\mathbb{E}}_t[\log \tilde{R}_{t+1}]\right]\right| - \mathbb{E}[\tilde{L}_t(\tilde{R}_{t+1})]. \tag{23}$$

Furthermore, if R is sufficiently close to growth-optimal, i.e., $\mathbb{E}[\delta_t(R)]$ is sufficiently small, then

$$\frac{L(\Delta B_{t+1})}{L(R^f_t \Delta S_{t+1})} \geq \frac{\left|\mathbb{E}[\log \tilde{R}_{t+1}] - \mathbb{E}\left[\tilde{\mathbb{E}}_t[\log \tilde{R}_{t+1}]\right]\right| - \mathbb{E}[\tilde{L}_t(\tilde{R}_{t+1})]}{\mathbb{E}\left[\log R_{t+1} - \log R^f_t\right]} \tag{24}$$

Proposition 3 shows how to find large subjective belief distortions: find an asset that performs either much better or much worse than investors expect on average. Furthermore, implied belief distortions are larger when investors perceive the asset in question to be relatively safe, meaning that $\tilde{L}_t(\tilde{R}_{t+1})$ is low. Intuitively, it is difficult to justify a large mean-return distortion for a low-volatility asset, unless beliefs are particularly far from rational. The proposition's assumption that $\mathbb{E}[\tilde{L}_t(\tilde{R}_{t+1})] \geq \mathbb{E}[\tilde{L}_t(\tilde{R}^{-1}_{t+1})]$ is only used to obtain the clean, compact expression (23). It is relatively innocuous and holds, for instance, in conditionally log-normal or left-skewed environments.[3]

3 Data and variable construction

Table F.1 in Appendix F.1 summarizes the survey and asset market data sources.

For the realized returns and yields, the data sources are standard. Realized returns on the aggregate value-weighted stock market are obtained from CRSP. For volatility, I compute realized variance from daily return series obtained from Ken French's website and the CBOE VIX obtained from OptionMetrics. Realized Treasury yields and returns

[3]Under conditional log-normality, we have $\tilde{L}_t(R_{t+1}) = \tilde{L}_t(R^{-1}_{t+1}) = \frac{1}{2}\tilde{\mathrm{Var}}_t[\log R_{t+1}]$. If a return is perceived to be left-skewed, then $\tilde{L}_t(R_{t+1})] \geq \tilde{L}_t(R^{-1}_{t+1})$ holds.

come from the St. Louis Fed (FRED), the Fed Board updates to Gürkaynak et al. (2007) (GSW), and CRSP. When there is overlapping data on par yields, I use the FRED yields as the first choice, followed by GSW second, followed by CRSP last. For zero-coupon yields, GSW is the only source. Appendix F.2 compares yields from these various sources. For bond returns, CRSP is the only source. From the fixed-term indexes file, I obtain return series corresponding to an approximately 10-year and 30-year bond, respectively. I also obtain returns on a bond index tracking bonds with maturities above 10 years.

Survey-based US stock market expected returns come from Nagel and Xu (2023), who compile three sources: (i) the individual investor expectations from Nagel and Xu (2022), which agglomerate the UBS/Gallup survey, the Conference Board survey, and the Michigan Survey of Consumers, plus a few smaller surveys; (ii) the Livingston Survey from the Philadelphia Fed; and (iii) the Graham-Harvey CFO survey (Ben-David et al., 2013). From these sources, Nagel and Xu (2023) construct proxies for annual expected excess returns on the stock market (or S&P 500, depending on the source). See Nagel and Xu (2023) for more details on this data.

I convert expected market returns $\tilde{\mathbb{E}}_t[R^m_{t+1}]$ to expected log returns $\tilde{\mathbb{E}}_t[\log R^m_{t+1}]$ via the identity $\tilde{\mathbb{E}}_t[\log R_{t+1}] = \log \tilde{\mathbb{E}}_t[R_{t+1}] - \tilde{L}_t(R_{t+1})$. I then use the log-normal approximation $\tilde{L}_t(R_{t+1}) \approx \frac{1}{2}\tilde{\mathrm{Var}}_t[\log R_{t+1}]$. Nagel and Xu (2023) construct a measure of $\tilde{\mathrm{Var}}_t[\log R^m_{t+1}]$ from the elicited 10th and 90th percentile return forecasts in the CFO survey. To extend this subjective variance to a longer history, I regress it on measures of uncertainty that are available further back in time. First, I regress the subjective variance $\tilde{\mathrm{Var}}_t[\log R^m_{t+1}]$ onto the contemporaneous squared VIX, its one-month lag, and its trailing-12-month moving average. This regression has an R-squared of 0.60 and generates the fitted value $\hat{y}_t = 0.012 + 0.52\mathrm{VIX}^2_t + 0.005\mathrm{VIX}^2_{t-1} + 0.071(\frac{1}{12}\sum_{j=0}^{11}\mathrm{VIX}^2_{t-j})$. I then backfill fitted values for months that pre-date the VIX by using the Nagel and Xu (2023) procedure of projecting VIX onto the news-implied volatility (NVIX) measure of Manela and Moreira (2017). The result of this is a time series of fitted subjective variances $\tilde{\mathrm{Var}}_t[\log R^m_{t+1}]$ that I use to construct expected log market returns via the right-hand-side of

$$\tilde{\mathbb{E}}_t[\log R^m_{t+1}] \approx \log \tilde{\mathbb{E}}_t[R^m_{t+1}] - \frac{1}{2}\tilde{\mathrm{Var}}_t[\log R^m_{t+1}]. \tag{25}$$

The results are displayed in Figure 1.

Figure 1 (left panel) displays the CFO's reported subjective volatility as well as its fitted value from the projection onto VIX and lags. Notice that the subjective volatility measure shares many of the same patterns as objective volatility measures. That said the spikes in the subjective volatility are milder than the objective volatility spikes. On

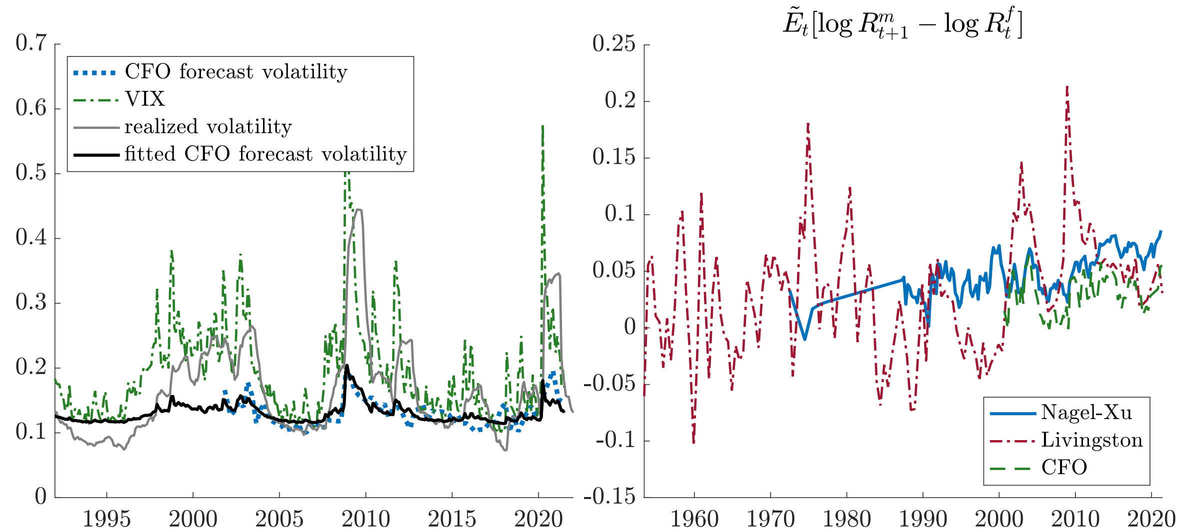

Figure 1. Subjective and objective stock volatilities and returns. *Left panel:* volatilities and volatility forecasts. The "CFO forecast volatility" (dotted blue) is CFO's next-year estimate of return volatility. The "fitted CFO forecast volatility" (solid black) comes from a regression of CFO's subjective variance forecasts on the squared VIX and lags of squared VIX. The "realized volatility" is the square root of the return variance from the past 252 trading days, annualized. *Right panel:* survey-based expected one-year log stock returns in excess of the one-year Treasury. Expected log returns are obtained from expected arithmetic returns via a variance adjustment described in equation (25).

average, the subjective CFO volatility (13.2%) is substantially below the VIX (20.9%) and slightly below realized volatility (16.6%). This is consistent with the UEI condition: recall that $\mathbb{E}[L_t(X)] \geq \mathbb{E}[\tilde{L}_t(X)]$ is a sufficient condition for UEI; taking variance as a good proxy for risk perception, the data suggest that this sufficient condition holds.

One potential concern is that I have a volatility measure only in the CFO survey. Consequently, I must use the CFO's subjective variance to convert expected arithmetic returns into expected log returns—as in equation (25)—not only for the CFO survey but also for the Nagel-Xu and Livingston surveys. Quantitatively, this is a relatively minor issue. To see this, suppose we were to take a conservative approach and approximate subjective variance $\tilde{\mathrm{Var}}_t[\log R^m_{t+1}]$ with objective volatility $\mathrm{Var}_t[\log R^m_{t+1}]$. On average, this would raise my subjective variance measure from $0.132^2 = 0.017$ to $0.166^2 = 0.028$, which, according to (25), would lower my subjective log returns by about 0.5%.

Figure 1 (right panel) displays the resulting expected excess log stock returns. There are some shared cyclical patterns but also discrepancies. Importantly, my results primarily make use of the average levels of these log returns rather than their volatilities or cyclical dynamics (except insofar as those second moments impact standard errors).

Treasury Bond expected returns are constructed from yield forecasts in the Blue Chip Financial Forecasts (BCFF) survey. BCFF has consensus (i.e., averaged across respondents) yield forecasts for Treasuries with maturities 6 months, 1 year, 2 years, 5 years, 10 years, and 30 years.[4] Using these par yield forecasts, I follow standard procedures in the literature to interpolate an expected par yield curve and then bootstrap an expected future zero-coupon yield curve $(\tilde{\mathbb{E}}_t[y^k_{t+1}])^{30}_{k=1}$.[5] With this in hand, I construct one-year subjective expected log returns via $\tilde{\mathbb{E}}_t[\log R^k_{t+1}] = ky^k_t - (k-1)\tilde{\mathbb{E}}_t[y^{k-1}_{t+1}]$ at the $k = 10$-year and $k = 30$-year maturities.

I also project the bond expectations back and forward in time by regressing them on various yields and their lags. I regress $\tilde{\mathbb{E}}_t y^9_{t+1}$ (resp., $\tilde{\mathbb{E}}_t y^{29}_{t+1}$) onto the time-$t$ 1-year and 10-year (30-year) par yields, as well as their one-month lagged values and a trailing-12-month average of their values. The in-sample R-squareds from these regressions are 0.9821 and 0.9612, respectively. I use this estimation to construct hypothetical values of $\tilde{\mathbb{E}}_t[\log R^{10}_{t+1}]$ and $\tilde{\mathbb{E}}_t[\log R^{30}_{t+1}]$ going back as far as I have actual yield data.

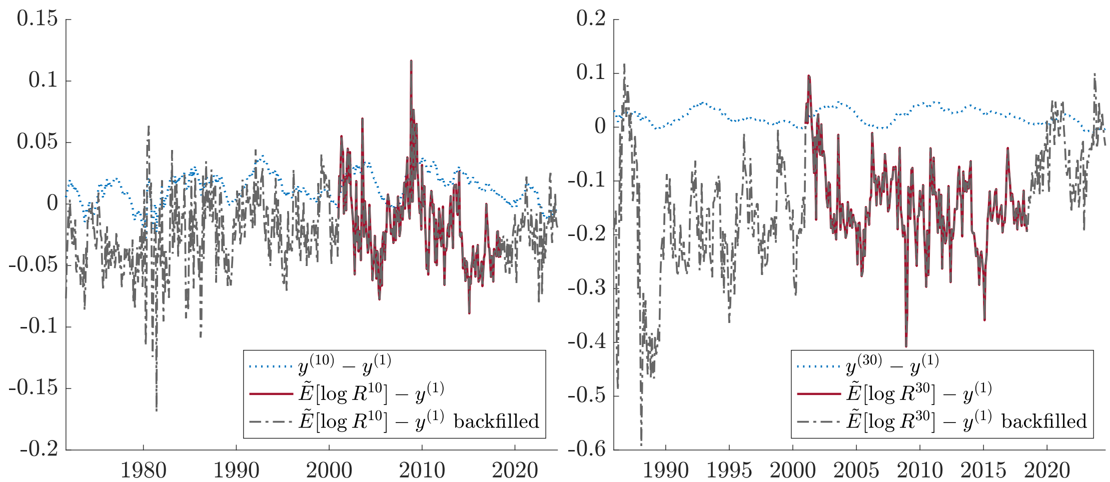

Figure 2. Subjective expected bond returns. Expected log (zero-coupon) bond returns constructed using BCFF consensus forecasts of the one-year ahead yield curve. "Backfilled" refers to expected returns constructed from projecting yield forecasts onto the contemporaneous and lagged yield curve.

[4]From March 2002 – February 2006, no 30-year bond was issued, so the survey records a forecast of the 20-year yield. I account for this in all results with both the survey expected returns and realized returns.

[5]I follow Kim and Orphanides (2012) in treating the BCFF forecasts as approximately mid-quarter forecasts, and so I use a weighted-average of the 3-, 4-, and 5-quarter ahead forecasts, with weights depending on the survey calendar month, to get a forecast at a one-year horizon. Since the BCFF surveys are taken nearer to the beginning of the month, I also shift the responses and align them with with previous-month yields and asset prices, which are measured at month-end (this shift does not materially affect any result). I then interpolate these yields to obtain a full term structure by linear interpolation for maturities up to 5 years and a fitted Nelson-Siegel model for maturities beyond 5 years (with the Nelson-Siegel model fitted to expectations data on maturities 2+ years). The result is a term structure of one-year-ahead expectations of the par yield curve. I then bootstrap the approximate expected continuously compounded zero-coupon yield curve from this expected par yield curve. Technically, this bootstrapping is a nonlinear transformation. In order to construct the zero-coupon yield expectations using this approximation, I am effectively assuming the amount of uncertainty in the par yield expectations is small. Appendix F.3 contains additional documentation of the transformation from the raw data (sparse par yield forecasts) to an interpolated par yield curve forecast and finally to a bootstrapped zero-coupon yield curve forecast.

The resulting subjective expected excess log bond returns, as well as their extended backfilled values, are depicted in Figure 2. Notice from the figure that the subjective expected returns tend to be below the long-maturity yield. Since the yield measures the long-term expected return, this implies that survey respondents are pessimistic about bond returns in this sample. The situation is particularly extreme in 30-year bonds. Pessimistic return expectations stem from beliefs about yields: both the 10-year and 30-year yield forecasts in the survey are consistently about 0.5% above yield levels (see Figure F.7). The high duration of the 30-year bond converts this survey expectation into a strongly negative expected return.

4 Estimating the bounds

4.1 The size of risks in the SDF

I turn to the principal question of the paper: where does the world live on the spectrum between risk-based volatility and belief-distortion-based volatility? Table 1 presents the main empirical result that risk accounts for at least half the volatility of the SDF. The table displays each unconditional means for the equity premium in the data and the survey, followed by ratios of the subjective-to-objective bounds, as in Proposition 2. Standard errors, shown in parentheses, are computed using a Newey and West (1987) window with 3 years of lags to account for the overlap in returns and persistence in spreads. Since the equity surveys feature different sampling frequencies, this means a different number of period lags in each panel (Nagel-Xu and CFO data use 12 quarter lags; the Livingston survey use 6 semi-annual lags). Standard errors for the volatility ratios are always computed using the Delta Method.

The main estimates, in the third column, compute the ratio of subjective to objective equity premia. Across all surveys, survey equity premia are over half of historical equity premia, leading to the conclusion that $L(R^f \Delta\tilde{S})/L(R^f \Delta S)$ is at least one-half. For econometric efficiency, I use the full-sample log equity premium, but the results are very similar if I estimate it within the same sample as the respective equity survey.[6]

According to Proposition 2, as long as the stock market is closer to growth-optimal than it is to perceived growth-optimal, the table provides valid lower bounds on the contribution of risk to various components of the SDF. A more modest interpretation of Table 1 is also permitted: if S represents an SDF that prices the stock market and riskless

[6]The full-sample estimate of $\mathbb{E}[\log R^m - \log R^f]$ is 0.0555, whereas the estimate using only dates for which I have the Nagel-Xu, Livingston, and CFO surveys are 0.0573, 0.0562, and 0.0459, respectively.

	Obj. Equity Premium $E[\log R^m/R^f]$	Subj. Equity Premium $E[\tilde{E}_t(\log R^m_{t+1}/R^f_t)]$	Size of Risks in SDF $L(R^f\Delta\tilde{S})/L(R^f\Delta S)$
A. Nagel-Xu.			
	0.0555	0.0476	0.8567
	(0.0161)	(0.0044)	(0.2458)
B. Livingston.			
	0.0555	0.0300	0.5412
	(0.0161)	(0.0081)	(0.1506)
C. CFO.			
	0.0555	0.0297	0.5349
	(0.0161)	(0.0038)	(0.1484)

Table 1. Size of risks in the SDF. This table displays various estimates of (20) using $R = \tilde{R} = R^m$. Panels A, B, and C use the Nagel-Xu, Livingston, and CFO survey expectations for the market, respectively. Newey-West asymptotic standard errors using 3 years of lags (i.e., 12 periods for Nagel-Xu, 6 periods for Livingston, 12 periods for CFO) are shown in parentheses.

	Obj. Equity minus Term Premium $E[\log R^m/R^{(k)}]$	Subj. Equity minus Term Premium $E[\tilde{E}_t(\log R^m_{t+1}/R^{(k)}_{t+1})]$	Size of Risks in Permanent Component $L(\Delta\tilde{H})/L(\Delta H)$
A. Nagel-Xu.			
$k = 10$ years	0.0455	0.0651	1.4290
	(0.0172)	(0.0081)	(0.5055)
$k = 30$ years	0.0464	0.1930	4.1568
	(0.0184)	(0.0138)	(1.6515)
B. Livingston.			
$k = 10$ years	0.0455	0.0476	1.0443
	(0.0172)	(0.0069)	(0.3251)
$k = 30$ years	0.0464	0.1755	3.7795
	(0.0184)	(0.0127)	(1.2749)
C. CFO.			
$k = 10$ years	0.0455	0.0472	1.0367
	(0.0172)	(0.0071)	(0.3554)
$k = 30$ years	0.0464	0.1751	3.7720
	(0.0184)	(0.0132)	(1.4892)

Table 2. Size of risks in the permanent component of the SDF. This table displays various estimates of (21) using $R = \tilde{R} = R^m$ and R^∞ proxied by R^{10} or R^{30}. Panels A, B, and C use the Nagel-Xu, Livingston, and CFO survey expectations for the market, respectively. All panels use BCFF expectations to compute subjective expected long-term bond returns. Newey-West asymptotic standard errors using 3 years of lags (i.e., 12 periods for Nagel-Xu, 6 periods for Livingston, 12 periods for CFO) are shown in parentheses.

rate, then all I require for the subjective-to-objective ratios to be meaningful is that the stock market is close to growth-optimal *among this subset of assets*. Of course, the stock market may be far from growth-optimal, an issue we address directly in Section 4.2.

Table 2 provides estimates of the size of risks in the permanent component of the SDF, that is, $L(\Delta\tilde{H})/L(\Delta H)$. The extreme pessimism in long bond forecasts, in the 10-year bond but especially the 30-year bond, results in enormous estimates of $L(\Delta\tilde{H})/L(\Delta H)$. Taking these estimates at face value, over 100% of the volatility of the permanent component of the SDF is risk-based.[7] Such a large estimate may be due partly to the specificity of the data sample, which features a negative subjective term premium, particularly for the 30-year bond. The very large standard errors also suggest that the large estimate may also be due to the tremendous volatility in long-term bond expectations.

4.2 Optimizing the returns

As mentioned, my bounds are more likely to be valid if an asset is closer to objective growth-optimal, which may be a poor assumption for the aggregate stock market. I therefore consult the literature and obtain a state-of-the-art proxy for the growth-optimal return. (Appendix E studies simpler growth-optimal alternatives that only utilize the stock market and risk-free bond, including constant leverage and volatility-timing.)

I follow Chen, Pelger and Zhu (2024a), which uses a neural network and an adversarial method to estimate jointly the SDF and relevant test assets—in particular, which weights to put on the cross-section of assets, and at which times. The result is an SDF which takes the form $\frac{S_{t+1}}{S_t} = 1 - \sum_{n=1}^{N} \omega(I_t, I_{n,t}) R^e_{n,t+1}$, where $R^e_{n,t+1}$ is the excess return of stock n, and where ω are portfolio weights. These weights are estimated as a function of $(I_t, I_{n,t})$, where I_t are macroeconomic time-series factors and $I_{n,t}$ is a vector of characteristics for stock n. Market timing is captured by the extent to which ω depends on macro time-series factors in I_t, while stock-picking is captured by the way ω depends on characteristics in $I_{n,t}$. Given the assumed structure of the SDF, I construct the proxy for the growth-optimal portfolio by[8]

$$R^*_{t+1} = \frac{S_t}{S_{t+1}} = \frac{1}{1 - \sum_{n=1}^{N} \omega(I_t, I_{n,t}) R^e_{n,t+1}}. \tag{26}$$

The implied growth-optimal return R^* performs extremely well, obtaining an average log excess return of 9.55% per annum (column 1 of Table 3). Or, in alternative units, this portfolio has an annualized Sharpe ratio of 1.61. This implies a volatile SDF, given the equality $L(R^f \Delta S) = \mathbb{E}[\log R^* - \log R^f]$.

[7] While Proposition 2 shows that the lower bounds technically cannot be above 1, the final column of Table 2 is documenting estimates of the right-hand-side of (21) ignoring the minimum operator.

[8] The estimation in Chen et al. (2024a) gives a monthly SDF, so I annualize these returns for my tests.

Not surprisingly, such a high-powered proxy for the SDF implies a much more conservative bound for risk-based volatility $L(R^f \Delta\tilde{S})/L(R^f \Delta S)$. The rows labeled $\tilde{R} = R^m$ in Table 3 show that, by comparing the objective growth-optimal portfolio to subjective returns of the market, the size of risks in the SDF is lower-bounded by between 31% and 50% (column 3 of Table 3).

To place the subjective returns on more equal footing, I augment them with market timing as well. Given a survey-based estimate of one-year expected market returns $\tilde{\mathbb{E}}_t[R^m_{t+1}]$ and variance $\tilde{\text{Var}}_t[R^m_{t+1}]$, I form the mean-variance optimal portfolio

$$\tilde{R}^*_{t+1} := \tilde{\theta}_t R^m_{t+1} + (1 - \tilde{\theta}_t) R^f_t, \quad \text{where} \quad \tilde{\theta}_t := \frac{\tilde{\mathbb{E}}_t[R^m_{t+1}] - R^f_t}{\tilde{\text{Var}}_t[R^m_{t+1}]}. \tag{27}$$

This approximately delivers the subjective growth-optimal portfolio when restricting trade to the aggregate stock market and short-term bonds. Because the subjective beliefs I use are one-year-ahead, this portfolio should be thought of as rebalanced annually, which is of course a conservative approach to dynamic trading. Furthermore, to prevent extreme trading, I truncate the optimal portfolio so that $\tilde{\theta}_t \in [0, 2]$, which prevents short-sales and leverage ratios above 2 and thereby dampens the subjective performance of this portfolio.[9] The resulting portfolio $\tilde{R}^*$ achieves a markedly better subjective performance than a portfolio that buys-and-holds the market (column 2 of Table 3). Under the fairer comparison of R^* against $\tilde{R}^*$, the results are much closer to our original results: a minimum of 45% to 79% of SDF volatility is due to risk (column 3 of Table 3).

One should regard these results as conservative. Indeed, $\tilde{R}^*$ uses only the stock market and riskless bond, imposes both a leverage and short-selling constraint, and rebalances annually. By contrast, the SDF estimation underlying R^* is substantially more complex: R^* is the optimal *unconstrained* portfolio formed by market timing with many time-series predictors, by investing in the entire cross-section of stocks, including many small stocks, and by monthly rebalancing. Given high turnover and substantial positions in illiquid stocks, one expects R^* to come with non-trivial transaction costs that we are not taking into account here. This relates to the findings of Jensen et al. (2024), in which many types of optimal portfolios prescribed by machine-learning methods are shown to feature large amounts of turnover and transactions costs, hence poor net-of-cost performance.

[9] This truncation has the additional benefit that the objective return series is well-defined and default-free in sample. That is, if I restrict $\tilde{\theta}_t \in [0, 2]$, then $\tilde{R}^*_{t+1} > 0$ for all one-year holding periods in sample.

	Objective Premium of R^* $E[\log R^*/R^f]$	Subjective Premium of $\tilde{R}$ $E[\tilde{E}_t(\log \tilde{R}_{t+1}/R^f_t)]$	Size of Risks in SDF $L(R^f\Delta\tilde{S})/L(R^f\Delta S)$
A. Nagel-Xu.			
$\tilde{R} = R^m$	0.0955	0.0476	0.4981
	(0.0118)	(0.0044)	(0.0447)
$\tilde{R} = \tilde{R}^*$	0.0955	0.0754	0.7897
	(0.0118)	(0.0082)	(0.0738)
B. Livingston.			
$\tilde{R} = R^m$	0.0955	0.0300	0.3146
	(0.0118)	(0.0081)	(0.0459)
$\tilde{R} = \tilde{R}^*$	0.0955	0.0639	0.6696
	(0.0118)	(0.0107)	(0.0704)
C. CFO.			
$\tilde{R} = R^m$	0.0955	0.0297	0.3110
	(0.0118)	(0.0038)	(0.0259)
$\tilde{R} = \tilde{R}^*$	0.0955	0.0434	0.4542
	(0.0118)	(0.0063)	(0.0392)

Table 3. Growth-optimal portfolio. This table provides estimates of (20), using a proxy for the growth-optimal portfolio from Chen et al. (2024a). All rows use $R = R^*$ to estimate $L(R^f\Delta S)$ but use different $\tilde{R}$ to estimate $L(R^f\Delta\tilde{S})$. The first row in each panel uses $\tilde{R} = R^m$. The second row uses $\tilde{R} = \tilde{R}^*$ defined in (27), with the additional constraint $\tilde{\theta}_t \in [0, 2]$. Panels A, B, and C use the Nagel-Xu, Livingston, and CFO survey expectations for the market, respectively. Newey-West asymptotic standard errors using 3 years of lags (i.e., 12 periods for Nagel-Xu, 6 periods for Livingston, 12 periods for CFO) are shown in parentheses.

4.3 The size of belief distortions

The bounds so far tell us about the role of marginal utility in the SDF, which indirectly informs us about belief distortions (since $S = B\tilde{S}$). Here, I instead provide direct estimates of the size of belief distortions.

Table 4 provides lower-bound estimates for $L(\Delta B)$, following Proposition 3. The inputs are the average forecast error about a candidate return, $\mathbb{E}[\log \tilde{R}_{t+1} - \tilde{\mathbb{E}}_t(\log \tilde{R}_{t+1})]$, and a variance adjustment, $\mathbb{E}[\tilde{L}_t(\tilde{R}_{t+1})]$. For the variance adjustment, I use the log-normal approximation $\tilde{L}_t(\tilde{R}_{t+1}) \approx \frac{1}{2}\tilde{\mathrm{Var}}_t[\log \tilde{R}_{t+1}]$. As candidate returns, I consider both the market portfolio, $\tilde{R} = R^m$, as well as the subjective growth-optimal proxy, $\tilde{R} = \tilde{R}^*$ in (27). Across surveys and candidate returns, the result is a relatively small lower-bound for $L(\Delta B)$. (The caveat is that these estimates are relatively imprecise, because estimating average forecast errors precisely requires a long sample.)

To assess magnitudes, I scale by the volatility of the SDF. Column 4 of Table 4 shows lower bounds for $L(\Delta B)/L(R^f\Delta S)$, where $L(R^f\Delta S) = \mathbb{E}[\log R^* - \log R^f]$ is the log premium on the Chen et al. (2024a) growth-optimal return R^* from (26). Across all surveys and candidate returns, $L(\Delta B)/L(R^f\Delta S)$ ranges from 0% to 35%. Contrast this with esti-

	Subj. Bias about $\tilde{R}$ $\lvert E[(\tilde{E}_t - E_t)\log \tilde{R}_{t+1}]\rvert$	Subj. Volatility about $\tilde{R}$ $E[\tilde{L}_t(\tilde{R}_{t+1})]$	Size of Belief Distortions $L(\Delta B)$	Size of Beliefs in SDF $L(\Delta B)/L(R^f \Delta S)$
A. Nagel-Xu.				
$\tilde{R} = R^m$	0.0085	0.0084	0.0000	0.0005
	(0.0228)	(0.0004)	(0.0229)	(0.1161)
$\tilde{R} = \tilde{R}^*$	0.0039	0.0322	-0.0283	-0.2961
	(0.0538)	(0.0014)	(0.0534)	(0.2746)
B. Livingston.				
$\tilde{R} = R^m$	0.0265	0.0082	0.0183	0.1917
	(0.0193)	(0.0002)	(0.0193)	(0.1008)
$\tilde{R} = \tilde{R}^*$	0.0114	0.0212	-0.0098	-0.1030
	(0.0308)	(0.0026)	(0.0303)	(0.1523)
C. CFO.				
$\tilde{R} = R^m$	0.0243	0.0086	0.0156	0.1637
	(0.0267)	(0.0006)	(0.0269)	(0.1036)
$\tilde{R} = \tilde{R}^*$	0.0599	0.0263	0.0336	0.3523
	(0.0422)	(0.0028)	(0.0407)	(0.1615)

Table 4. Size of beliefs in the SDF. This table displays various estimates of (23) and (24). To estimate $L(\Delta B)$ from (23), the first row in each panel uses $\tilde{R} = R^m$, while second row uses $\tilde{R} = \tilde{R}^*$ defined in (27), with the additional constraint $\tilde{\theta}_t \in [0, 2]$. The variance adjustment in column 2 is estimated using the log-normal approximation $\mathbb{E}[\tilde{L}_t(\tilde{R}_{t+1})] \approx \frac{1}{2}\tilde{\mathrm{Var}}_t[\log \tilde{R}_{t+1}]$. To estimate $L(\Delta B)/L(R^f \Delta S)$ from (24), the table uses $L(R^f \Delta S) = \mathbb{E}[\log R^* - \log R^f]$, where R^* is the growth-optimal portfolio from Chen et al. (2024a). Panels A, B, and C use the Nagel-Xu, Livingston, and CFO survey expectations for the market, respectively. Newey-West asymptotic standard errors using 3 years of lags (i.e., 12 periods for Nagel-Xu, 6 periods for Livingston, 12 periods for CFO) are shown in parentheses.

mates of $L(R^f \Delta\tilde{S})/L(R^f \Delta S)$ between 45% and 79% in Table 3. Thus, taken together, my estimates suggest that risk dominates belief distortions in accounting for SDF volatility.

4.4 Long-term risks

At this point, we have established that risks play a prominent role in SDF volatility. I now demonstrate that long-term risks are the quantitatively relevant ones.

To start, Table G.1 in the appendix reviews and refreshes the original Alvarez and Jermann (2005) results on the size of the permanent component in the SDF. Overall, the message from the point estimates is that at least 2/3 of the volatility of S stems from H. The reason: the equity premium (5.55%) is substantially larger than the term premium (between 1% and 2%, depending on the method), and so $\mathbb{E}[\log R^m - \log R^\infty]$ is nearly as large as the entire equity premium $\mathbb{E}[\log R^m - \log R^f]$.

Now, I perform a similar analysis in the subjective measure. Table 5 presents estimates for the average size of the permanent component $\tilde{H}$ in marginal utility $\tilde{S}$. The estimates in the final column are the time-series average of the conditional subjective

	Subj. Equity Premium $E[\tilde{E}_t(\log R^m_{t+1}/R^f_t)]$	Subj. Term Premium $E[\tilde{E}_t(\log R^{(k)}_{t+1}/R^f_t)]$	Size of Subj. Permanent Component $E[\tilde{L}_t(\Delta\tilde{H}_{t+1})/\tilde{L}_t(\Delta\tilde{S}_{t+1})]$
A. Nagel-Xu.			
$k = 10$ years	0.0476	-0.0175	0.8817
	(0.0044)	(0.0075)	(0.0547)
$k = 30$ years		-0.1454	0.9740
		(0.0137)	(0.0234)
B. Livingston.			
$k = 10$ years	0.0300	-0.0175	0.9080
	(0.0081)	(0.0075)	(0.0383)
$k = 30$ years		-0.1454	0.9719
		(0.0137)	(0.0271)
C. CFO.			
$k = 10$ years	0.0297	-0.0175	0.8686
	(0.0038)	(0.0075)	(0.0561)
$k = 30$ years		-0.1454	0.9848
		(0.0137)	(0.0149)

Table 5. Size of permanent risks in marginal utility. This table displays estimates of the subjective volatility bound from Corollary 1. Panels A, B, and C use the Nagel-Xu, Livingston, and CFO survey expectations for the market, respectively. Newey-West asymptotic standard errors using 3 years of lags (i.e., 12 periods for Nagel-Xu, 6 periods for Livingston, 12 periods for CFO) are shown in parentheses.

bound from Corollary 1. The results, across the three equity surveys and four different proxies for the long bond expected return, broadly echo the message of Alvarez and Jermann (2005): investors perceive that the lion's share of marginal utility volatility (90% or more) emanates from permanent risks.

To summarize, most SDF volatility is due to marginal utility volatility, and the overwhelming majority of marginal utility volatility is due to permanent risks.

5 An upper bound for the direct role of beliefs

In this section, I show how to bound the direct role of belief distortions on SDF volatility. A novel input is a particular forecast error predictability regression that estimates the level and cyclicality of investor pessimism. If investors are pessimism on average or pessimistic in bad time, as suggested by behavioral finance evidence, then the direct role of belief distortions can be sharply upper-bounded.

For this section, I assume a bit more than the previous ones: I assume a conditionally log-normal environment, which nests most macro-finance models, including the examples I illustrate in later sections. The empirical implementation with forecast error regressions also effectively assumes that belief distortions about the market return are good proxies for belief distortions more generally. With these slightly stronger assumptions, I can strengthen my conclusions about beliefs.

5.1 A theoretical upper bound and the role of pessimism

Starting from the relationship $S = B\tilde{S}$, I use conditional log-normality and my existing lower bounds for $L(R^f\Delta\tilde{S})/L(R^f\Delta S)$ in order to obtain the following upper bound on $L(\Delta B)/L(R^f\Delta S)$:

Proposition 4. *Let R and $\tilde{R}$ be any two returns such that, on average, R is closer to growth-optimal than $\tilde{R}$ is to perceived growth-optimal, in the sense that $\mathbb{E}[\delta_t(R)] \leq \mathbb{E}[\tilde{\delta}_t(\tilde{R})]$. Suppose $(B, S, \tilde{S}, R, \tilde{R})$ are conditionally log-normal. Then,*

$$\frac{L(\Delta B_{t+1})}{L(R_t^f \Delta S_{t+1})} \leq \max\left\{0,\, 1 - \frac{\mathbb{E}\left[\tilde{\mathbb{E}}_t[\log \tilde{R}_{t+1} - \log R_t^f]\right]}{\mathbb{E}[\log R_{t+1} - \log R_t^f]} - \frac{\Sigma(B,\tilde{S})\Phi(R)}{\mathbb{E}[\log R_{t+1} - \log R_t^f]}\right\}, \quad (28)$$

where $\Sigma(B, \tilde{S})$ is given by the following average covariance

$$\Sigma(B,\tilde{S}) := \mathbb{E}\Big[Cov_t[\log \Delta B_{t+1}, \log \Delta\tilde{S}_{t+1}]\Big]$$

and where $\Phi(R) \leq 1$ is a measure of how close R is to growth-optimal, defined by

$$\Phi(R) := \frac{\mathbb{E}[\log R_{t+1} - \log R_t^f]}{\mathbb{E}[\log R_{t+1}^* - \log R_t^f]}.$$

If subjective expected returns are close to their objective counterparts, as I find empirically, then belief distortions cannot play a large role in this direct sense, unless the covariance term is strongly negative. By contrast, a positive covariance term further reduces the scope for belief distortions. Thus, the next step in quantifying (28) is to understand and quantify the covariance term $\Sigma(B, \tilde{S})$.

The covariance between B and $\tilde{S}$ captures pessimism. Why? When this covariance is positive, beliefs tend to overweight high marginal utility states—investors believe bad times are more likely than they actually are. When the covariance is negative, investors are instead optimistic.

Two sets of existing evidence in behavioral finance suggest that $\Sigma(B, \tilde{S}) \geq 0$. First, evidence from subjective expected returns, as in this paper, suggests "pessimism-on-average," since subjective expected returns tend to be below objective expected returns. Second, the existing literature argues that belief distortions tend to be procyclical (De Bondt, 1993; Amromin and Sharpe, 2014; Greenwood and Shleifer, 2014; Nagel and Xu, 2023), meaning pessimism increases in bad times.[10] I use an example to illustrate why both pessimism-on-average and procyclical belief distortions induce $\Sigma(B, \tilde{S}) \geq 0$.

Example 1 (Understanding pessimism)**.** Consider an IID log return $\log R \sim N(r, v^2)$. Investor beliefs about this return are $\log R \sim N(r + \delta_t, v^2)$, with δ_t capturing their belief distortion. We consider two cases below: (i) pessimism-on-average, which is captured by $\delta_t = \bar{\delta} < 0$, and (ii) procyclical distortions, which is captured by $\delta_t > 0$ in good times and $\delta_t < 0$ in bad times. For simplicity, suppose investors are rational about everything else in the economy.

In this example, the investor belief distortion is a likelihood ratio between the objective and subjective return distributions, namely a ratio of normal densities. That is,

$$\frac{B_{t+1}}{B_t} = \frac{\exp\left(-\frac{(\log R_{t+1} - r - \delta_t)}{v^2}\right)}{\exp\left(-\frac{(\log R_{t+1} - r)}{v^2}\right)} = \exp\left[-\frac{1}{2}\left(\frac{\delta_t}{v}\right)^2 + \frac{\delta_t}{v}\left(\frac{\log R_{t+1} - r}{v}\right)\right].$$

At the same time, suppose the shocks to R are priced by the investor, in the sense that their marginal utility responds to them:

$$\log \tilde{S}_{t+1} - \log \tilde{S}_t = -\tilde{\pi}_t \left(\frac{\log R_{t+1} - r}{v}\right) + \text{other terms independent of } R.$$

In this case, $\tilde{\pi}_t$ is known as the investor-perceived risk price for the shock to R. I assume that $\tilde{\pi}_t > 0$ so that positive returns reduce investor marginal utility.

Since $\frac{\log R_{t+1} - r}{v} \sim N(0, 1)$, we can compute the covariance of interest as

$$\Sigma(B, \tilde{S}) = \mathbb{E}\Big[\mathrm{Cov}_t\big[\log \Delta B_{t+1}, \log \Delta \tilde{S}_{t+1}\big]\Big] = -\mathbb{E}\Big[\frac{\delta_t \tilde{\pi}_t}{v}\Big]. \tag{29}$$

Now, let us revisit the two cases.

1. Under pessimism-on-average, we have $\delta_t = \bar{\delta} < 0$. Along with $\tilde{\pi}_t > 0$, this immediately implies that $\Sigma(B, \tilde{S}) > 0$.

2. Under procyclical distortions, pessimism is countercyclical. Let us make the typical assumption that marginal utility volatility is also countercyclical. Therefore, $\tilde{\pi}_t$ and $-\delta_t$ are both high in bad times, and both low in good times. For theoretical clarity,

[10] Importantly, I am talking about the procyclicality of belief *distortions*, i.e., the gap between subjective beliefs and their objective counterpart. Even in papers that argue for acyclical beliefs (e.g. Nagel and Xu, 2023), the belief distortion is procyclical because objective risk premia are countercyclical.

suppose $\mathbb{E}[\delta_t] = 0$ so that agents are neither optimistic nor pessimistic on average. Then, we have $\Sigma(B, \tilde{S}) = \frac{1}{v}\text{Cov}[\tilde{\pi}_t, -\delta_t] > 0$.

Consequently, pessimism-on-average and procyclical distortions induce $\Sigma(B, \tilde{S}) > 0$. $\square$

The example models I study in Sections 6-7 are also consistent with $\Sigma(B, \tilde{S}) \geq 0$. Section 6 features procyclical belief distortions due to growth extrapolation (and one can show $\Sigma(B, \tilde{S}) = 0$), while Section 7 features pessimism-on-average hard-wired into one agent's beliefs (and one can show $\Sigma(B, \tilde{S}) = \eta\beta\sigma_c > 0$).

5.2 Estimating pessimism

As mentioned, existing evidence suggests that $\Sigma(B, \tilde{S}) \geq 0$. I go beyond this evidence and provide a direct estimate of $\Sigma(B, \tilde{S})$. The next lemma shows how to estimate this object in after projecting down to a "single-factor economy."

Lemma 2. *Suppose $(B, S, \tilde{S}, R)$ are conditionally log-normal. Consider the projection of $(B, S, \tilde{S})$ onto the space spanned by a single return R. Define the forecast error $FE_{t+1} := R_{t+1} - \tilde{\mathbb{E}}_t[R_{t+1}]$, subjective volatility $\tilde{v}_t := \tilde{Var}_t[R_{t+1}]^{1/2}$, and subjective Sharpe ratio $\tilde{\pi}_t := \frac{1}{\tilde{v}_t}(\tilde{\mathbb{E}}_t[R_{t+1}] - R^f_t)$. Then, the covariance $\Sigma(B, \tilde{S})$ is given by*

$$\Sigma(B, \tilde{S}) = \mathbb{E}\Big[\frac{FE_{t+1}}{\tilde{v}_t} \times \tilde{\pi}_t\Big]. \tag{30}$$

The expression for $\Sigma(B, \tilde{S})$ in Lemma 2 can be understood as a slight generalization of (29). The scaled forecast error $\frac{FE_{t+1}}{\tilde{v}_t}$ is an unbiased estimator of the scaled gap in expected returns, $\frac{1}{\tilde{v}_t}\big(\mathbb{E}_t[R_{t+1}] - \tilde{\mathbb{E}}_t[R_{t+1}]\big)$. This gap in conditional expected returns, in turn, measures the degree of investor pessimism. The subjective Sharpe ratio $\tilde{\pi}_t$ measures marginal utility sensitivity to shocks. Therefore, the second moment $\mathbb{E}[\frac{FE_{t+1}}{\tilde{v}_t}\tilde{\pi}_t]$ measures the average covariance between pessimism and marginal utility growth, $\Sigma(B, \tilde{S})$.

I implement Lemma 2, using $R = R^m$ as the single return, to directly estimate the degree of pessimism. The bottom row of Table 6 displays the results.

Across all surveys, I fail to reject pessimism $\Sigma(B, \tilde{S}) \geq 0$. For interpretability, I display estimates of the regression coefficient $b = \mathbb{E}[\frac{FE_{t+1}}{\tilde{v}_t}\tilde{\pi}_t]/\mathbb{E}[\tilde{\pi}_t^2]$, which is a scaled version of $\Sigma(B, \tilde{S}) = \mathbb{E}[\frac{FE_{t+1}}{\tilde{v}_t}\tilde{\pi}_t]$. Because forecast errors are extremely noisy, and the survey samples are short, the standard errors are quite large. In the Nagel-Xu and Livingston surveys, the point estimates are statistically indistinguishable from zero. In the CFO survey, $b = 0.80$ is statistically significant and implies that the market's objective risk premium is, on average, 1.80 times as large as the subjective risk premium—roughly consistent with the objective-to-subjective ratio of log premia $\frac{5.55\%}{2.97\%} \approx 1.87$.

assumption on $\Sigma(B,\tilde{S}),\Phi(R)$	**A. Nagel-Xu.** $L(\Delta B)/L(R^f\Delta S)$	**B. Livingston.** $L(\Delta B)/L(R^f\Delta S)$	**C. CFO.** $L(\Delta B)/L(R^f\Delta S)$
$\Sigma(B,\tilde{S})=0$ (benchmark)	0.1433	0.4588	0.4651
	(0.2456)	(0.1813)	(0.1440)
$\Sigma(B,\tilde{S})$ estimated, $\Phi(R)=1$	-0.1870	1.2744	-1.0780
	(0.5293)	(0.5478)	(0.5296)
$\Sigma(B,\tilde{S})$ estimated, $\Phi(R)=\frac{1}{2}$	-0.0219	0.8666	-0.3065
	(0.3136)	(0.2899)	(0.3308)
$\Sigma(B,\tilde{S})$ estimated, $\Phi(R)=\frac{1}{3}$	0.0332	0.7307	-0.0493
	(0.2640)	(0.2206)	(0.2659)
regression coef. $E[\frac{FE_{t+1}}{\tilde{v}_t}\tilde{\pi}_t]/E[\tilde{\pi}_t^2]$	0.0785	-0.1597	0.8016
	(0.3234)	(0.2318)	(0.4217)

Table 6. Upper bound on beliefs. This table displays upper bounds on the importance of belief volatility, using $R=R^m$. Columns A, B, and C use the Nagel-Xu, Livingston, and CFO survey expectations for the market, respectively. The bottom panel displays estimates of the forecast error regression coefficient $\mathbb{E}\left[\frac{FE_{t+1}}{\tilde{v}_t}\tilde{\pi}_t\right]/\mathbb{E}\left[\tilde{\pi}_t^2\right]$, where $\tilde{v}_t$ is the survey measure of subjective volatility. For all estimates, Newey-West asymptotic standard errors using 3 years of lags (i.e., 12 periods for Nagel-Xu, 6 periods for Livingston, 12 periods for CFO) are shown in parentheses.

5.3 Estimating the upper bound on beliefs

Next, I implement Proposition 4 and Lemma 2 to bound the direct role of beliefs. Table 6 displays the results, with $R=\tilde{R}=R^m$. That is, I use the market return for two purposes—to construct the ratio of subjective-to-objective premia, which serves as a lower bound for $L(R^f\Delta\tilde{S})/L(R^f\Delta S)$, and to estimate the pessimism statistic $\Sigma(B,\tilde{S})$.

In the first row, I impose $\Sigma(B,\tilde{S})=0$. For the purposes of estimating an upper bound for $L(\Delta B)/L(R^f\Delta S)$, this is a conservative assumption. Indeed, the existing literature suggests $\Sigma(B,\tilde{S})\geq 0$, as discussed above, and my own evidence has failed to reject this hypothesis. By assuming $\Sigma(B,\tilde{S})=0$ in (28), we are taking the largest upper bound consistent with a prior that $\Sigma(B,\tilde{S})\geq 0$. In this benchmark case, I estimate that at most 14% to 47% of SDF volatility can be attributed to belief distortions.

The subsequent rows of Table 6 instead use my empirical estimates for $\Sigma(B,\tilde{S})$, noisy though they might be. To implement (28), I must make an assumption on $\Phi(R)$, the ratio of the log risk premia of the market and the growth-optimal portfolio. My earlier results in Table 3 showed that the Chen et al. (2024a) proxy for R^* performs about 1.7 times as well as R^m ($1.7\approx\frac{9.55\%}{5.55\%}$), which would imply $\Phi(R)\approx 1/1.7$. Table 6 considers a range of assumptions around this value: $\Phi(R)=1$, $\Phi(R)=\frac{1}{2}$, and $\Phi(R)=\frac{1}{3}$. It turns out that the assumed value for $\Phi(R)$ is not the critical factor.

Instead, the key factor driving the bound for beliefs is the estimated covariance $\Sigma(B,\tilde{S})$. For the Nagel-Xu and CFO surveys, in which $\Sigma(B,\tilde{S})$ is estimated to be positive,

$L(\Delta B)/L(R^f \Delta S)$ is essentially always upper-bounded by zero, regardless of the value for $\Phi(R)$. According to these point estimates, belief distortions cannot directly contribute *any* volatility to the SDF. The reason is that $\Sigma(B, \tilde{S}) > 0$ in these two surveys, which tightens the bounds. The results are much weaker for the Livingston survey which features $\Sigma(B, \tilde{S}) < 0$.

In summary, this section provides suggestive evidence that belief distortions likely account for a small fraction of overall SDF volatility. This finding is not ironclad, for two reasons. First, the surveys are not completely consistent with regards to their degree of pessimism and belief cyclicality, as measured by $\Sigma(B, \tilde{S})$. Second, standard errors are large in all these estimates, because estimates of $\Sigma(B, \tilde{S})$ are extremely noisy. Thus, better estimates of pessimism, measured by $\Sigma(B, \tilde{S})$, would be valuable in future research.

6 Example structural models with long-term risks

I provide three example models to help interpret the general theory above. All models are exchange economies but with differing endowment and belief structures. The first model features rational expectations about a time-varying long-run growth rate (Bansal and Yaron, 2004). The second model features extrapolative beliefs about growth (Collin-Dufresne et al., 2017; Nagel and Xu, 2022). In these two settings, there is a representative agent with recursive preferences a la Epstein and Zin (1989) and elasticity of intertemporal substitution (EIS) equal to one. These preferences satisfy the Bellman equation

$$U_t = (1-\beta)\log C_t + \beta \frac{\log \tilde{\mathbb{E}}_t[\exp((1-\gamma)U_{t+1})]}{1-\gamma}, \tag{31}$$

where $\tilde{\mathbb{E}}$ is the agent's subjective expectation, γ is her risk aversion, and β is her discount factor. The third model also features growth extrapolation but with CRRA preferences for comparison. After presenting the models, Section 6.4 compares them across various measures of SDF volatility. Appendix B contains the derivations.

6.1 Long-run risks

In this first example, the agent has rational expectations: $\tilde{\mathbb{E}} = \mathbb{E}$. There are two shocks which are jointly Normal, $W := (W^{(1)}, W^{(2)})' \sim N(0, I)$. Aggregate consumption follows

$$\log C_{t+1} = \log C_t + x_t + \sigma_c \left(\begin{smallmatrix}1\\0\end{smallmatrix}\right) \cdot W_{t+1},$$

and its *expected growth rate* x_t has AR(1) dynamics

$$x_{t+1} = (1-\rho)\bar{x} + \rho x_t + \sigma_x \left(\begin{smallmatrix}(1-\kappa^2)^{1/2} \\ \kappa\end{smallmatrix}\right) \cdot W_{t+1} \tag{32}$$

The parameter ρ captures the persistence of expected consumption, and κ controls the correlation between consumption and growth shocks. A typical specification sets $\kappa = 1$, but this flexibility will be helpful going forward. Appendix B.1 follows standard methods to solve the model and derive the permanent-transitory decomposition of the SDF. For the quantitative analysis below, I adopt a relatively typical calibration, setting $\gamma = 10$, $\beta = 0.99$, $\rho = 0.95$, $\sigma_c = 0.02$, $\sigma_x = 0.002$, and $\kappa = 0.9$ (Bansal et al., 2012).

6.2 Growth extrapolation

Now, consider a model in which actual growth is IID but agents perceive a stochastic growth rate. It turns out, with the appropriate modeling of beliefs, we can just reinterpret the previous example as x representing the *subjective expected growth rate*, which remains constant at $\bar{x}$ under the rational expectation. This will be a model with long-run risks only in investors' heads.

In particular, consider the following enriched version of an extrapolative growth model. True consumption dynamics are

$$\log C_{t+1} = \log C_t + \bar{x} + \sigma_c \left(\begin{smallmatrix}1 \\ 0\end{smallmatrix}\right) \cdot W_{t+1}.$$

While conditional mean growth is constant at $\bar{x}$, agents are not convinced of this fact, and they do not observe the objective shocks W either. Building on many extrapolation models, agents will learn about the mean consumption growth via *constant-gain learning*. I will also allow the learning process to be subject to *sentiment shocks*.

Mathematically, let x_t denote the agent's perceived expected growth rate, i.e., $x_t := \tilde{\mathbb{E}}_t[\log C_{t+1} - \log C_t]$. Then, the agent's perception of consumption dynamics are

$$\log C_{t+1} = \log C_t + x_t + \sigma_c \left(\begin{smallmatrix}1 \\ 0\end{smallmatrix}\right) \cdot \tilde{W}_{t+1},$$

where $\tilde{W}$ represent perceived shocks, i.e., a normally-distributed variable which has zero mean under the agent's belief, $\tilde{\mathbb{E}}_t[\tilde{W}_{t+1}] = 0$. The posterior expected growth is given by

$$x_t = (1-\phi)\bar{x} + \phi z_t, \tag{33}$$

where z_t is a "sentiment" variable that follows

$$z_{t+1} - z_t = \underbrace{\lambda[\log C_{t+1} - \log C_t - z_t]}_{\text{constant gain learning}} + \underbrace{\left(\begin{smallmatrix} \nu_1 - \lambda\sigma_c \\ \nu_2 \end{smallmatrix}\right) \cdot \tilde{W}_{t+1}}_{\text{sentiment shocks}} \tag{34}$$

In words, sentiment increases with excess growth realizations $\log C_{t+1} - \log C_t - z_t$, at a fixed learning rate λ, and is also subject to additional shocks. After sentiment is determined, posterior expected growth is determined by a type of "model averaging" between the correct time-invariant growth $\bar{x}$ and the growth sentiment z_t. The parameter ϕ governs the degree of rationality: $\phi = 0$ is fully rational, while $\phi = 1$ is fully extrapolative. Alternatively, as shown in Nagel and Xu (2022), $1 - \phi$ captures the weight a constant-gain learner puts on his prior.

Combining equations (33)-(34) with the perceived consumption law of motion yields the belief evolution

$$x_{t+1} = \lambda(1-\phi)\bar{x} + (1 - \lambda(1-\phi))x_t + \phi\nu \cdot \tilde{W}_{t+1}, \tag{35}$$

where $\nu := (\nu_1, \nu_2)'$. From equation (35), we can see the analogy to the long-run risks model. By inspection, the dynamics in this equation are identical to those of equation (32), provided we set the prior dependence parameter to any interior value $\phi \in (0,1)$ and then put $\lambda = \frac{1-\rho}{1-\phi}$, $\nu_1 = \phi^{-1}(1-\kappa^2)^{1/2}\sigma_x$, and $\nu_2 = \phi^{-1}\kappa\sigma_x$. In that case, the impact of λ is entirely through "perceived growth persistence" that I refer to going forward:

$$\tilde{\rho} := 1 - (1-\phi)\lambda.$$

Notice also that setting $\phi < 1$ serves a technical convenience by keeping $\tilde{\rho} < 1$ so that x_t is stationary under the subjective probability. Given this isomorphism, the long-run risks solution for S_t and H_t are isomorphic to investor marginal utility $\tilde{S}_t$ and its permanent component $\tilde{H}_t$ under growth extrapolation.

It remains to characterize the belief distortion B_t, in order to map this solution back into the objective probability. In this conditionally log-normal environment, all belief distortions are characterized by a mean distortion to the shocks. That is, while W are the true zero-mean shocks, the agent perceives them to have non-zero mean, $\tilde{\mathbb{E}}_t[W_{t+1}] = \mu_t$, and instead views $\tilde{W}_{t+1} = W_{t+1} - \mu_t$ as the relevant disturbance. Hence,

$$\frac{B_{t+1}}{B_t} = \exp\left[\mu_t \cdot W_{t+1} - \frac{1}{2}\|\mu_t\|^2\right] \tag{36}$$

for some μ_t. To identify μ_t, compare true and perceived consumption dynamics to get

$$\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right) \cdot \mu_t = \frac{x_t - \bar{x}}{\sigma_c}.$$

This determines the first element of μ_t. The second element is not pinned down, because any value for it leads to the same perceived growth dynamics. Intuitively, pure sentiment shocks are not tied down by any objective variable. For theoretical clarity, I set

$$\left(\begin{smallmatrix}0\\1\end{smallmatrix}\right) \cdot \mu_t = 0,$$

so that all results are generated via mechanisms that are tied down. (This can be generalized. Allowing a non-zero second component of μ_t would work like animal spirits, whereby agents can be optimistic/pessimistic about future sentiment shocks.)

We compute the SDF and its permanent component via the formulas $S_t = B_t \tilde{S}_t$ and $H_t = B_t \tilde{H}_t$. From here, we may compute various volatility ratios. In the interest of space, these volatility ratios are derived and presented in Appendix B.2.

As a baseline, I adopt a standard calibration without sentiment shocks (Nagel and Xu, 2022). I set $\gamma = 4$, $\beta = 0.99$, $\sigma_c = 0.02$, $\phi = 0.9$ (almost fully extrapolative), $\lambda = 0.075$ (interpretable as roughly 13 years of effective information retention), and $\nu_1 - \lambda\sigma_c = \nu_2 = 0$ (no sentiment shocks).[11] This calibration generates perceived conditional growth rate volatility $\phi\|\nu\| = 0.0013$ and a perceived persistence $\tilde{\rho} = 0.993$. Relative to the long-run risk model calibration, the extrapolative calibration features a lower short-run volatility of perceived growth but a higher persistence hence higher long-run volatility.

6.3 Growth extrapolation (CRRA)

For comparison, I also consider a version of the previous extrapolative growth model under CRRA utility with risk aversion γ. The sentiment and belief processes are identical to the previous setting. But now, marginal utility growth is simply

$$\frac{\tilde{S}_{t+1}}{\tilde{S}_t} = \beta \Big(\frac{C_{t+1}}{C_t} \Big)^{-\gamma}.$$

There are two main differences in the CRRA model, relative to the Epstein-Zin version. First, marginal utility has no exposure to shocks to perceived growth x_t. This implies

[11] Nagel and Xu (2022) set $\lambda = 0.018$ at a quarterly frequency, which is approximately 4×0.018 annually. Next, in their model $1 - \phi$ can be interpreted as the weight agents place on their prior. Their baseline sets $\phi = 1$, but this would induce a belief that $\tilde{x}_t$ follows a random walk, which causes some technical issues for unconditional subjective volatilities. Thus, I pick a value ϕ close to 1 but slightly below.

lower marginal utility volatility in this CRRA setting, which then implies that much more of SDF volatility is attributable to beliefs. Second, all the volatility in x_t is captured by the conditional mean of $\tilde{S}_{t+1}/\tilde{S}_t$, i.e., through the risk-free rate. A volatile risk-free rate implies much more SDF volatility comes from transitory fluctuations.

The expressions for the SDF and its various components are contained in Appendix B.3. We calibrate this model exactly as the Epstein-Zin version, with the exception that the EIS with CRRA utility is γ^{-1}, the inverse of the risk aversion parameter.

6.4 Comparing the models

Table 7 compares the three models in terms of various volatilities and volatility ratios.

	Long-Run Risk	Extrapolation (EZ)	Extrapolation (CRRA)
Annual Avg SDF Volatility			
$E[2L_t(\Delta S_{t+1})]^{1/2}$	0.4265	0.3574	0.1948
Size of Permanent Component			
$L(\Delta H)/L(\Delta S)$	1.1786	2.1275	17.5983
Size of Transitory Component			
$L(\Delta G)/L(\Delta S)$	0.0090	0.2635	14.1140
Risk Component			
$L(\Delta\tilde{S})/L(\Delta S)$	1.0000	0.7736	0.1677
Permanent Risk Component			
$L(\Delta\tilde{H})/L(\Delta H)$	1.0000	0.9118	0.9831
Belief Component			
$L(\Delta B)/L(\Delta S)$	0.0000	0.2471	0.8270

Table 7. Volatilities and volatility ratios. This table displays volatilities across the three example models, in their benchmark calibrations. The "Long-Run Risk" model has $\gamma = 10$, $\beta = 0.99$, $\rho = 0.95$, $\sigma_c = 0.02$, $\sigma_x = 0.002$, $\kappa = 0.9$. The "Extrapolation (EZ)" model has $\gamma = 4$, $\beta = 0.99$, $\phi = 0.9$, $\lambda = 0.075$, $\sigma_c = 0.02$, and $\nu = (\lambda\sigma_c, 0)'$. "Extrapolation (CRRA)" uses the same parameters except with CRRA preferences.

Start with the long-run risk model (column 1). This model can generate a large amount of SDF volatility via a very prominent permanent component, consistent with Hansen and Jagannathan (1991) and Alvarez and Jermann (2005). Indeed, the amount of SDF volatility is $\sqrt{2L(\Delta S)} = \|\pi\| = 0.43$, while the size of the permanent and transitory components are $L(\Delta H)/L(\Delta S) = 1.18$ and $L(\Delta G)/L(\Delta S) = 0.01$, respectively. This class of models obtains all these features via a highly persistent growth rate (high ρ), combined with an investor who is both patient (high β) and risk averse (high γ).

Recall that the growth extrapolation model with EZ preferences (column 2) is very much like a long-run risks model but where the growth rate fluctuates merely in investors' heads. Therefore, it should not be surprising that this model succeeds in generating high SDF volatility, a large permanent component, and a modest transitory component. Because of the relatively high calibrated persistence of perceived growth ($\tilde{\rho} = 0.993$), the permanent component is even larger here than in the long-run risk model, $L(\Delta H)/L(\Delta S) = 2.13$. It turns out the transitory component is also larger, i.e., $L(\Delta G)/L(\Delta S) = 0.26$, because of negative correlation between G and H. In particular, optimism about perceived growth (high x_t) raises the transitory component of marginal utility, while simultaneously reducing the permanent component.

Perhaps more surprisingly, this EZ growth extrapolation model generates sizeable risk-based volatility. In fact, $L(\Delta\tilde{S})/L(\Delta S) = 0.7736$ and $L(\Delta\tilde{H})/L(\Delta H) = 0.9118$ are close to the empirical estimates in Tables 1 and 2. Interestingly, only 24.71% of total SDF volatility comes directly from belief distortions in this model, roughly in line with Table 4. This is because the mechanism of this model, leading to the large value of $L(\Delta S)$, arises via an *interaction* between preferences and perceived growth dynamics. Overall, this model thus generates significant SDF volatility, and other desirable properties, because marginal utility and its permanent component are sensitive to perceived growth shocks. I refer to this mechanism as *subjective risk*, because beliefs indirectly create large risk.

By comparing the growth extrapolation models with EZ versus CRRA preferences, we see the critical importance of subjective risk. In contrast to the EZ model, the CRRA model does not have any mechanism whereby beliefs fluctuations can impact marginal utility, which is purely a function of consumption. That is why most of its SDF volatility stems from beliefs rather than risks: the CRRA model features $L(\Delta\tilde{S})/L(\Delta S) = 0.1677$ and $L(\Delta B)/L(\Delta S) = 0.8270$, in stark contrast to my empirical estimates. It turns out that this counterfactually large direct role for beliefs is tightly related to another failure of this CRRA model—it features far too much transitory SDF volatility, with $L(\Delta G)/L(\Delta S) = 14.1$. This connection between direct belief-based volatility and transitory SDF volatility is further highlighted by the next exercise.

In a final exercise, I add sentiment shocks to the extrapolation model, as a way to increase the volatility of belief distortions. The question is how this extra belief volatility affects the contributions to the SDF of risk, beliefs, and permanent/transitory shocks. Specifically, the extrapolation model is solved for different levels of ν_2 (the sensitivity of perceived growth to a shock orthogonal to consumption). To keep these various settings on a level playing field, I recalibrate risk aversion γ to hold fixed the model-implied average SDF volatility $\mathbb{E}[2L_t(\Delta S_{t+1})]^{1/2}$. That is, since sentiment shocks increase SDF volatility, I reduce γ as I increase ν_2. Figure 3 displays results for the size of the risk-based component $L(\Delta\tilde{S})$, transitory component $L(\Delta G)$, and belief component $L(\Delta B)$, all scaled by the total volatility of the SDF $L(\Delta S)$. The shaded regions are those where volatility ratios conform with empirical evidence both in my paper and in Alvarez and

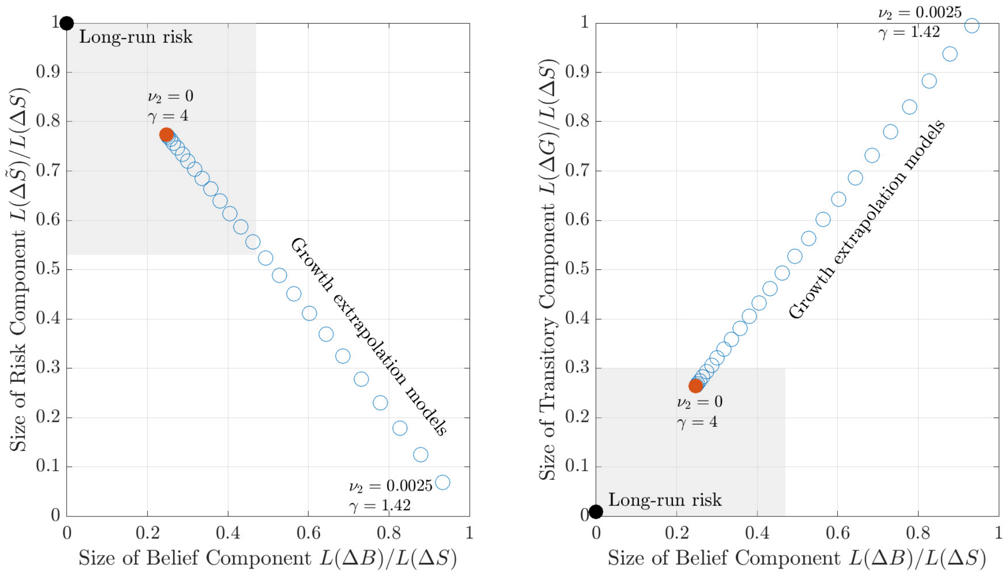

Figure 3. Volatility ratios with sentiment. This figure compares the baseline long-run risk model (solid black dot), the baseline growth extrapolation model (solid red dot), and various growth extrapolation models with sentiment shocks $\nu_2 > 0$ (hollow blue circles). As ν_2 is increased, γ is decreased to keep the one-year average SDF volatility $\mathbb{E}[2L_t(\Delta S_{t+1})]^{1/2}$ unchanged at 0.3574. The baseline calibrations are those in Table 7. The shaded areas denote empirically-valid regions for the volatility ratios: $L(\Delta\tilde{S})/L(\Delta S) \geq 0.53$ (the smallest value in Table 1), $L(\Delta B)/L(\Delta S) \leq 0.47$ (the largest value in Table 6, row 1), and $L(\Delta G)/L(\Delta S) \leq 0.30$ (the largest value in Alvarez and Jermann (2005)).

Jermann (2005). Sentiment shocks raise the contribution of beliefs to SDF volatility, and reduce the contribution of risk, but they do so by increasing the transitory component. In this class of models, there is a tension: attributing large amounts of SDF volatility to beliefs necessitates a counterfactually transitory SDF. Thus, if distorted beliefs are to be taken seriously, their *direct impact* on SDF volatility must not be too large, and their impact must instead come *indirectly via subjective risk*.

To demonstrate that this conclusion is not an artifact of the particular calibration, I provide a theoretical result formalizing the tension between adding belief-based volatility while keeping transitory volatility low. Thus, the conclusion that SDF volatility must come via the subjective risk channel is general to this class of models.

Proposition 5. *Consider the Epstein-Zin growth extrapolation model. For each parameter $\theta \in \{\gamma, \beta, \nu_2\}$, we have that $L(\Delta G)/L(\Delta S)$ and $L(\Delta B)/L(\Delta S)$ respond in the same direction to θ:*

$$sign\Big(\frac{\partial}{\partial\theta}\frac{L(\Delta G)}{L(\Delta S)}\Big) = sign\Big(\frac{\partial}{\partial\theta}\frac{L(\Delta B)}{L(\Delta S)}\Big).$$

Furthermore, assuming either ϕ or $\tilde{\rho}$ are small enough, then the same conclusion holds for $\theta = \phi$.

Proposition 5 extends beyond comparative statics to γ and ν_2, as in the numerical example above, to β and ϕ as well. These preference and sentiment parameters are somewhat difficult to pin down directly without asset prices, and so Proposition 5 can also be thought of as a *partial identification* result: certain parts of the parameter space—such as low γ, high ν_2, low β, low ϕ, or low $\tilde{\rho}$—are ruled out by the requirement that we simultaneously match modest values of $L(\Delta B)/L(\Delta S)$ and $L(\Delta G)/L(\Delta S)$.

7 Example structural models with belief heterogeneity

In this section, I provide an alternative class of example models with heterogeneity in beliefs. My model has two agent types who disagree about growth rates, experience sentiment shocks, have identical CRRA utility, trade in dynamically complete markets, and live on an infinite horizon. The closest paper is Dumas et al. (2009). I have two objectives. First, I illustrate how to aggregate beliefs and marginal utilities in this world to correspond to what is measured in the survey. Second, the model helps reinforce the key finding that subjective risks are critically important drivers of SDF volatility.

Now to the details. I present this model in continuous time, because it adds significant tractability. The aggregate endowment follows a geometric Brownian motion,

$$\frac{dC_t}{C_t} = \mu_c dt + \sigma_c \left(\begin{smallmatrix}1\\0\end{smallmatrix}\right) \cdot dW_t,$$

where W is a two-dimensional Brownian motion. The first shock drives aggregate consumption, while the second shock will drive beliefs directly and is independent of consumption growth. By no-arbitrage, the objective SDF follows

$$\frac{dS_t}{S_t} = -r_t dt - \pi_t \cdot dW_t,$$

for an endogenous interest rate r_t and risk price vector π_t.

For simplicity, I assume only two types of agents, indexed by $\theta \in \{0, 1\}$. Agents of type $\theta = 1$ hold pessimistic views about expected growth. By contrast, the beliefs of agents of type $\theta = 0$ are fully rational: she knows that the expected growth rate is the constant μ_c and observes all shocks correctly. Appendix C.2 generalizes the model to an arbitrary number of types θ.

We nest agent types and their beliefs as follows. Define agent θ's conditional expected growth rate $\mu_c^\theta := \frac{1}{dt}\mathbb{E}_t^\theta[dC_t/C_t] = \mu_c - \theta\sigma_c\beta$, with $\beta > 0$ capturing the degree of pessimism bias. That is, agent θ believes that

$$\frac{dC_t}{C_t} = \mu_c^\theta dt + \sigma_c \left(\begin{smallmatrix}1\\0\end{smallmatrix}\right) \cdot d\tilde{W}_t^\theta,$$

where $\tilde{W}_t^\theta$ is a Brownian motion under agent θ's belief. The assumption that agents' beliefs about growth are constant is a simplification. Appendix C.2 generalizes the setup to allow time-varying optimism and pessimism.

To introduce sentiment risk in a way that economizes on state variables, I assume furthermore that the biases with regard to the second shock (sentiment) are proportional to that of the first shock (consumption). Given this setting, Girsanov's theorem then implies that the perceived shocks are characterized by

$$d\tilde{W}_t^\theta := dW_t + \theta\beta\left(\begin{smallmatrix}1\\\chi\end{smallmatrix}\right)dt \tag{37}$$

for some parameter χ that captures the volatility of sentiment shocks. Since sentiment shocks are orthogonal to all fundamentals, the parameter χ only serves to create trade but can influence endogenous prices through this mechanism. Summarizing this setup, the martingale belief distortion for agent θ is given by

$$B_t^\theta = \exp\left[-\theta\beta\left(\begin{smallmatrix}1\\\chi\end{smallmatrix}\right) \cdot W_t - \frac{1}{2}(\theta\beta)^2(1+\chi^2)t\right]. \tag{38}$$

In many models of belief heterogeneity, the "market selection hypothesis" holds, in that the long-run equilibrium is dominated by agents with the most accurate beliefs (Kogan et al., 2006; Yan, 2008; Dumas et al., 2009; Borovička, 2020). To avoid this issue and generically ensure survival of all agents, I embed an overlapping generations structure as in Gârleanu and Panageas (2015), Ehling et al. (2018), and Panageas (2020). In particular, I assume that agents die idiosyncratically at rate δ and are replaced by a mass δ of newborns. The time-t population of surviving agents that were born at time $u \leq t$ is thus $\delta e^{-\delta(t-u)}$, while the total population is constant at $\int_{-\infty}^t \delta e^{-\delta(t-u)} du = 1$. I assume that $\eta^1 := \eta$ fraction of newborns pessimists, and $\eta^0 = 1 - \eta$ are optimists To keep financial markets dynamically complete, I follow the standard practice and allow all agents to trade fairly priced annuity contracts to insure their death shocks—agents will optimally fully insure in these markets. Finally, to ensure that all agents start with some wealth, I assume an agent born at time u inherits a tree whose time $t \geq u$ dividends are given by $C_{t,u} := \kappa e^{-\kappa(t-u)} C_t$. The time-$t$ dividends of all trees ever born add up to the aggregate endowment, since $\int_{-\infty}^t C_{t,u} du = \int_{-\infty}^t \kappa e^{-\kappa(t-u)} C_t du = C_t$.

All agents have CRRA utility with common risk aversion γ and common discount rate ρ. Let $c^\theta_{u,t}$ be the time-t consumption flow for a type-θ agent born at time $u \leq t$. Given the details above, agents solve

$$\max_{(c^\theta_{t,u})_{u\geq t}} \mathbb{E}\Big[\int_t^\infty e^{-(\rho+\delta)(u-t)} \frac{B^\theta_u}{B^\theta_t} \frac{(c^\theta_{u,t})^{1-\gamma}}{1-\gamma} du\Big] \tag{39}$$

subject to the lifetime budget constraint

$$\mathbb{E}\Big[\int_t^\infty e^{-\delta(u-t)} \frac{S_u}{S_t} c^\theta_{u,t} du\Big] = \frac{P_{t,t}}{\delta}, \tag{40}$$

where $P_{t,t} = \mathbb{E}_t\big[\int_t^\infty \frac{S_u}{S_t} C_{u,t} du\big] = \mathbb{E}_t\big[\int_t^\infty \kappa e^{-\kappa(u-t)} \frac{S_u}{S_t} C_u du\big]$ denotes the (endogenous) time-t value of the trees born at time t, and the division by δ converts this into per-capita units. Notice that the death rate δ only enters the optimization problem as an augmented discount rate. Also notice that belief distortions are simply captured by B^θ_t inside the expectation of the utility function—this converts the agent's subjective expectation $\mathbb{E}^\theta$ to the objective expectation $\mathbb{E}$.

The equilibrium market clearing conditions are as follows. Define the aggregate consumption of type-θ agents, $c^\theta_t := \eta^\theta \int_{-\infty}^t \delta e^{-\delta(u-t)} c^\theta_{u,t} du$. The goods market clearing condition says $\sum_\theta c^\theta_t = C_t$. The asset market clearing condition says that $\sum_\theta n^\theta_t = P_t$, where n^θ_t is the total net worth of type-θ agents and P_t is the total wealth in the economy, namely the total present value of all trees alive at time t. Because of the exponential decay of dividends, total wealth is equal to $P_t = \mathbb{E}_t\big[\int_t^\infty e^{-\kappa u} \frac{S_u}{S_t} C_u du\big] = P_{t,t}/\kappa$.[12]

I characterize the entire equilibrium in terms of a single state variable $x_t := c^1_t / C_t$, the consumption share of the pessimists. The dynamics of x_t are endogenous. Write

$$dx_t = \mu_{x,t} dt + \sigma_{x,t} \cdot dW_t$$

for some drift $\mu_{x,t}$ and diffusion $\sigma_{x,t}$ to be determined. As shown in Proposition C.1, this model can be solved up to a set of two ODEs, that are given in Appendix C.

The next question is how one should average across the heterogeneous agents in this model. To maintain consistency with the surveys, which obtain a "consensus expectation" by averaging across respondents, I define the consensus belief by the population-weighted average belief. As shown in Appendix D, this type of consensus belief pre-

[12]Recall that the time-u dividends for cohort-s trees are $\kappa e^{-\kappa(u-s)} C_u$. Hence, a time-$t$ claim to all the alive trees delivers total time-u dividends equal to $\int_{-\infty}^t \kappa e^{-\kappa(u-s)} C_u ds = e^{-\kappa(u-t)} C_u$, i.e., summing over trees born before t. The present-value of these dividends is aggregate wealth, so $P_t := \mathbb{E}_t\big[\int_t^\infty e^{-\kappa u} \frac{S_u}{S_t} C_u du\big]$.

serves all the volatility bounds in this paper, so long as we interpret B as the cross-sectional average belief distortion, $\tilde{S}$ as the belief-weighted-average marginal utility, and $\tilde{H}$ as its permanent component. In other words, we first define B_t by

$$\frac{dB_t}{B_t} = \sum_\theta \eta^\theta \frac{dB_t^\theta}{B_t^\theta}. \tag{41}$$

Given our SDF S and its permanent component H, one can then define $\tilde{S}_t := S_t/B_t$ and $\tilde{H}_t := H_t/B_t$. At this point, we have all the various components of the SDF: S, H, G, B, $\tilde{S}$, and $\tilde{H}$. Appendix C presents the calculations of these components.

To compute various volatilities in this continuous-time model, I use "localized" versions of the entropy statistics. Localized entropies deliver clean expressions that permit analytical results as well. For a positive process X, define the local volatility operator

$$\mathcal{L}_t(X) := \lim_{\Delta \to 0} \Delta^{-1} L_t(X_{t+\Delta}). \tag{42}$$

Given the Brownian environment, this localized volatility equals one-half the quadratic variation of $\log X$, i.e., $\mathcal{L}_t(X) = \frac{1}{2}\frac{1}{dt} d[\log X]_t$ (see Appendix A.8). To average over all states, I then compute unconditional versions $\mathbb{E}[\mathcal{L}_t(X)]$.

Table 8 presents these volatilities and their ratios in a benchmark calibration of the model. Pessimists have 3% lower expectations about output growth ($\beta = 0.03$). This calibration features equal population shares of rational agents and pessimists ($\eta = 0.5$), a moderate amount of sentiment shocks ($\chi = 7$), and typical level of risk aversion ($\gamma = 8.44$). The calibration produces SDF volatilities that are roughly in line with the data: an annualized Sharpe ratio of 0.29, a permanent component that explains most of SDF volatility, and a dominant role for risk as opposed to beliefs.

SDF Volatility $E[2\mathcal{L}_t(S)]^{1/2}$	Permanent $\frac{E[\mathcal{L}_t(H)]}{E[\mathcal{L}_t(S)]}$	Transitory $\frac{E[\mathcal{L}_t(G)]}{E[\mathcal{L}_t(S)]}$	Risk $\frac{E[\mathcal{L}_t(\tilde{S})]}{E[\mathcal{L}_t(S)]}$	Permanent Risk $\frac{E[\mathcal{L}_t(\tilde{H})]}{E[\mathcal{L}_t(H)]}$	Beliefs $\frac{E[\mathcal{L}_t(B)]}{E[\mathcal{L}_t(S)]}$
0.2856	0.7868	0.1247	0.7828	1.0569	0.1379

Table 8. Volatilities and volatility ratios (localized). This table displays entropies and entropy ratios in the disagreement model. The calibration is $\gamma = 8.44$, $\rho = 0.01$, $\mu_c = 0.02$, $\sigma_c = 0.03$, $\delta = 0.02$, $\kappa = 0.033$, $\beta = 0.03$, $\chi = 7$, $\eta = 0.5$. The local entropies are calculated from a 5,000-year simulation of the equilibrium.

Interestingly, despite the fact that a non-trivial share of agents are quite pessimistic, the direct role of belief distortions in SDF volatility is quite small at only 13.79%. This occurs for two reasons. First, remember that the "consensus belief" is defined as the cross-sectional average, which is in between the pessimistic and rational agents. Second, in contrast to a direct role, belief distortions also play an indirect role here: *disagreement creates risk*. Disagreement risk arises because rational agents must hold whatever assets the pessimistic agents sell them, and this requires equilibrium risk compensation. While the precise mechanism is different, disagreement risk here thus parallels the subjective risk channel from Section 6.

In this class of models, disagreement risk must be the key role played by belief heterogeneity. To demonstrate this, Figure 4, considers alternative amounts of sentiment shocks (χ ranging from 0 to 18). With larger sentiment shocks, agents trade more in a market that is orthogonal to aggregate consumption. While this trade does increase SDF volatility, it does so almost purely via belief volatility itself. To make a fair comparison across calibrations, I also recalibrate risk aversion to keep the overall level of Sharpe ratios roughly constant as I increase χ (i.e., I reduce γ from 10 to 6 as I increase χ from 0 to 18). The results show that large sentiment shocks raises the direct role of beliefs, reduces the risk-based component of the SDF, and reduces the persistence of the SDF. These results—a large direct role for beliefs and a large role for transitory shocks—are at odds with the data.

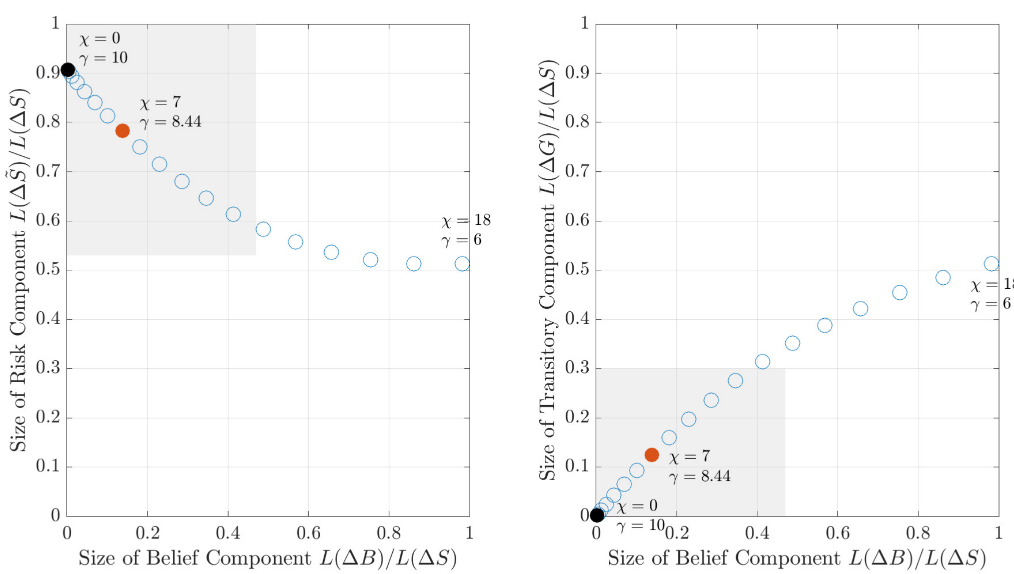

Figure 4. Volatility ratios with sentiment. This figure compares models with different magnitudes of sentiment shocks (χ). Plotted are the model with $\chi = 0$ (solid black dot), the benchmark model with $\chi = 7$ (solid red dot), and models with $\chi \in \{1, 2, \ldots, 18\}$ (hollow blue circles). As χ is increased, γ is decreased to keep the one-year average SDF volatility $\mathbb{E}[2L_t(\Delta S_{t+1})]^{1/2}$ roughly unchanged at 0.29. The shaded areas denote empirically-valid regions for the volatility ratios: $L(\Delta\tilde{S})/L(\Delta S) \geq 0.53$ (the smallest value in Table 1), $L(\Delta B)/L(\Delta S) \leq 0.47$ (the largest value in Table 6, row 1), and $L(\Delta G)/L(\Delta S) \leq 0.30$ (the largest value in Alvarez and Jermann (2005)).

I complement these results with an analytical calculation. This calculation examines the conditional local entropies $\mathcal{L}_t$ and therefore ignores the (quantitatively minor) effects of a parameter on the stationary consumption distribution. As suggested by the calibration results, one can show that sentiment shocks always amplify the belief component and the transitory component of the SDF, whereas risk aversion always mitigates both components. Surprisingly, a similar story holds for every parameter: anything that increases the belief component $\mathcal{L}_t(B)$ also increases the transitory component $\mathcal{L}_t(G)$, and vice versa. This reinforces my claim that beliefs must impact equilibrium largely through risk, because otherwise they create too much transitory volatility in the SDF.

Proposition 6. *In the heterogeneous belief model, we have*

$$\mathcal{L}_t(G) = \Big(\frac{x_t}{\eta}\Big)^2 \mathcal{L}_t(B), \tag{43}$$

so $\mathcal{L}_t(G)$ and $\mathcal{L}_t(B)$ respond in the same direction to any parameter besides η. Meanwhile, $\mathcal{L}_t(H) = \frac{1}{2}(\gamma\sigma_c)^2$ is independent of all parameters governing beliefs.

The result that $\mathcal{L}_t(H)$ is independent of belief distortions is a very general property. Indeed, Appendix C.2 enriches the model to include (i) a richer consumption process with stochastic growth and volatility, (ii) arbitrary optimism/pessimism and underreaction/overreaction to various shocks, and (iii) arbitrary belief heterogeneity. So long as there exists one agent type with rational beliefs, indexed by $\theta = \theta^*$, his marginal utility serves as the SDF, and so

$$\frac{S_t}{S_0} = e^{-\rho t}\Big(\frac{x_t^{\theta^*}}{x_0^{\theta^*}}\Big)^{-\gamma}\Big(\frac{C_t}{C_0}\Big)^{-\gamma}. \tag{44}$$

This equation immediately implies that H coincides with the permanent component of $C^{-\gamma}$, which is exogenous to all belief distortions. All the additional richness in heterogeneity and beliefs increases SDF volatility entirely through G. Consequently, the message that $\mathcal{L}_t(G)$ and $\mathcal{L}_t(B)$ increase together persists in a much more general setting. Given pre-existing evidence that $\mathcal{L}_t(G)/\mathcal{L}_t(S)$ cannot be too large, we therefore also know that $\mathcal{L}_t(B)/\mathcal{L}_t(S)$ cannot be too large in a general class of disagreement models.

8 Conclusion

This paper seeks to quantify how much of asset market dynamics are attributable to risk versus beliefs. In particular, combining a volatility bounds approach with survey data on stocks and bonds, I estimate that at least 50% of SDF volatility must stem from risk.

Some estimates for this risk-based fraction exceed 100%. At the same time, estimates for the belief-based fraction, while noisier, are in some cases near 0%. I obtain all of these results despite taking financial surveys at face value—I do what behavioral finance does.

How should researchers proceed knowing that SDF volatility is primarily driven by marginal utility? One option is to abandon non-rational beliefs for the purpose of explaining aggregate asset-pricing puzzles. This approach would give up on matching the minutiae in financial surveys, taking the stance that belief distortions are more of a cumbersome distraction than a productive solution. Perfectly consistent with this perspective on aggregate asset pricing is a popular view that subjective beliefs about idiosyncratic components of cash flows create "excess volatility." This paper is silent on the importance of belief distortions about asset-specific cash flows.

A second option, that I elaborate on in this paper, is to refine the way in which beliefs can matter. Using some example structural models, I show how belief distortions must matter primarily by creating marginal utility volatility. What does this mean? In the growth extrapolation framework, marginal utility inherits important volatility from *subjective risk*. In the heterogeneous belief framework, marginal utility volatility is amplified by *disagreement-induced risk*. Having isolated these channels, further investigation into belief-induced risks is needed to bring additional nuance and clarity to a behavioral finance view of asset markets.

Finally, while my approach is assumption-light, the strongest assumption concerns the absence of significant frictions and constraints. I assume that survey-takers are marginal in asset markets—or at least that their reported beliefs roughly coincide with marginal traders' beliefs. Almost all quantitative behavioral asset-pricing models share this assumption. That being said, it would be worthwhile to explore to what extent my results can be generalized to accommodate financial frictions. Intuition suggests that asset prices depend less on the beliefs of constrained agents and more on those of unconstrained agents. Does this imply that the subjective belief distortions measured in surveys are less important than I find here?

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Appendix

A Proofs

A.1 Proof of Lemma 1

We prove only the subjective versions; the objective bounds follow from those by taking $B \equiv 1$ (no belief distortion). Note by the pricing equation (4), the permanent-transitory factorization (7), the fact $\tilde{G} = G$, and the long bond result (9) that

$$\begin{aligned} 0 \leq \tilde{L}_t\Big(\frac{\tilde{S}_{t+1}}{\tilde{S}_t}R_{t+1}\Big) &= -\tilde{\mathbb{E}}_t \log\Big(\frac{\tilde{S}_{t+1}}{\tilde{S}_t}\Big) - \tilde{\mathbb{E}}_t \log R_{t+1} \\ &= -\tilde{\mathbb{E}}_t \log\Big(\frac{G_{t+1}}{G_t}\Big) - \tilde{\mathbb{E}}_t \log\Big(\frac{\tilde{H}_{t+1}}{\tilde{H}_t}\Big) - \tilde{\mathbb{E}}_t \log R_{t+1} \\ &= \tilde{\mathbb{E}}_t \log R^{\infty}_{t+1} + \tilde{L}_t\Big(\frac{\tilde{H}_{t+1}}{\tilde{H}_t}\Big) - \tilde{\mathbb{E}}_t \log R_{t+1} \end{aligned}$$

Rearranging leads to (15). To derive (13), start with the identity

$$\tilde{L}_t\Big(\frac{\tilde{S}_{t+1}}{\tilde{S}_t}\Big) = \tilde{L}_t\Big(\frac{\tilde{S}_{t+1}}{\tilde{S}_t}R^f_t\Big) = \tilde{\mathbb{E}}_t[\log R^{\infty}_{t+1}] - \log R^f_t + \tilde{L}_t\Big(\frac{\tilde{H}_{t+1}}{\tilde{H}_t}\Big) \tag{A.1}$$

and then use (15). □

A.2 Proof of Proposition 1

To prove (19), use the UEI property to get $L(\frac{\tilde{H}_{t+1}}{\tilde{H}_t}) \geq \mathbb{E}[\tilde{L}_t(\frac{\tilde{H}_{t+1}}{\tilde{H}_t})]$. Combining this result with the bound in (15), we obtain (19).

The same argument with $\frac{\tilde{S}_{t+1}}{\tilde{S}_t}R^f_t$ in place of $\frac{\tilde{H}_{t+1}}{\tilde{H}_t}$ leads to $L(\frac{\tilde{S}_{t+1}}{\tilde{S}_t}R^f_t) \geq \mathbb{E}[\tilde{L}_t(\frac{\tilde{S}_{t+1}}{\tilde{S}_t}R^f_t)]$. Because R^f_t is conditionally risk-free, we have $\tilde{L}_t(\frac{\tilde{S}_{t+1}}{\tilde{S}_t}R^f_t) = \tilde{L}_t(\frac{\tilde{S}_{t+1}}{\tilde{S}_t})$. Combining these results with the bound in (13), we obtain (18).

Finally, we can deduce (16)-(17) from the subjective versions by considering $B \equiv 1$ (i.e., no belief distortion), so that all objects and expectations with tildes become objective versions, and then applying the law of iterated expectations. □

A.3 Proof of Proposition 2

First, make note of the following key result: the inequalities in (12) and (14) become equalities with $R_{t+1} = R^*_{t+1}$. Similarly, the subjective versions in (13) and (15) become equalities with $R_{t+1} = \tilde{R}^*_{t+1}$.

Now, let us characterize the "wedges" between entropies and expected returns. For the objective entropies, we obtain

$$\begin{aligned} \omega(R) &:= L\Big(\frac{S_{t+1}}{S_t}R^f_t\Big) - \mathbb{E}[\log R_{t+1} - \log R^f_t] \\ &= \mathbb{E}[\log R^*_{t+1} - \log R^f_t] - \mathbb{E}[\log R_{t+1} - \log R^f_t] = \mathbb{E}[\delta_t(R)] \end{aligned}$$

For the subjective entropies, we have analogously

$$\begin{aligned}\tilde{\omega}(R) &:= \mathbb{E}\Big[\tilde{L}_t\Big(\frac{\tilde{S}_{t+1}}{\tilde{S}_t}R_t^f\Big)\Big] - \mathbb{E}\big[\tilde{\mathbb{E}}_t[\log R_{t+1} - \log R_t^f]\big] \\ &= \mathbb{E}\big[\tilde{\mathbb{E}}_t[\log R_{t+1}^* - \log R_t^f]\big] - \mathbb{E}\big[\tilde{\mathbb{E}}_t[\log R_{t+1} - \log R_t^f]\big] = \mathbb{E}[\tilde{\delta}_t(R)]\end{aligned}$$

Given two returns $R, \tilde{R}$, we use the assumption that R is closer to objective growth-optimal than $\tilde{R}$ is to subjective growth-optimal, i.e., $\mathbb{E}[\delta_t(R)] \leq \mathbb{E}[\tilde{\delta}_t(\tilde{R})]$, to obtain

$$\omega(R) \leq \tilde{\omega}(\tilde{R}).$$

Therefore, compute

$$\frac{\mathbb{E}\big[\tilde{L}_t(\frac{\tilde{S}_{t+1}}{\tilde{S}_t}R_t^f)\big]}{L(\frac{S_{t+1}}{S_t}R_t^f)} = \frac{\mathbb{E}\big[\tilde{\mathbb{E}}_t[\log \tilde{R}_{t+1} - \log R_t^f]\big] + \tilde{\omega}(\tilde{R})}{\mathbb{E}[\log R_{t+1} - \log R_t^f] + \omega(R)} \geq \frac{\mathbb{E}\big[\tilde{\mathbb{E}}_t[\log \tilde{R}_{t+1} - \log R_t^f]\big] + \omega(R)}{\mathbb{E}[\log R_{t+1} - \log R_t^f] + \omega(R)}$$

Note that the latter ratio is increasing (decreasing) in $\omega(R)$ if the ratio is smaller (larger) than one. This leads to

$$\frac{\mathbb{E}\big[\tilde{L}_t(\frac{\tilde{S}_{t+1}}{\tilde{S}_t}R_t^f)\big]}{L(\frac{S_{t+1}}{S_t}R_t^f)} \geq \min\left\{1, \frac{\mathbb{E}\big[\tilde{\mathbb{E}}_t[\log \tilde{R}_{t+1} - \log R_t^f]\big]}{\mathbb{E}[\log R_{t+1} - \log R_t^f]}\right\} \tag{A.2}$$

Finally, we use the UEI assumption on the left-hand-side, namely $L(\frac{\tilde{S}_{t+1}}{\tilde{S}_t}R_t^f) \geq \mathbb{E}\tilde{L}_t(\frac{\tilde{S}_{t+1}}{\tilde{S}_t}R_t^f)$, in order to obtain the final result (20). To obtain (21), we repeat a very similar argument to obtain

$$\frac{L(\frac{\tilde{H}_{t+1}}{\tilde{H}_t})}{L(\frac{H_{t+1}}{H_t})} \geq \frac{\mathbb{E}\big[\tilde{\mathbb{E}}_t[\log \tilde{R}_{t+1} - \log R_t^\infty]\big] + \omega(R)}{\mathbb{E}[\log R_{t+1} - \log R_t^\infty] + \omega(R)},$$

because the permanent component wedge $\omega_H(R) := L\big(\frac{H_{t+1}}{H_t}\big) - \mathbb{E}[\log R_{t+1} - \log R_t^\infty] = \omega(R)$ equals the overall wedge studied above, and similarly $\tilde{\omega}_H(R) = \tilde{\omega}(R)$. □

A.4 Proof of Corollary 1

Throughout, suppose R is a return such that $\tilde{\mathbb{E}}_t \log R_{t+1} > \log R_t^f$. From (A.1), we have

$$\frac{\tilde{L}_t(\frac{\tilde{H}_{t+1}}{\tilde{H}_t})}{\tilde{L}_t(\frac{\tilde{S}_{t+1}}{\tilde{S}_t})} = \frac{\tilde{L}_t(\frac{\tilde{H}_{t+1}}{\tilde{H}_t})}{\tilde{\mathbb{E}}_t \log R_{t+1}^\infty - \log R_t^f + \tilde{L}_t(\frac{\tilde{H}_{t+1}}{\tilde{H}_t})}$$

Note that this ratio is increasing (decreasing) in $\tilde{L}_t(\frac{\tilde{H}_{t+1}}{\tilde{H}_t})$ if it is smaller (larger) than one. So if $\tilde{\mathbb{E}}_t \log R_{t+1}^\infty > \log R_t^f$, we use (15) to get

$$1 \geq \frac{\tilde{L}_t(\frac{\tilde{H}_{t+1}}{\tilde{H}_t})}{\tilde{L}_t(\frac{\tilde{S}_{t+1}}{\tilde{S}_t})} \geq \frac{\tilde{\mathbb{E}}_t[\log R_{t+1} - \log R_{t+1}^\infty]}{\tilde{\mathbb{E}}_t \log R_{t+1} - \log R_t^f}$$

On the other hand, if $\tilde{\mathbb{E}}_t \log R_{t+1}^\infty < \log R_t^f$, we use (15) to get

$$1 \leq \frac{\tilde{L}_t(\frac{\tilde{H}_{t+1}}{\tilde{H}_t})}{\tilde{L}_t(\frac{\tilde{S}_{t+1}}{\tilde{S}_t})} \leq \frac{\tilde{\mathbb{E}}_t[\log R_{t+1} - \log R_{t+1}^\infty]}{\tilde{\mathbb{E}}_t \log R_{t+1} - \log R_t^f}$$

Combine these results with the non-negativity of entropies to obtain the result. □

A.5 Proof of Proposition 3

We have

$$0 \le L_t\Big(\frac{B_{t+1}}{B_t}R_{t+1}\Big) = \log \tilde{\mathbb{E}}_t[R_{t+1}] - \mathbb{E}_t\Big[\log \frac{B_{t+1}}{B_t}\Big] - \mathbb{E}_t[\log R_{t+1}]$$
$$= \tilde{L}_t(R_{t+1}) + \tilde{\mathbb{E}}_t[\log R_{t+1}] - \mathbb{E}_t[\log R_{t+1}] + L_t\Big(\frac{B_{t+1}}{B_t}\Big)$$

and similarly

$$0 \le L_t\Big(\frac{B_{t+1}}{B_t}R_{t+1}^{-1}\Big) = \log \tilde{\mathbb{E}}_t[R_{t+1}^{-1}] - \mathbb{E}_t\Big[\log \frac{B_{t+1}}{B_t}\Big] + \mathbb{E}_t[\log R_{t+1}]$$
$$= \tilde{L}_t(R_{t+1}^{-1}) - \tilde{\mathbb{E}}_t[\log R_{t+1}] + \mathbb{E}_t[\log R_{t+1}] + L_t\Big(\frac{B_{t+1}}{B_t}\Big).$$

We rearrange these two inequalities to get the following:

$$L_t\Big(\frac{B_{t+1}}{B_t}\Big) \ge \mathbb{E}_t[\log R_{t+1}] - \tilde{\mathbb{E}}_t[\log R_{t+1}] - \tilde{L}_t(R_{t+1})$$
$$\text{and} \quad L_t\Big(\frac{B_{t+1}}{B_t}\Big) \ge \tilde{\mathbb{E}}_t[\log R_{t+1}] - \mathbb{E}_t[\log R_{t+1}] - \tilde{L}_t(R_{t+1}^{-1})$$

Taking unconditional expectations of these two lower bounds, then combining them, and finally using the assumption $\mathbb{E}[\tilde{L}_t(R_{t+1})] \ge \mathbb{E}[\tilde{L}_t(R_{t+1}^{-1})]$, we obtain (23).

Finally, note that if $R = R^*$ is objective growth-optimal, then $\mathbb{E}[\log R_{t+1} - \log R_t^f] = L(R_t^f \Delta S_{t+1})$. Therefore, if $\mathbb{E}[\delta_t(R)]$ is sufficiently small, we obtain the ratio lower-bound (24). □

A.6 Proof of Proposition 4

Using $S = B\tilde{S}$, we have the following conditional variance decomposition:

$$\mathrm{Var}_t\Big[\log \frac{S_{t+1}}{S_t}R_t^f\Big] = \mathrm{Var}_t\Big[\log \frac{B_{t+1}}{B_t}\Big] + \mathrm{Var}_t\Big[\log \frac{\tilde{S}_{t+1}}{\tilde{S}_t}R_t^f\Big] + 2\mathrm{Cov}_t\Big[\log \frac{B_{t+1}}{B_t}, \log \frac{\tilde{S}_{t+1}}{\tilde{S}_t}R_t^f\Big]$$

We apply the following facts to this decomposition. First, recall that under conditional log-normality $L_t(X_{t+1}) = \frac{1}{2}\mathrm{Var}_t[\log X_{t+1}]$. Conditional log-normality also implies that beliefs are only distorted about conditional means, and so $L_t(X_{t+1}) = \tilde{L}_t(X_{t+1})$. Using these results, we have

$$L_t\Big[\frac{S_{t+1}}{S_t}R_t^f\Big] = L_t\Big[\frac{B_{t+1}}{B_t}\Big] + \tilde{L}_t\Big[\frac{\tilde{S}_{t+1}}{\tilde{S}_t}R_t^f\Big] + \mathrm{Cov}_t\Big[\log \frac{B_{t+1}}{B_t}, \log \frac{\tilde{S}_{t+1}}{\tilde{S}_t}R_t^f\Big]$$

Taking unconditional expectations, dividing by $L[\frac{S_{t+1}}{S_t}R_t^f]$, and rearranging, we can rewrite this as the following:

$$\frac{L(\frac{B_{t+1}}{B_t})}{L(\frac{S_{t+1}}{S_t}R_t^f)} = 1 - \frac{\mathbb{E}\tilde{L}_t(\frac{\tilde{S}_{t+1}}{\tilde{S}_t}R_t^f)}{L(\frac{S_{t+1}}{S_t}R_t^f)} - \frac{\mathbb{E}\mathrm{Cov}_t[\log \frac{B_{t+1}}{B_t}, \log \frac{\tilde{S}_{t+1}}{\tilde{S}_t}R_t^f]}{L(\frac{S_{t+1}}{S_t}R_t^f)}$$

Note we have used $\mathbb{E}L_t[\frac{B_{t+1}}{B_t}] = L[\frac{B_{t+1}}{B_t}]$ and $\mathbb{E}L_t[\frac{S_{t+1}}{S_t}R_t^f] = L[\frac{S_{t+1}}{S_t}R_t^f]$, since $\mathbb{E}[\frac{B_{t+1}}{B_t}] = \mathbb{E}[\frac{S_{t+1}}{S_t}R_t^f] = 1$. Using the bound in (A.2) on the second term on the right-hand-side, we obtain the result (28). (Notice that the bound (A.2) has not used the UEI condition. That said, UEI holds automatically under the conditions stated for this proposition, in particular joint conditional log-normality of all variables.) □

A.7 Proof of Lemma 2

Under conditional log-normality, no-arbitrage implies that S and B take the form

$$\begin{aligned}\frac{S_{t+1}}{S_t} &= (R_t^f)^{-1} \exp\Big[-\frac{1}{2}\|\pi_t\|^2 - \pi_t \cdot W_{t+1} \Big] \\ \frac{B_{t+1}}{B_t} &= \exp\Big[-\frac{1}{2}\|\mu_t\|^2 + \mu_t \cdot W_{t+1} \Big]\end{aligned}$$

where W_{t+1} is a vector of standard normal shocks. If we recall that $S = B\tilde{S}$, then investor marginal utility is given by

$$\frac{\tilde{S}_{t+1}}{\tilde{S}_t} = (R_t^f)^{-1} \exp\Big[-\frac{1}{2}\|\pi_t\|^2 + \frac{1}{2}\|\mu_t\|^2 - \tilde{\pi}_t \cdot W_{t+1} \Big]$$

where $\tilde{\pi}_t = \pi_t + \mu_t$ is the investor-perceived market price of risk (or, put differently, $\mu_t = \tilde{\pi}_t - \pi_t$ is the deviation between perceived and actual risk prices). By no-arbitrage and the investor Euler equation, each of these risk prices is related to the vector of basis returns R by

$$\begin{aligned}\pi_t &= \mathrm{Var}_t[R_{t+1}]^{-1/2}\Big(\mathbb{E}_t[R_{t+1}] - R_t^f\Big) \\ \tilde{\pi}_t &= \tilde{\mathrm{Var}}_t[R_{t+1}]^{-1/2}\Big(\tilde{\mathbb{E}}_t[R_{t+1}] - R_t^f\Big)\end{aligned}$$

At this level of generality, the covariance of interest can be written

$$\Sigma(B, \tilde{S}) = -\mathbb{E}[\mu_t \cdot \tilde{\pi}_t]. \tag{A.3}$$

Next, we project $(B, S, \tilde{S})$ into the space spanned by a one-dimensional return R. In that case, it is without loss of generality to assume a single dimension for the shock vector as well. We also use the fact that, under conditional log-normality, conditional return variances are equal under $\mathbb{P}$ and $\tilde{\mathbb{P}}$, so we have $\mathrm{Var}_t[R_{t+1}] = \tilde{\mathrm{Var}}_t[R_{t+1}] = \tilde{v}_t^2$. Using these results,

$$\begin{aligned}\Sigma(B, \tilde{S}) = -\mathbb{E}[(\tilde{\pi}_t - \pi_t)\tilde{\pi}_t] &= \mathbb{E}\Big[(\mathbb{E}_t[R_{t+1}] - \tilde{\mathbb{E}}_t[R_{t+1}])\tilde{\pi}_t/\tilde{v}_t\Big] \\ &= \mathbb{E}\Big[(R_{t+1} - \tilde{\mathbb{E}}_t[R_{t+1}])\tilde{\pi}_t/\tilde{v}_t\Big]\end{aligned}$$

where the second line uses the law of iterated expectations. Noting that $\tilde{\pi}_t$ is the perceived Sharpe ratio and recognizing $FE_{t+1} = R_{t+1} - \tilde{\mathbb{E}}_t[R_{t+1}]$, we have the result. □

A.8 UEI holds in continuous-time Brownian environments

The Unconditional Entropy Inequality (UEI) assumption says that all higher moments beyond the mean are perceived equally by the subjective and objective probability. In this section, I show, as claimed in the text, that the UEI automatically holds in continuous-time Brownian environments. I model the dynamics of a variable $\log X_t$ via an Itô process and prove in such case that

$$\lim_{\Delta \to 0} \Delta^{-1}\tilde{L}_t(X_{t+\Delta}) = \lim_{\Delta \to 0} \Delta^{-1}L_t(X_{t+\Delta}).$$

i.e., local versions of the entropy operator coincide.

Suppose

$$d \log X_t = \alpha_t dt + \nu_t \cdot dW_t,$$

where W is a Brownian motion under $\mathbb{P}$. By Girsanov's theorem, the dynamics under the subjective measure are

$$d \log X_t = \tilde{\alpha}_t dt + \nu_t \cdot d\tilde{W}_t,$$

where $\tilde{W}$ is a Brownian motion under $\tilde{\mathbb{P}}$ and $\tilde{\alpha}_t$ is another adapted process. Compute

$$L_t(X_{t+\Delta}) = \log(\mathbb{E}_t[\exp(\int_t^{t+\Delta} (\alpha_s + \frac{1}{2}|\nu_s|^2) ds)]) - \int_t^{t+\Delta} \alpha_s ds$$
$$\tilde{L}_t(X_{t+\Delta}) = \log(\tilde{\mathbb{E}}_t[\exp(\int_t^{t+\Delta} (\tilde{\alpha}_s + \frac{1}{2}|\nu_s|^2) ds)]) - \int_t^{t+\Delta} \tilde{\alpha}_s ds,$$

where $\mathbb{E}$ and $\tilde{\mathbb{E}}$ are alternative expectation operators. Dividing both metrics by Δ and taking $\Delta \to 0$, and using L'Hôpital's rule, yields

$$\lim_{\Delta \to 0} \Delta^{-1} L_t(X_{t+\Delta}) = \alpha_t + \frac{1}{2}|\nu_t|^2 - \alpha_t = \frac{1}{2}|\nu_t|^2 = \tilde{\alpha}_t + \frac{1}{2}|\nu_t|^2 - \tilde{\alpha}_t = \lim_{\Delta \to 0} \Delta^{-1} \tilde{L}_t(X_{t+\Delta})$$

Thus, the two metrics coincide locally. □

A.9 Proof of Proposition 5

Using the calculations in Appendix B, using the notation $\Delta X = \frac{X_{t+1}}{X_t}$ for any variable X, and using the definition of the perceived persistence $\tilde{\rho} = 1 - (1 - \phi)\lambda$, we have the following connection between the two volatility ratios

$$\frac{L(\Delta G)}{L(\Delta S)} = K \times \frac{L(\Delta B)}{L(\Delta S)}, \quad \text{where} \quad K := \frac{2(1 - \tilde{\rho} + \phi\nu_1/\sigma_c)}{(1 - \tilde{\rho})^2 \sigma_c^2}$$

This automatically proves the result that $\frac{L(\Delta G)}{L(\Delta S)}$ and $\frac{L(\Delta B)}{L(\Delta S)}$ move in the same direction in response to any parameter not explicitly included in K. Thus, the result is proved for γ, β, ν_2.

For the remaining parameter ϕ, we proceed directly as follows. First, from the fact that K is increasing in ϕ, holding $\tilde{\rho}$ fixed, it suffices to show that $\frac{L(\Delta B)}{L(\Delta S)}$ is increasing in ϕ, because that will imply that $\frac{L(\Delta G)}{L(\Delta S)}$ is also increasing in ϕ. Second, using the expression in Appendix B and noting $\text{Var}[r_t] = \text{Var}[x_t]$, we may write

$$\frac{L(\Delta B)}{L(\Delta S)} = \frac{1/\sigma_c^2}{\|\tilde{\pi}\|^2/\text{Var}[x_t] + 1 + 1/\sigma_c^2}.$$

Consequently, $\frac{L(\Delta B)}{L(\Delta S)}$ is increasing in ϕ if and only if $\|\tilde{\pi}\|^2/\text{Var}[x_t]$ is decreasing in ϕ. Using the expressions $\text{Var}[x_t] = \frac{\phi^2\|\nu\|^2}{1-(\tilde{\rho}-\phi\nu_1/\sigma_c)^2}$ and $\tilde{\pi} = \gamma\sigma_c\binom{1}{0} + \frac{(\gamma-1)\beta}{1-\beta\tilde{\rho}}\phi\nu$, this ratio is

$$\frac{\|\tilde{\pi}\|^2}{\text{Var}[x_t]} = \left[1 - (\tilde{\rho} - \frac{\phi\nu_1}{\sigma_c})^2\right] \frac{[\gamma\sigma_c + \frac{(\gamma-1)\beta\phi\nu_1}{1-\beta\tilde{\rho}}]^2 + [\frac{(\gamma-1)\beta\phi\nu_2}{1-\beta\tilde{\rho}}]^2}{\phi^2\|\nu\|^2}$$

Differentiate $\|\tilde{\pi}\|^2/\text{Var}[x_t]$ with respect to ϕ, holding $\tilde{\rho}$ fixed, to get

$$\frac{\partial}{\partial\phi} \frac{\|\tilde{\pi}\|^2}{\text{Var}[x_t]}\Big|_{\tilde{\rho}\text{ fixed}} = 2\frac{1}{\phi^3\|\nu\|^2}\Big\{\Big(\tilde{\rho} - \frac{\phi\nu_1}{\sigma_c}\Big)\frac{\phi\nu_1}{\sigma_c}\|\tilde{\pi}\|^2 - \Big[1 - (\tilde{\rho} - \frac{\phi\nu_1}{\sigma_c})^2\Big]\Big(\gamma\sigma_c + \frac{(\gamma-1)\beta\phi\nu_1}{1-\beta\tilde{\rho}}\Big)\gamma\sigma_c\Big\}$$

This expression is clearly negative for ϕ small enough or for $\tilde{\rho}$ small enough. □

A.10 Proof of Proposition 6

See Appendix C for the expressions for the conditional entropies to obtain (43). □

B Derivations for the models in Section 6

B.1 Baseline long-run risk model of Section 6.1

First, guess that the value function is given by the form $U_t = \log C_t + u_0 + u_x x_t$, and obtain the values of u_0 and u_x by substituting this guess into the Bellman equation (31) and using the method of undetermined coefficients. This delivers

$$\begin{aligned} u_x &= \frac{\beta}{1-\beta\rho} \\ u_0 &= \frac{\beta}{1-\beta}\Big[(1-\rho)\bar{x}u_x + \frac{1}{2}(1-\gamma)\big([\sigma_c + (1-\kappa^2)^{1/2}\sigma_x u_x]^2 + \kappa^2\sigma_x^2 u_x^2\big)\Big] \end{aligned}$$

From this solution, we can write the SDF explicitly as

$$\frac{S_{t+1}}{S_t} = \beta\frac{C_t}{C_{t+1}}\frac{\exp[(1-\gamma)U_{t+1}]}{\mathbb{E}_t[\exp[(1-\gamma)U_{t+1}]]} = \exp\Big\{-r_t - \pi\cdot W_{t+1} - \frac{1}{2}\|\pi\|^2\Big\}, \tag{B.1}$$

where the one-period interest rate $r_t = \log R_t^f$ and market price of risk π are given by

$$r_t := -\log(\beta) + x_t + \frac{1}{2}\sigma_c^2 - \gamma\sigma_c^2 - \frac{(\gamma-1)\beta}{1-\beta\rho}(1-\kappa^2)^{1/2}\sigma_x\sigma_c \tag{B.2}$$

$$\pi := \gamma\sigma_c \left(\begin{smallmatrix}1\\0\end{smallmatrix}\right) + \frac{(\gamma-1)\beta}{1-\beta\rho}\sigma_x\left(\begin{smallmatrix}(1-\kappa^2)^{1/2}\\ \kappa\end{smallmatrix}\right). \tag{B.3}$$

Now, we factorize the SDF. Guess that the transitory piece is given by $G_t = e^{\eta t - \zeta(x_t - x_0)}$ for some constants (η,ζ), and then substitute the guess $S_t = G_t H_t$ into the SDF formula. This yields

$$\begin{aligned} \frac{H_{t+1}}{H_t} &= \exp(-\eta + \zeta(x_{t+1}-x_t))\frac{S_{t+1}}{S_t} \\ &= \exp\Big\{-\eta - r_t - (\pi - \zeta\sigma_x\left(\begin{smallmatrix}(1-\kappa^2)^{1/2}\\ \kappa\end{smallmatrix}\right))\cdot W_{t+1} - \frac{1}{2}\|\pi\|^2 + \zeta(1-\rho)(\bar{x}-x_t)\Big\} \end{aligned}$$

Using the conjecture that H is a martingale allows us to solve for η and ζ with the method of undetermined coefficients. These values are then

$$\begin{aligned} \zeta &= -(1-\rho)^{-1} \\ \eta &= \log(\beta) - \frac{1}{2}\sigma_c^2 + \gamma\sigma_c^2 + \frac{(\gamma-1)\beta}{1-\beta\rho}(1-\kappa^2)^{1/2}\sigma_x\sigma_c + \zeta(1-\rho)\bar{x} + \frac{1}{2}\zeta^2\sigma_x^2 - \zeta\sigma_x\left(\begin{smallmatrix}(1-\kappa^2)^{1/2}\\ \kappa\end{smallmatrix}\right)\cdot\pi. \end{aligned}$$

Thus, we obtain

$$\frac{H_{t+1}}{H_t} = \exp\Big\{-\frac{1}{2}\Big\|\pi + \frac{\sigma_x}{1-\rho}\left(\begin{smallmatrix}(1-\kappa^2)^{1/2}\\ \kappa\end{smallmatrix}\right)\Big\|^2 - \Big(\pi + \frac{\sigma_x}{1-\rho}\left(\begin{smallmatrix}(1-\kappa^2)^{1/2}\\ \kappa\end{smallmatrix}\right)\Big)\cdot W_{t+1}\Big\} \tag{B.4}$$

$$\frac{G_{t+1}}{G_t} = \exp\Big[\eta + \frac{1}{1-\rho}(x_{t+1}-x_t)\Big]. \tag{B.5}$$

Finally, we compute volatilities of these objects. The annual average SDF volatility is

$$\mathbb{E}\Big[2L_t\Big(\frac{S_{t+1}}{S_t}\Big)\Big]^{-1/2} = \|\pi\|$$

The sizes of the permanent and transitory components are given by

$$\frac{L(\frac{H_{t+1}}{H_t})}{L(\frac{S_{t+1}}{S_t})} = \frac{\left\|\pi + \frac{\sigma_x}{1-\rho}\left(\begin{smallmatrix}(1-\kappa^2)^{1/2}\\ \kappa\end{smallmatrix}\right)\right\|^2}{\|\pi\|^2 + \mathrm{Var}[r_t]}$$

$$\frac{L(\frac{G_{t+1}}{G_t})}{L(\frac{S_{t+1}}{S_t})} = \frac{\frac{2}{1-\rho}\mathrm{Var}[r_t]}{\|\pi\|^2 + \mathrm{Var}[r_t]}.$$

For all of these expressions, note that $\mathrm{Var}[r_t] = \mathrm{Var}[x_t]$, i.e., the objective variance of agent's growth perceptions. Theoretically, one can show that that $L(\Delta H)/L(\Delta S)$ is always greater than 1 if $\gamma \geq 1$.[13]

B.2 Growth-extrapolation model of Section 6.2

In the text, we have shown that the growth-extrapolation model is isomorphic, in investors' heads, to a long-run risks model with $\rho = \tilde{\rho} := 1 - (1-\phi)\lambda$, $\phi\nu_1 = (1-\kappa^2)^{1/2}\sigma_x$, and $\phi\nu_2 = \kappa\sigma_x$ Therefore, the investor marginal utility and its permanent component are identical to (B.1) and (B.4), with the appropriate replacement of parameters. These are given by

$$\frac{\tilde{S}_{t+1}}{\tilde{S}_t} = \exp\Big\{ - r_t - \tilde{\pi}\cdot\tilde{W}_{t+1} - \frac{1}{2}\|\tilde{\pi}\|^2\Big\} \tag{B.6}$$

$$\frac{\tilde{H}_{t+1}}{\tilde{H}_t} = \exp\Big\{ - \frac{1}{2}\Big\|\tilde{\pi} + \frac{\phi\nu}{1-\tilde{\rho}}\Big\|^2 - \Big(\tilde{\pi} + \frac{\phi\nu}{1-\tilde{\rho}}\Big)\cdot\tilde{W}_{t+1}\Big\} \tag{B.7}$$

where r_t and $\tilde{\pi}$ are the riskless rate and perceived risk prices, respectively,

$$r_t := -\log(\beta) + x_t + \frac{1}{2}\sigma_c^2 - \gamma\sigma_c^2 - \frac{(\gamma-1)\beta}{1-\beta\tilde{\rho}}\sigma_c\phi\nu_1$$

$$\tilde{\pi} := \gamma\sigma_c\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right) + \frac{(\gamma-1)\beta}{1-\beta\tilde{\rho}}\phi\nu.$$

Similarly, the transitory component of the SDF, which recall is identical under both the subjective and objective probability, is given by

$$\frac{G_{t+1}}{G_t} = \exp\Big[\tilde{\eta} + \frac{1}{1-\tilde{\rho}}(x_{t+1} - x_t)\Big],$$

for some constant $\tilde{\eta}$.

We compute the SDF and its permanent component via the formulas $S_t = B_t\tilde{S}_t$ and $H_t = B_t\tilde{H}_t$. Combining equations (B.6)-(B.7) with the belief distortion (36), we have

$$\frac{S_{t+1}}{S_t} = \exp\Big\{ - r_t - (\tilde{\pi} - \mu_t)\cdot W_{t+1} - \frac{1}{2}\|\tilde{\pi} - \mu_t\|^2\Big\} \tag{B.8}$$

$$\frac{H_{t+1}}{H_t} = \exp\Big\{ - \frac{1}{2}\Big\|\tilde{\pi} - \mu_t + \frac{\phi\nu}{1-\tilde{\rho}}\Big\|^2 - \Big(\tilde{\pi} - \mu_t - \frac{\phi\nu}{1-\tilde{\rho}}\Big)\cdot W_{t+1}\Big\} \tag{B.9}$$

[13] The unconditional variance of r_t is computed as $\sigma_x^2/(1-\rho^2)$ using its AR(1) dynamics inherited from x_t. Then, using the fact that $\pi \geq 0$ when $\gamma \geq 1$, followed by the assumption $0 < \rho < 1$, we have $\|\pi + \frac{\sigma_x}{1-\rho}\left(\begin{smallmatrix}(1-\kappa^2)^{1/2}\\ \kappa\end{smallmatrix}\right)\|^2 \geq \|\pi\|^2 + (\frac{\sigma_x}{1-\rho})^2 \geq \|\pi\|^2 + \frac{\sigma_x^2}{1-\rho^2} = \|\pi\|^2 + \mathrm{Var}[r_t]$.

Thus, objective risk prices are lowered by μ_t relative to perceived risk prices. For example, if agents are pessimistic about growth, then the first component of μ_t is negative, which raises risk prices.

Next, we compute the volatilities of all these objects. The annual average SDF volatility is

$$\mathbb{E}\Big[2L_t\Big(\frac{S_{t+1}}{S_t}\Big)\Big]^{1/2} = \Big(\mathbb{E}\|\tilde{\pi} - \mu_t\|^2\Big)^{1/2} = \Big(\|\tilde{\pi}\|^2 + \frac{\mathrm{Var}[r_t]}{\sigma_c^2}\Big)^{1/2}.$$

The sizes of the permanent and transitory components are given by

$$\frac{L(\frac{H_{t+1}}{H_t})}{L(\frac{S_{t+1}}{S_t})} = \frac{\|\tilde{\pi} + \frac{\phi\nu}{1-\tilde{\rho}}\|^2 + \frac{1}{\sigma_c^2}\mathrm{Var}[r_t]}{\|\tilde{\pi}\|^2 + (1+\frac{1}{\sigma_c^2})\mathrm{Var}[r_t]}$$

$$\frac{L(\frac{G_{t+1}}{G_t})}{L(\frac{S_{t+1}}{S_t})} = \frac{\frac{2(1-\tilde{\rho}+\frac{\phi\nu_1}{\sigma_c})}{(1-\tilde{\rho})^2}\mathrm{Var}[r_t]}{\|\tilde{\pi}\|^2 + (1+\frac{1}{\sigma_c^2})\mathrm{Var}[r_t]}$$

Finally, the ratio of subjective-to-objective SDF volatility is given by

$$\frac{L(\frac{\tilde{S}_{t+1}}{\tilde{S}_t})}{L(\frac{S_{t+1}}{S_t})} = \frac{\|\tilde{\pi}\|^2 + (1-\frac{\tilde{\pi}_1}{\sigma_c})^2\mathrm{Var}[r_t]}{\|\tilde{\pi}\|^2 + (1+\frac{1}{\sigma_c^2})\mathrm{Var}[r_t]}$$

and its permanent counterpart by

$$\frac{L(\frac{\tilde{H}_{t+1}}{\tilde{H}_t})}{L(\frac{H_{t+1}}{H_t})} = \frac{\|\tilde{\pi} + \frac{\phi\nu}{1-\tilde{\rho}}\|^2 + (\tilde{\pi}_1 + \frac{\phi\nu_1}{1-\tilde{\rho}})^2\frac{1}{\sigma_c^2}\mathrm{Var}[r_t]}{\|\tilde{\pi} + \frac{\phi\nu}{1-\tilde{\rho}}\|^2 + \frac{1}{\sigma_c^2}\mathrm{Var}[r_t]}$$

Finally, the size of belief volatility in the SDF is given by

$$\frac{L(\frac{B_{t+1}}{B_t})}{L(\frac{S_{t+1}}{S_t})} = \frac{\frac{1}{\sigma_c^2}\mathrm{Var}[r_t]}{\|\tilde{\pi}\|^2 + (1+\frac{1}{\sigma_c^2})\mathrm{Var}[r_t]}$$

To compute all these volatility ratios, we can calculate them directly or use Lemma B.1 below with $M \in \{\tilde{S}, S, \tilde{H}, H, G, B\}$. For all of these expressions, note that $\mathrm{Var}[r_t] = \mathrm{Var}[x_t]$. Furthermore, using $\tilde{W}_{t+1} = W_{t+1} - \mu_t$, we can show that x_t also follows an AR(1) under the objective measure, but with modified persistence:

$$x_{t+1} - \bar{x} = \Big(1 - \lambda(1-\phi) - \frac{\phi\nu_1}{\sigma_c}\Big)(x_t - \bar{x}) + \phi\nu\cdot W_{t+1},$$

and so its unconditional moments are $\mathbb{E}[x_t] = \bar{x}$ and $\mathrm{Var}[x_t] = \frac{\phi^2\|\nu\|^2}{1-(1-\lambda(1-\phi)-\frac{\phi\nu_1}{\sigma_c})^2}$.

B.3 CRRA growth-extrapolation model of Section 6.3

In the CRRA model, marginal utility growth is simply

$$\frac{\tilde{S}_{t+1}}{\tilde{S}_t} = \beta\Big(\frac{C_{t+1}}{C_t}\Big)^{-\gamma} = \exp\Big\{-r_t - \frac{1}{2}\|\tilde{\pi}\|^2 - \tilde{\pi}\cdot\tilde{W}_{t+1}\Big\},$$

where the riskless rate and perceived risk prices are given by

$$r_t := -\log(\beta) + \gamma x_t - \frac{1}{2}(\gamma\sigma_c)^2$$
$$\tilde{\pi} := \gamma\sigma_c\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)$$

Following the same procedures as above, we can compute the other objects as

$$\begin{aligned}
\frac{\tilde{H}_{t+1}}{\tilde{H}_t} &= \exp\left\{-\frac{1}{2}\left\|\tilde{\pi} + \frac{\gamma\phi\nu}{1-\tilde{\rho}}\right\|^2 - \left(\tilde{\pi} + \frac{\gamma\phi\nu}{1-\tilde{\rho}}\right)\cdot \tilde{W}_{t+1}\right\}\\
\frac{S_{t+1}}{S_t} &= \exp\left\{-r_t - \frac{1}{2}\|\tilde{\pi} - \mu_t\|^2 - (\tilde{\pi} - \mu_t)\cdot W_{t+1}\right\}\\
\frac{H_{t+1}}{H_t} &= \exp\left\{-\frac{1}{2}\left\|\tilde{\pi} - \mu_t + \frac{\gamma\phi\nu}{1-\tilde{\rho}}\right\|^2 - \left(\tilde{\pi} - \mu_t + \frac{\gamma\phi\nu}{1-\tilde{\rho}}\right)\cdot W_{t+1}\right\}\\
\frac{G_{t+1}}{G_t} &= \exp\left[\tilde{\eta} + \frac{\gamma}{1-\tilde{\rho}}(x_{t+1} - x_t)\right],
\end{aligned}$$

for some constant $\tilde{\eta}$. Note that the belief distortion B is identical in this CRRA model as the Epstein-Zin version above. To compute volatility ratios, we again use Lemma B.1 for a generic $M \in \{\tilde{S}, S, \tilde{H}, H, G, B\}$.

B.4 Lemma to compute volatility ratios

The volatility ratios in these models are all easily computable analytically, because of the log-normality of the equilibrium solution. We have the following tool for this purpose. Every object of interest in this model takes the form of some M in (B.10) below. Therefore, applying this lemma to the specific forms of S, H, $\tilde{S}$, $\tilde{H}$, G, and B, we can obtain any volatility ratio we like. Note that to apply this to objects like $\tilde{S}$ and $\tilde{H}$, which are written in the text as being driven by the subjective shocks $\tilde{W}$, we first need to write them in terms of the objective shocks by using the identity $\tilde{W}_{t+1} = W_{t+1} - \mu_t$.

Lemma B.1. *For some scalars a, b, q and vector p, consider the process*

$$\frac{M_{t+1}}{M_t} = \exp\left\{a + bx_t - [p + q\mu_t]\cdot W_{t+1} - \frac{1}{2}\|p + q\mu_t\|^2\right\}, \tag{B.10}$$

where x_t follows an AR(1) with long-run mean $\bar{x}$, and where $\mu_t = \frac{x_t - \bar{x}}{\sigma_c}\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)$. Then,

$$L\left(\frac{M_{t+1}}{M_t}\right) = \log\mathbb{E}\left[e^{a+bx_t}\right] - (a + b\bar{x}) + \frac{1}{2}\mathbb{E}\|p + q\mu_t\|^2 = \frac{1}{2}\left(b^2 + \frac{q^2}{\sigma_c^2}\right)Var[x_t] + \frac{1}{2}\|p\|^2. \tag{B.11}$$

Proof. This follows from direct calculation of $L(\Delta M) = \log\mathbb{E}[\Delta M] - \mathbb{E}[\log\Delta M]$ using the fact that $\mathbb{E}[\mu_t] = 0$, $\mathrm{Var}[\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)\cdot\mu_t] = \frac{1}{\sigma_c^2}\mathrm{Var}[x_t]$, and $\mathrm{Var}[x_t]$ is the stationary variance of the AR(1) x_t. □

C Derivations for the models in Section 7

C.1 Baseline two-agent model

Equilibrium derivation. First, we solve the model up to a set of ODEs.

Proposition C.1. *Assume a solution exists to the system of ODEs (C.8) below. Then, there exists a bubble-free equilibrium that is Markovian in the pessimists' consumption share x_t. The equilibrium market price of risk and interest rate are given by*

$$\begin{aligned}
\pi_t &= \gamma\sigma_c\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right) + x_t\beta\left(\begin{smallmatrix}1\\\chi\end{smallmatrix}\right) \\
r_t &= \rho + \gamma\delta\Big(1 - \sum_\theta \eta^\theta \frac{c^\theta_{t,t}}{C_t}\Big) + \gamma(\mu_c - \sigma_c^2) - \frac{1}{2}\frac{1+\gamma}{\gamma}\|\pi_t\|^2 + \sigma_c\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)\cdot\pi_t \\
&\quad - x_t\frac{\beta}{\gamma}\left(\begin{smallmatrix}1\\\chi\end{smallmatrix}\right)\cdot\Big[\gamma\sigma_c\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right) + \frac{1}{2}(1-\gamma)\beta\left(\begin{smallmatrix}1\\\chi\end{smallmatrix}\right) - \pi_t\Big].
\end{aligned}$$

The drift and diffusion of pessimist's consumption share are given by

$$\begin{aligned}
\mu_{x,t} &= \delta\Big[(1-x_t)\frac{\eta^1 c^1_{t,t}}{C_t} - x_t\frac{\eta^0 c^0_{t,t}}{C_t}\Big] + x_t(1-x_t)\frac{\beta}{\gamma^2}\left(\begin{smallmatrix}1\\\chi\end{smallmatrix}\right)\cdot\Big[\gamma\sigma_c\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right) + \frac{1}{2}(1-\gamma)\beta\left(\begin{smallmatrix}1\\\chi\end{smallmatrix}\right) - \pi_t\Big] \\
\sigma_{x,t} &= -\frac{1}{\gamma}x_t(1-x_t)\beta\left(\begin{smallmatrix}1\\\chi\end{smallmatrix}\right)
\end{aligned}$$

The expressions for the newborn consumption shares $c^\theta_{t,t}/C_t$ are given in the proof.

Proof of Proposition C.1. The FOC for a type-θ agent born at time u is

$$\frac{c^\theta_{t,u}}{c^\theta_{u,u}} = e^{-\frac{\rho}{\gamma}(t-u)}\Big(\frac{B^\theta_t}{B^\theta_u}\Big)^{\frac{1}{\gamma}}\Big(\frac{S_t}{S_u}\Big)^{-\frac{1}{\gamma}} \tag{C.1}$$

Plug this into the definition of aggregate type-θ consumption $c^\theta_t := \eta^\theta\int_{-\infty}^t \delta e^{-\delta(t-u)} c^\theta_{t,u}du$ to obtain the following expression for the consumption share $x^\theta_t := c^\theta_t/C_t$:

$$x^\theta_t = \frac{\eta^\theta\int_{-\infty}^t \delta e^{-(\delta+\frac{\rho}{\gamma})(t-u)} c^\theta_{u,u}\big(\frac{B^\theta_t}{B^\theta_u}\big)^{\frac{1}{\gamma}}\big(\frac{S_t}{S_u}\big)^{-\frac{1}{\gamma}}du}{C_t} \tag{C.2}$$

Applying Itô's formula to this expression, we obtain

$$\begin{aligned}
dx^\theta_t &= \delta\frac{\eta^\theta c^\theta_{t,t}}{C_t}dt - \Big(\delta + \frac{\rho}{\gamma}\Big)x^\theta_t dt + \frac{1}{\gamma}x^\theta_t\Big(\frac{dB^\theta_t}{B^\theta_t} - \frac{dS_t}{S_t}\Big) \\
&\quad + x^\theta_t\Big[\frac{1}{2\gamma^2}\Big((1-\gamma)\frac{d[B^\theta]_t}{(B^\theta_t)^2} + (1+\gamma)\frac{d[S]_t}{S_t^2} - 2\frac{d[B^\theta,S]_t}{B^\theta_t S_t}\Big) - \frac{dC_t}{C_t} + \frac{d[C]_t}{C_t^2} - \frac{1}{\gamma}\Big(\frac{d[B^\theta,C]_t}{B^\theta_t C_t} - \frac{d[S,C]_t}{S_t C_t}\Big)\Big]
\end{aligned} \tag{C.3}$$

Substituting $dB^\theta_t = B^\theta_t\theta\beta\left(\begin{smallmatrix}1\\\chi\end{smallmatrix}\right)\cdot dW_t$, $dS_t = -S_t[r_t dt + \pi_t\cdot dW_t]$, and $dC_t = C_t[\mu_c dt + \sigma_c\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)\cdot dW_t]$, and then using the market clearing condition $\sum dx^\theta_t = 0$, we obtain the equations for π_t and r_t given in the proposition. Substituting those results back into (C.3) for $\theta = 1$, we obtain the expressions for $\mu_{x,t}$ and $\sigma_{x,t}$ as well.

Next, substitute the FOC (C.1) into the lifetime budget constraint (40) to obtain

$$\frac{c^\theta_{t,t}}{C_t} = \frac{\kappa p_t}{\delta q_{\theta,t}}, \tag{C.4}$$

where we have defined

$$p_t := \mathbb{E}_t\Big[\int_t^\infty e^{-\kappa(u-t)} \frac{S_u}{S_t}\frac{C_u}{C_t} du\Big] \tag{C.5}$$

$$q_{\theta,t} := \mathbb{E}\Big[\int_t^\infty e^{-(\delta+\frac{\rho}{\gamma})(u-t)} \big(\frac{B^\theta_u}{B^\theta_t}\big)^{\frac{1}{\gamma}} \big(\frac{S_u}{S_t}\big)^{1-\frac{1}{\gamma}} du\Big] \tag{C.6}$$

As these are present values, we are already imposing a conjecture that the equilibrium is bubble-free, as stated in the proposition. Note that $p_t = P_t/C_t$ is the aggregate price-dividend ratio in the economy, if the equilibrium is bubble-free. Given the homogeneity of preferences and budget sets, all alive type-θ agents consume the same fraction of their wealth. Hence, the type-θ wealth-to-consumption ratio n^θ_t/c^θ_t is equal to the wealth-to-consumption ratio of a type-θ newborn. A newborn's initial wealth is $P_{t,t}/\delta = \kappa p_t C_t/\delta$ and their consumption is $c^\theta_{t,t}$, which by (C.4) implies that the type-θ wealth-consumption ratio is $q_{\theta,t}$. With this result, we use asset market clearing $\sum_\theta n^\theta_t = P_t$ to obtain that

$$\sum_\theta x^\theta_t q_{\theta,t} = p_t. \tag{C.7}$$

Since $x^0_t = 1 - x^1_t = 1 - x_t$, we can obtain the price-dividend ratio p_t if we know the state variable and the wealth-consumption ratios $q_{\theta,t}$.

Finally, we now show that q_0, q_1 solve an ODE system. We guess-and-verify that the equilibrium objects $r_t, \pi_t, \mu_{x,t}, \sigma_{x,t}, p_t, q_{\theta,t}$ are solely functions of x_t (i.e., a Markov equilibrium). Observe that

$$e^{-(\delta+\frac{\rho}{\gamma})t}(B^\theta_t)^{\frac{1}{\gamma}} S_t^{1-\frac{1}{\gamma}} q_{\theta,t} + \int_0^t e^{-(\delta+\frac{\rho}{\gamma})u}(B^\theta_u)^{\frac{1}{\gamma}} S_u^{1-\frac{1}{\gamma}} du$$

is a local martingale and must have zero drift. Applying Itô's formula, we have

$$\begin{aligned}0 = 1 - \frac{1}{\gamma}\Big[\gamma\delta + \rho + (\gamma-1)r(x) - \frac{1}{2}\frac{1-\gamma}{\gamma}(\theta\beta)^2(1+\chi^2) - \frac{1}{2}\frac{\gamma-1}{\gamma}\|\pi(x)\|^2 - \frac{\gamma-1}{\gamma}(\theta\beta)\big(\begin{smallmatrix}1\\ \chi\end{smallmatrix}\big)\cdot\pi(x)\Big]q_\theta(x) \\ + \Big[\mu_x(x) + \sigma_x(x)\cdot\Big(\frac{1}{\gamma}\theta\beta\big(\begin{smallmatrix}1\\ \chi\end{smallmatrix}\big) - \frac{\gamma-1}{\gamma}\pi(x)\Big)\Big]q'_\theta(x) + \frac{1}{2}\|\sigma_x(x)\|^2 q''_\theta(x), \quad \theta\in\{0,1\}.\end{aligned} \tag{C.8}$$

Equation (C.8) is an ODE for q_θ. Because the equilibrium risk-free rate r and the drift μ_x both depend on $c^\theta_{t,t}/C_t$, hence on p and q_θ, these ODEs are nonlinear and they form a coupled system. As the proposition states, we assume a solution to this system of ODEs, in which case the entire equilibrium is indeed Markovian in x. □

Remark C.1 (Log utility special case)**.** *With log utility ($\gamma = 1$), we have an analytical solution. In particular, from the definition of q_θ, and using the change-of-measure B^θ to convert from $\mathbb{E}$ to $\tilde{\mathbb{E}}^\theta$, we have $q_\theta(x) = (\delta+\rho)^{-1}$. By (C.7), we then have $p(x) = (\delta+\rho)^{-1}$. Plugging these back into (C.4), we have $c^\theta_{t,t}/C_t = \kappa/\delta$ for each θ. The expressions for the interest rate, market price of risk, and state dynamics become*

$$r(x) = \rho + \delta - \kappa + \mu_c - \sigma_c^2 - x\beta\sigma_c \qquad \mu_x(x) = \kappa\big[(1-x)\eta^1 - x\eta^0\big] - x^2(1-x)\beta^2(1+\chi^2)$$

$$\pi(x) = \sigma_c\big(\begin{smallmatrix}1\\ 0\end{smallmatrix}\big) + x\beta\big(\begin{smallmatrix}1\\ \chi\end{smallmatrix}\big) \qquad \sigma_x(x) = -x(1-x)\beta\big(\begin{smallmatrix}1\\ \chi\end{smallmatrix}\big)$$

Decomposition of the SDF. In this model, since the $\theta = 0$ agents are rational, their marginal utility is equated to the SDF, and so

$$\frac{S_t}{S_0} = e^{-\rho t}\Big(\frac{1-x_t}{1-x_0}\Big)^{-\gamma}\Big(\frac{C_t}{C_0}\Big)^{-\gamma}. \tag{C.9}$$

Since the state variable x_t is stationary, due to the OLG structure, the permanent component of S_t is precisely the permanent component of $C_t^{-\gamma}$. Using Itô's formula, we can write

$$\frac{C_t^{-\gamma}}{C_0^{-\gamma}} = \exp\Big\{-\frac{1}{2}(\gamma\sigma_c)^2 t - \gamma\sigma_c\big(\begin{smallmatrix}1\\0\end{smallmatrix}\big)\cdot W_t + \Big(\frac{1}{2}\gamma(\gamma+1)\sigma_c^2 - \gamma\mu_c\Big)t\Big\},$$

and so the martingale component of this process is

$$\frac{H_t}{H_0} = \exp\Big\{-\frac{1}{2}(\gamma\sigma_c)^2 t - \gamma\sigma_c\big(\begin{smallmatrix}1\\0\end{smallmatrix}\big)\cdot W_t\Big\}.$$

The transitory component of the SDF is thus $G_t = S_t/H_t$, or

$$\frac{G_t}{G_0} = \Big(\frac{1-x_t}{1-x_0}\Big)^{-\gamma}\exp\Big\{-\Big(\rho + \gamma\mu_c - \frac{1}{2}\gamma(\gamma+1)\sigma_c^2\Big)t\Big\}.$$

Next, we use (41) to define the average belief distortion. Noting that $dB^0/B^0 = 0$, we thus have $dB/B = \eta dB^1/B^1$. From here, we may apply Itô's formula to obtain $d\log B_t = \eta dB_t^1/B_t^1 - \frac{1}{2}\eta^2 d[B^1]_t/(B_t^1)^2 = \eta\beta\big(\begin{smallmatrix}1\\ \chi\end{smallmatrix}\big)\cdot dW_t - \frac{1}{2}\eta^2\beta^2(1+\chi^2)dt$. Integrating this result, we have

$$\frac{B_t}{B_0} = \exp\Big\{-\frac{1}{2}\eta^2\beta^2(1+\chi^2)t - \eta\beta\big(\begin{smallmatrix}1\\ \chi\end{smallmatrix}\big)\cdot W_t\Big\}.$$

Defining $\tilde{S}_t := S_t/B_t$ and $\tilde{H}_t := H_t/B_t$, as required by the aggregation results given in Appendix D, we then obtain

$$\begin{aligned}
\frac{\tilde{S}_t}{\tilde{S}_0} &= \exp\Big\{-\int_0^t \Big(r_u + \frac{1}{2}\|\pi_u\|^2 - \frac{1}{2}\eta^2\beta^2(1+\chi^2)\Big)du - \int_0^t \Big(\pi_u - \eta\beta\big(\begin{smallmatrix}1\\ \chi\end{smallmatrix}\big)\Big)\cdot dW_u\Big\}\\
\frac{\tilde{H}_t}{\tilde{H}_0} &= \exp\Big\{-\frac{1}{2}\Big((\gamma\sigma_c)^2 - \eta^2\beta^2(1+\chi^2)\Big)t - \Big(\gamma\sigma_c\big(\begin{smallmatrix}1\\0\end{smallmatrix}\big) - \eta\beta\big(\begin{smallmatrix}1\\ \chi\end{smallmatrix}\big)\Big)\cdot W_t\Big\}
\end{aligned}$$

Volatilities of the SDF components. We use the localized entropy definition given in equation (42) to compute

$$\begin{aligned}
\mathcal{L}_t(S) &= \frac{1}{2}\|\pi_t\|^2\\
\mathcal{L}_t(H) &= \frac{1}{2}(\gamma\sigma_c)^2\\
\mathcal{L}_t(G) &= \frac{1}{2}\beta^2(1+\chi^2)x_t^2\\
\mathcal{L}_t(B) &= \frac{1}{2}\eta^2\beta^2(1+\chi^2)\\
\mathcal{L}_t(\tilde{S}) &= \mathcal{L}_t(S) + \mathcal{L}_t(B) - \eta\beta\big(\gamma\sigma_c + \beta(1+\chi^2)x_t\big)\\
\mathcal{L}_t(\tilde{H}) &= \mathcal{L}_t(H) + \mathcal{L}_t(B) - \eta\beta\gamma\sigma_c
\end{aligned}$$

C.2 Generalization: arbitrary heterogeneity and arbitrary beliefs

Objective setting. The objective environment is as follows. Let K be a 3×3 full-rank matrix, and let K_i denote the ith column of K, which has unit length. The aggregate endowment follows a geometric process with time-varying growth and uncertainty,

$$\begin{aligned}\frac{dC_t}{C_t} &= g_t dt + \sqrt{v_t} K_1 \cdot dW_t \\ dg_t &= -\omega_g (g_t - \bar{g}) dt + \sigma_g \sqrt{v_t} K_2 \cdot dW_t \\ dv_t &= -\omega_v (v_t - \bar{v}) dt + \sigma_v \sqrt{v_t} K_3 \cdot dW_t,\end{aligned}$$

where W is a three-dimensional Brownian motion. One can think of $K_1 \cdot dW_t$ as the consumption level shock, $K_2 \cdot dW_t$ as the consumption growth rate shock, and $K_3 \cdot dW_t$ as the uncertainty shock. By no-arbitrage, the objective SDF follows

$$\frac{dS_t}{S_t} = -r_t dt - \pi_t \cdot dW_t,$$

for an endogenous interest rate r_t and risk price vector π_t.

Agent types and beliefs. An agent's type is indexed by $\theta := (\tilde{\omega}_g, \tilde{\omega}_v, \tilde{\bar{g}}, \tilde{\bar{v}}, \chi)$, which are the variables that characterize an agent's belief distortions in this economy. Let the population distribution of agents be given by the probability density $\eta(\theta)$. Let Θ denote the domain of θ.

We nest all agent types and their beliefs as follows. An arbitrary agent perceives the dynamics to be

$$\begin{aligned}\frac{dC_t}{C_t} &= \tilde{g}_t^\theta dt + \sqrt{v_t} K_1 \cdot d\tilde{W}_t^\theta \\ d\tilde{g}_t^\theta &= -\tilde{\omega}_g^\theta (\tilde{g}_t^\theta - \tilde{\bar{g}}^\theta) dt + \sigma_g \sqrt{v_t} K_2 \cdot d\tilde{W}_t^\theta \\ dv_t &= -\tilde{\omega}_v^\theta (v_t - \tilde{\bar{v}}^\theta) dt + \sigma_v \sqrt{v_t} K_3 \cdot d\tilde{W}_t^\theta\end{aligned}$$

All variables with tildes are the perceived counterparts to the objective variables.

By Girsanov's theorem, we have that the agent's belief distortion follows

$$\frac{dB_t^\theta}{B_t^\theta} = m_t^\theta \cdot dW_t$$

and that the perceived Brownian motion is linked to the original shocks via

$$d\tilde{W}_t^\theta = dW_t - m_t^\theta.$$

Substituting this into the perceived dynamics, we have

$$\begin{aligned}\frac{dC_t}{C_t} &= \Big(\tilde{g}_t^\theta - \sqrt{v_t} K_1 \cdot m_t^\theta\Big) dt + \sqrt{v_t} K_1 \cdot dW_t \\ d\tilde{g}_t^\theta &= \Big(-\tilde{\omega}_g^\theta (\tilde{g}_t^\theta - \tilde{\bar{g}}^\theta) - \sigma_g \sqrt{v_t} K_2 \cdot m_t^\theta\Big) dt + \sigma_g \sqrt{v_t} K_2 \cdot dW_t \\ dv_t &= \Big(-\tilde{\omega}_v^\theta (v_t - \tilde{\bar{v}}^\theta) - \sigma_v \sqrt{v_t} K_3 \cdot m_t^\theta\Big) dt + \sigma_v \sqrt{v_t} K_3 \cdot dW_t\end{aligned}$$

Equating the expressions for dC_t and dv_t to their original objective counterparts, we have

$$K_1 \cdot m_t^\theta = \frac{\tilde{g}_t^\theta - g_t}{\sqrt{v_t}}$$
$$K_3 \cdot m_t^\theta = \frac{\omega_v(v_t - \bar{v}) - \tilde{\omega}_v^\theta(v_t - \bar{\tilde{v}}^\theta)}{\sigma_v \sqrt{v_t}}$$

Since $K_2 \cdot m_t^\theta$ is not restricted, I set it to $\frac{\chi^\theta}{\sqrt{v_t}}$ for some type-specific constant χ^θ—this functions almost exactly like "sentiment shocks" from our other models. Thus, the belief distortion is completely characterized by

$$m_t^\theta = \frac{1}{\sqrt{v_t}}(K')^{-1} \begin{bmatrix} \tilde{g}_t^\theta - g_t \\ \chi^\theta \\ \frac{\omega_v(v_t - \bar{v}) - \tilde{\omega}_v^\theta(v_t - \bar{\tilde{v}}^\theta)}{\sigma_v} \end{bmatrix} \tag{C.10}$$

Note that this is only a function of exogenous variables $(\tilde{g}_t^\theta, g_t, v_t)$, and therefore the system (g_t, v_t) and $(\tilde{g}_t^\theta)_{\theta \in \Theta}$ forms an exogenous Markovian system characterizing beliefs and objective dynamics.

Aggregation. Finally, I assume that $\eta(\theta)$ is the density of newborns are of type-θ, which implies that population shares are constant. Other details of the model are identical to Section 7, with the exception that all market clearing conditions now involve integrals rather than sums.

Indeed, the equilibrium market clearing conditions are as follows. Define the aggregate consumption of type-θ agents, $c_t^\theta := \eta(\theta) \int_{-\infty}^t \delta e^{-\delta(u-t)} c_{u,t}^\theta du$. The goods market clearing condition says $\int c_t^\theta d\theta = C_t$. Rewrite this in terms of consumption shares by defining $x_t^\theta := c_t^\theta / C_t$ to be the consumption share of the type-θ agents and then noting that $\int x_t^\theta d\theta = 1$. Additionally, the asset market clearing condition says that $\int n_t^\theta d\theta = P_t$, where n_t^θ is the total net worth of type-θ agents and P_t is the total wealth in the economy, namely the total present value of all trees alive at time t. Recall that, because of the exponential decay of dividends, total wealth is simply equal to $P_t = \mathbb{E}_t\big[\int_t^\infty e^{-\kappa u} \frac{S_u}{S_t} C_u du\big] = P_{t,t}/\kappa$.

For technical convenience, I assume there is at least one rational agent, which I denote by $\theta = \theta^*$. This makes calculations easier, since the SDF can be determined from their marginal utility process.

Assumption C.1. *There exists at least one agent type, denoted by $\theta = \theta^*$, which has $\tilde{\omega}_g^\theta = \omega_g$, $\tilde{\omega}_v^\theta = \omega_v$, $\bar{\tilde{g}}^\theta = \bar{g}$, $\bar{\tilde{v}}^\theta = \bar{v}$, and $\chi^\theta = 0$. Hence, this agent has $m_t^{\theta^*} = 0$.*

Model solution. I characterize the entire equilibrium in terms of the objective state variables (g_t, v_t), the belief distortions $(m_t^\theta : \theta \in \Theta)$, and the consumption shares $(x_t^\theta : \theta \in \Theta)$. The dynamics of x_t^θ are endogenous. Write

$$dx_t^\theta = \mu_{x,t}^\theta dt + \sigma_{x,t}^\theta \cdot dW_t$$

for some drift $\mu_{x,t}^\theta$ and diffusion $\sigma_{x,t}^\theta$ to be determined.

Proposition C.2. *In a bubble-free equilibrium, the following hold. Define the consumption-weighted mean and second moment of individual optimisms:*

$$M_t^x := \int_\Theta x_t^\theta m_t^\theta d\theta$$
$$M_t^{x,2} := \int_\Theta x_t^\theta \|m_t^\theta\|^2 d\theta$$

The equilibrium market price of risk and interest rate are given by

$$\pi_t = \gamma\sqrt{v_t}K_1 - M_t^x$$
$$r_t = \rho + \gamma\delta\Big(1 - \int \eta(\theta)\frac{c_{t,t}^{\theta}}{C_t}d\theta\Big) + \gamma(g_t - v_t) - \frac{1+\gamma}{2}\|\pi_t\|^2 + \sqrt{v_t}K_t \cdot (\pi_t + M_t^x) - \pi_t \cdot M_t^x - \frac{1-\gamma}{2}M_t^{x,2}$$

The drift and diffusion of the type-θ consumption share is given by

$$\mu_{x,t}^{\theta} = \delta\Big(\eta(\theta)\frac{c_{t,t}^{\theta}}{C_t} - x_t^{\theta}\int \eta(\theta')\frac{c_{t,t}^{\theta'}}{C_t}d\theta'\Big) - \sigma_{x,t}^{\theta} \cdot (\sqrt{v_t}K_t - \pi_t) - \frac{1-\gamma}{2\gamma}x_t^{\theta}\Big(M_t^{x,2} - \|m_t^{\theta}\|^2\Big)$$
$$\sigma_{x,t}^{\theta} = \frac{1}{\gamma}x_t^{\theta}(m_t^{\theta} - M_t^x)$$

Finally, note that x_t^{θ} never reaches $\{0\}$ with probability 1.

Proof of Proposition C.2. The FOC for a type-θ agent born at time u is still (C.1). And therefore, equations (C.1), (C.2), and (C.3) continue to hold. Substituting dB_t^{θ}, dS_t, and dC_t, and using the market clearing condition $\int_{\Theta} dx_t^{\theta}d\theta = 0$, we obtain the equations for π_t and r_t given in the proposition. Substituting those results back into (C.3) for any given θ, we obtain the expressions for $\mu_{x,t}^{\theta}$ and $\sigma_{x,t}^{\theta}$ as well.

Next, as before, we substitute the FOC (C.1) into the lifetime budget constraint (40) to obtain the newborn consumption share (C.4). We can also follow the same steps to obtain (C.7), which says that a consumption-weighted average of the wealth-consumption ratios $q_{\theta,t}$ equals the aggregated price-dividend ratio p_t. It remains, then, to determined the wealth-consumption ratios $(q_{\theta,t})_{\theta\in\Theta}$. These objects are determined as fixed points to their present-value equations (C.6). These equations are fixed point equations, rather than explicit solutions, because $q_{\theta,t}$ and p_t appear in the SDF via the riskless rate r_t, which has the newborn consumption term $c_{t,t}^{\theta}/C_t = \kappa p_t/\delta q_{\theta,t}$. Additionally, the drift $\mu_{x,t}^{\theta}$ also contains $c_{t,t}^{\theta}/C_t$.

Finally, the claim that x_t^{θ} never reaches $\{0\}$ is proved by examining its dynamics. Notice that as $x_t^{\theta} \to 0$, we have that $\sigma_{x,t}^{\theta} \to 0$ and $\mu_{x,t}^{\theta} \to \delta\eta(\theta)\frac{c_{t,t}^{\theta}}{C_t}$, which is positive since $c_{t,t}^{\theta} > 0$. Given these limits, we have that $\{0\}$ is an inaccessible boundary. □

C.3 Belief distortions do not impact H in the general model

Now, I prove the main result of interest for the richer model. In particular, we can decompose the SDF S into permanent and transitory components, H and G, respectively. Then,

Proposition C.3. *The local entropy of the permanent component, $\mathcal{L}_t(H)$, is exogenous to belief distortions.*

Proof of Proposition C.3. Given the SDF process, I first obtain its permanent-transitory factorization. Recall there is a rational agent, denoted by type $\theta = \theta^*$ (Assumption C.1). The marginal utility of this agent is equated to the SDF, and so equation (44) holds, i.e.,

$$\frac{S_t}{S_0} = e^{-\rho t}\Big(\frac{x_t^{\theta^*}}{x_0^{\theta^*}}\Big)^{-\gamma}\Big(\frac{C_t}{C_0}\Big)^{-\gamma}. \tag{C.11}$$

Note that the state variables x_t^{θ} are stationary, due to fact that $\{0\}$ is an inaccessible boundary for all x_t^{θ} (Proposition C.2). And so H_t is precisely the permanent component of $C_t^{-\gamma}$. Since $C_t^{-\gamma}$ is exogenous to beliefs, we therefore also have that H_t is exogenous to beliefs.

More formally, Lemma C.1 below shows that the permanent component H_t follows

$$d\log H_t = -\frac{1}{2}\|z_t\|^2 dt + z_t \cdot dW_t, \quad \text{where} \quad z_t := -\sqrt{v_t}\Big(K_1 - \alpha_g\sigma_g K_2 - \alpha_v\sigma_v K_3\Big), \tag{C.12}$$

where $\alpha_g = -\gamma\omega_g^{-1}$ and α_v is the smaller root of the following quadratic equation[14]

$$0 = \frac{1}{2}(\alpha_v\sigma_v)^2 - \Big(\omega_v + \gamma\sigma_v K_1\cdot K_3 - \alpha_g\sigma_g\sigma_v K_2\cdot K_3\Big)\alpha_v - \gamma\alpha_g\sigma_g K_1\cdot K_2 + \frac{1}{2}\gamma(\gamma+1) + \frac{1}{2}(\alpha_g\sigma_g)^2.$$

Importantly, notice that (α_g, α_v) are independent of all the parameters characterizing belief distortions $(\tilde{\omega}_g^\theta, \tilde{\omega}_v^\theta, \tilde{\bar{g}}^\theta, \tilde{\bar{v}}^\theta, \chi^\theta)$. Therefore, H_t is independent of all belief distortions. □

Lemma C.1. *The martingale component of S in equation (C.11) is given by a process H that follows (C.12).*

Proof of Proposition C.1. Because $x_t^{\theta^*}$ is stationary, it suffices to obtain the martingale component of $C_t^{-\gamma}$. Let H be a martingale and write its dynamics as

$$d\log H_t = -\frac{1}{2}\|z_t\|^2 dt + z_t \cdot dW_t,$$

for some process z_t to be determined. First, notice from Itô's formula that

$$\frac{dC_t^{-\gamma}}{C_t^{-\gamma}} = \Big(a_g g_t + a_v v_t\Big)dt + \sqrt{v_t}\, b\cdot dW_t \tag{C.13}$$

where

$$a_g = -\gamma, \quad a_v = \frac{1}{2}\|b\|^2 + \frac{\gamma}{2}, \quad b = -\gamma K_1.$$

Conjecture that

$$\frac{C_t^{-\gamma}}{C_0^{-\gamma}} = \underbrace{\exp\Big[-\alpha_0 t - \alpha_g(g_t - g_0) - \alpha_v(v_t - v_0)\Big]}_{\text{stationary component of } C^{-\gamma}}\frac{H_t}{H_0}, \tag{C.14}$$

[14] As shown in Hansen and Scheinkman (2009b), a unique permanent-transitory factorization renders the state dynamics stationary under the probability measure induced by H. We follow their theory in picking this unique factorization. For any H, the dynamics under the induced measure are given by

$$\begin{aligned} dg_t &= \Big[\omega_g(\bar{g} - g_t) - \sigma_g v_t K_2 \cdot \Big(K_1 - \alpha_g\sigma_g K_2 - \alpha_v\sigma_v K_3\Big)\Big]dt + \sigma_g\sqrt{v_t}K_2\cdot dW_t^H \\ dv_t &= \Big[\omega_v(\bar{v} - v_t) - \sigma_v v_t K_3 \cdot \Big(K_1 - \alpha_g\sigma_g K_2 - \alpha_v\sigma_v K_3\Big)\Big]dt + \sigma_v\sqrt{v_t}K_3\cdot dW_t^H \end{aligned}$$

Stacking these states into a vector $X_t = (g_t, v_t)'$, the drift is given by $\mathbb{E}_t[dX_t] = (A_0 + AX_t)dt$, where

$$A = \begin{bmatrix} -\omega_g & -\sigma_g K_2\cdot(K_1 + \sigma_g\omega_g^{-1}K_2 - \alpha_v\sigma_v K_3) \\ 0 & -\omega_v - \sigma_v K_3\cdot(K_1 + \sigma_g\omega_g^{-1}K_2 - \alpha_v\sigma_v K_3) \end{bmatrix}$$

Both eigenvalues of A are negative, hence the dynamics stable, if and only if $\omega_v + \sigma_v K_3\cdot(K_1 + \sigma_g\omega_g^{-1}K_2) > \alpha_v\sigma_v^2$, which proves that we should always take the smaller root for α_v.

for some $\alpha_0, \alpha_g, \alpha_v$. If so, then by matching the conjectured dynamics in (C.14) to the actual dynamics in (C.13), we have

$$z_t = \sqrt{v_t}\Big(b + \alpha_g \sigma_g K_2 + \alpha_v \sigma_v K_3\Big)$$
$$-\frac{1}{2}\|z_t\|^2 = a_g g_t + a_v v_t - \frac{1}{2}\|b\|^2 v_t + \alpha_0 + \alpha_g \omega_g (\bar{g} - g_t) + \alpha_v \omega_v (\bar{v} - v_t)$$

Combining these two equations to eliminate z yields

$$\begin{aligned} 0 &= a_g g_t + a_v v_t - \frac{1}{2}\|b\|^2 v_t + \alpha_0 + \alpha_g \omega_g (\bar{g} - g_t) + \alpha_v \omega_v (\bar{v} - v_t) \\ &\quad + \frac{1}{2} v_t \Big(\|b\|^2 + (\alpha_g \sigma_g)^2 + (\alpha_v \sigma_v)^2 + 2\alpha_g \sigma_g \alpha_v \sigma_v K_2 \cdot K_3 + 2\alpha_g \sigma_g K_2 \cdot b + 2\alpha_v \sigma_v K_3 \cdot b \Big) \end{aligned}$$

In order for this equation to hold for all possible values of (g_t, v_t), we require

$$\begin{aligned} 0 &= \alpha_0 + \alpha_g \omega_g \bar{g} + \alpha_v \omega_v \bar{v} \\ 0 &= a_g - \alpha_g \omega_g \\ 0 &= a_v - \omega_v \alpha_v + \frac{1}{2}\sigma_v^2 \alpha_v^2 + \frac{1}{2}\sigma_g^2 \alpha_g^2 + \alpha_g \sigma_g \alpha_v \sigma_v K_2 \cdot K_3 + \alpha_g \sigma_g K_2 \cdot b + \alpha_v \sigma_v K_3 \cdot b \end{aligned}$$

The first two equations immediately deliver solutions for α_0 and α_g, which are

$$\begin{aligned} \alpha_0 &= -\big(\alpha_g \omega_g \bar{g} + \alpha_v \omega_v \bar{v}\big) \\ \alpha_g &= \frac{a_g}{\omega_g} \end{aligned}$$

The third equation is a quadratic equation for α_v, which is

$$0 = \frac{1}{2}\sigma_v^2 \alpha_v^2 - \Big(\omega_v - \sigma_v K_3 \cdot b - \alpha_g \sigma_g \sigma_v K_2 \cdot K_3\Big)\alpha_v + a_v + \frac{1}{2}\sigma_g^2 \alpha_g^2 + \alpha_g \sigma_g K_2 \cdot b$$

As explained in footnote 14, we must take the smaller root of this quadratic equation. Plugging $a_g = -\gamma$, $a_v = \frac{1}{2}\|b\|^2 + \frac{\gamma}{2}$, and $b = -\gamma K_1$ into all these equations delivers the results claimed. □

Entropies of all components of the SDF. For completeness, we calculate the local volatilities of all components of the SDF. Given the permanent component H of the SDF, we obtain the transitory component G from (C.11) and (C.14) as

$$\frac{G_t}{G_0} = e^{-\rho t}\Big(\frac{x_t^{\theta^*}}{x_0^{\theta^*}}\Big)^{-\gamma} \exp\Big[-\alpha_0 t - \alpha_g (g_t - g_0) - \alpha_v (v_t - v_0)\Big]$$

The middle term satisfies $-\gamma d\log(x_t^{\theta^*}) = -\gamma[\mu_{x,t}^{\theta^*}/x_t^{\theta^*} - \frac{1}{2}\|\sigma_{x,t}^{\theta^*}/x_t^{\theta^*}\|^2]dt - \gamma(\sigma_{x,t}^{\theta^*}/x_t^{\theta^*}) \cdot dW_t$. Using Proposition C.2 to substitute $\mu_x^{\theta^*}$ and $\sigma_x^{\theta^*}$, and using Assumption C.1 to substitute $m_t^{\theta^*} = 0$, we obtain that $d\log G_t$ follows

$$\begin{aligned} d\log G_t &= -\rho dt - \alpha_0 dt - \alpha_g dg_t - \alpha_v dv_t - \gamma d\log(x_t^{\theta^*}) \\ &= \Big[\alpha_g \omega_g g_t + \alpha_v \omega_v v_t - \rho - \delta\gamma\Big(\frac{\eta(\theta)}{x_t^{\theta}}\frac{c_{t,t}^{\theta}}{C_t} - \int \eta(\theta')\frac{c_{t,t}^{\theta'}}{C_t}d\theta'\Big)\Big]dt \\ &\quad + \Big[\frac{1-\gamma}{2}M_t^{x,2} + \frac{1}{2\gamma}\|M_t^x\|^2 - M_t^x \cdot (\sqrt{v_t}K_1 - \pi_t)\Big]dt + \Big[M_t^x - \sqrt{v_t}\Big(\alpha_g \sigma_g K_2 + \alpha_v \sigma_v K_3\Big)\Big] \cdot dW_t \end{aligned}$$

Next, recall that the average belief distortion is defined by

$$\frac{dB_t}{B_t} = \int \eta(\theta)\frac{dB_t^\theta}{B_t^\theta}d\theta = \Big(\int \eta(\theta)m_t^\theta d\theta\Big) \cdot dW_t$$

The localized entropies of S, H, G, and B are thus given by

$$\begin{aligned}
\mathcal{L}_t(S) &= \frac{1}{2}v_t + \frac{1}{2}\|M_t^x\|^2 - \sqrt{v_t}K_1 \cdot M_t^x \\
\mathcal{L}_t(H) &= \frac{1}{2}v_t\Big(1 - [\alpha_g\sigma_g K_2 + \alpha_v\sigma_v K_3] \cdot K_1 + (\alpha_g\sigma_g)^2 + (\alpha_v\sigma_v)^2 + 2\alpha_g\alpha_v\sigma_g\sigma_v K_2 \cdot K_3\Big) \\
\mathcal{L}_t(G) &= \frac{1}{2}v_t\Big((\alpha_g\sigma_g)^2 + (\alpha_v\sigma_v)^2 + 2\alpha_g\alpha_v\sigma_g\sigma_v K_2 \cdot K_3\Big) + \frac{1}{2}\|M_t^x\|^2 - \sqrt{v_t}\Big(\alpha_g\sigma_g K_2 + \alpha_v\sigma_v K_3\Big) \cdot M_t^x \\
\mathcal{L}_t(B) &= \frac{1}{2}\Big\| \int \eta(\theta)m_t^\theta d\theta\Big\|^2
\end{aligned}$$

D Heterogeneity and aggregation

In reality, survey data do not contain a single measure of beliefs, but rather an entire cross-section. A common practice is to compute summary statistics like the cross-sectional average belief about an asset return. If we use such an average belief, then whose marginal utility are we bounding via Lemma 1?

Suppose there are $i \in \{1, \dots, N\}$ individuals making return forecasts. All the analysis from the previous sections goes through, i-by-i. In particular, each of their beliefs can be characterized by a $\mathbb{P}$-martingale B_t^i, and each of them have a marginal utility process $\tilde{S}_t^i$. Assuming perfect risk-sharing for the shocks that are spanned by asset payoffs, we can treat $\tilde{S}_t^i$ as the projection of true individual marginal utility onto the return space. In that case, we have the generalization of formula (5) to

$$S_t = \tilde{S}_t^i B_t^i, \quad \forall i. \tag{D.1}$$

Following the permanent-transitory factorization arguments as above, we then also have the generalization of formula (8),

$$H_t = \tilde{H}_t^i B_t^i, \quad \forall i, \tag{D.2}$$

where $\tilde{H}^i$ is the permanent component of $\tilde{S}^i$, i.e., $\tilde{H}^i$ is a martingale under $\tilde{\mathbb{P}}^i$. All agents share the same transitory component $\tilde{G}^i = G$.

Averaging beliefs amounts to the following aggregation procedure. Consider B defined by

$$\frac{B_{t+1}}{B_t} := \frac{1}{N} \sum_{i=1}^{N} \frac{B_{t+1}^i}{B_t^i}. \tag{D.3}$$

Then, B is a $\mathbb{P}$-martingale as in the initial analysis. Computing the cross-sectional average belief is equivalent to using B:

$$\frac{1}{N} \sum_{i=1}^{N} \tilde{\mathbb{E}}_t^i [X_{t+1}] = \frac{1}{N} \sum_{i=1}^{N} \mathbb{E}_t [\frac{B_{t+1}^i}{B_t^i} X_{t+1}] = \mathbb{E}_t [\frac{B_{t+1}}{B_t} X_{t+1}] =: \tilde{\mathbb{E}}_t [X_{t+1}].$$

In that case, to satisfy the original relation (8), we must define

$$\tilde{S}_t := \frac{1}{N} \sum_{i=1}^{N} \Big(\frac{B_t^i}{B_t} \Big) \tilde{S}_t^i. \tag{D.4}$$

and

$$\tilde{H}_t := \frac{1}{N} \sum_{i=1}^{N} \Big(\frac{B_t^i}{B_t} \Big) \tilde{H}_t^i. \tag{D.5}$$

One can think of $\tilde{S}$ and $\tilde{H}$ as the "aggregated marginal utility" and its permanent component, and equations (D.4)-(D.5) shows that these are really belief-weighted-averages of individual variables. Furthermore, it is straightforward to check that $\tilde{H}$ is a $\tilde{\mathbb{P}}$-martingale, where $\tilde{\mathbb{P}}$ is defined via B (i.e., through the averaging procedure). With this definition, the same bounds on $\tilde{S}$ and $\tilde{H}$ hold as the previous analysis (e.g., Lemma 1 and subsequent results), i.e., we have

$$\tilde{L}_t\Bigg(\underbrace{\frac{1}{N}\sum_{i=1}^{N}\Big(\frac{B^i_{t+1}/B^i_t}{B_{t+1}/B_t}\Big)\frac{\tilde{S}^i_{t+1}}{\tilde{S}^i_t}}_{=\tilde{S}_{t+1}/\tilde{S}_t}\Bigg) \geq \underbrace{\frac{1}{N}\sum_{i=1}^{N}\tilde{\mathbb{E}}^i_t}_{:=\tilde{\mathbb{E}}_t}\big[\log R_{t+1} - \log R^f_t\big] \tag{D.6}$$

$$\tilde{L}_t\Bigg(\underbrace{\frac{1}{N}\sum_{i=1}^{N}\Big(\frac{B^i_{t+1}/B^i_t}{B_{t+1}/B_t}\Big)\frac{\tilde{H}^i_{t+1}}{\tilde{H}^i_t}}_{=\tilde{H}_{t+1}/\tilde{H}_t}\Bigg) \geq \underbrace{\frac{1}{N}\sum_{i=1}^{N}\tilde{\mathbb{E}}^i_t}_{:=\tilde{\mathbb{E}}_t}\big[\log R_{t+1} - \log R^\infty_{t+1}\big] \tag{D.7}$$

Bounds (D.6)-(D.7) show that we may average the survey expected returns, but we must interpret this average belief as a lower bound on the volatility of *belief-weighted-averages* of individual-level variables like $\tilde{S}^i$ and $\tilde{H}^i$.

D.1 Forecast horizons and consensus beliefs

To illustrate the difficulties in defining a consensus belief distortion, and to simultaneously justify our definition for the continuous-time model, I now jump to a continuous-time Brownian model.

Let W be a vector of Brownian motions driving all information. Consider agents of types $\theta \in \Theta$, and let $\eta(\theta)$ denote the population density of type-θ agents. Type-θ agents' belief distortions follow

$$\frac{dB^\theta_t}{B^\theta_t} = m^\theta_t \cdot dW_t$$

for some process m^θ_t.

The problem: we need to define an aggregate belief distortion B. Given the Brownian information structure, and the assumption that B is a Radon-Nikodym derivative, it is without loss of generality to let

$$\frac{dB_t}{B_t} = m_t \cdot dW_t$$

for some m_t. All local martingales have this structure in the present environment (and under weak conditions, B will in fact be a true martingale).

We want to figure out how to define m_t in terms of $(m^\theta_t)_{\theta\in\Theta}$ in order to obtain

$$\frac{B_{t+\Delta}}{B_t} = \int_\Theta \frac{B^\theta_{t+\Delta}}{B^\theta_t}\eta(\theta)d\theta \tag{D.8}$$

for every t and some horizon $\Delta > 0$. If we succeed in defining such a B, then all conditional expectations at horizon Δ will be "consensus beliefs." To see this, just check that for any variable $X \in \mathcal{F}_{t+\Delta}$, we have

$$\mathbb{E}_t\Big[\frac{B_{t+\Delta}}{B_t}X\Big] = \mathbb{E}\Big[\Big(\int_\Theta \frac{B^\theta_{t+\Delta}}{B^\theta_t}\eta(\theta)d\theta\Big)X\Big] = \int_\Theta \mathbb{E}_t\Big[\frac{B^\theta_{t+\Delta}}{B^\theta_t}X\Big]\eta(\theta)d\theta$$

This satisfies our objective, since I am using consensus beliefs from the surveys, which are on the right-hand-side.

For (D.8) to hold, a requirement is that

$$m_T = \int_\Theta \frac{B_T^\theta / B_t^\theta}{B_T / B_t} m_T^\theta \eta(\theta) d\theta, \quad \forall T \in [t, t+\Delta). \tag{D.9}$$

This implies that the aggregate belief cannot be well-defined and internally consistent for any horizon except the local horizon $\Delta \to 0$. To see this, consider replacing t with $t+s < t+\Delta$ (which intuitively means the aggregate belief is trying to forecast from the perspective of the later time $t+s$ instead of time t), and notice that the expression for m_T cannot remain the same.

Therefore, we must focus on the local horizon $\Delta \to 0$. Taking this limit in (D.9), the requirement on aggregation collapses to

$$m_t = \int_\Theta m_t^\theta \eta(\theta) d\theta \tag{D.10}$$

or equivalently

$$\frac{dB_t}{B_t} = \int_\Theta \frac{dB_t^\theta}{B_t^\theta} \eta(\theta) d\theta. \tag{D.11}$$

In other words, we define the increments of B_t by the cross-sectional population-weighted average of individual belief distortion increments. This ensures that over short horizons—but only over short horizons—the belief defined by B is indeed the "consensus expectation." Notice that (D.11) coincides with my proposed aggregation in equation (41) for the two-type disagreement model.

Remark D.1. *Importantly, straight time-averaging of belief distortions, which may sound very natural, does not produce the appropriate consensus conditional expectation. To see this, consider defining B instead by*

$$B_t = \int_\Theta B_t^\theta \eta(\theta) d\theta.$$

By Itô's formula, this is equivalent to

$$m_t = \int_\Theta \frac{B_t^\theta}{B_t} m_t^\theta \eta(\theta) d\theta$$

Clearly, this generically differs from condition (D.10), which as we proved above is a requirement for B to define a consensus belief at short horizons. Thus, simple averaging of the individual distortions does not work.

As an example, consider a two-agent framework where one agent is optimistic all shocks to a constant degree ($m_t^o \equiv \bar{m}^o > 0$) while the other agent is rational ($m_t^r = 0$). Suppose there are fraction $\bar{\eta}_o$ optimists and fraction $\bar{\eta}_r = 1 - \bar{\eta}_o$ rational agents. In that case, $m_t = \frac{B_t^o}{\bar{\eta}_o B_t^o + \bar{\eta}_r} \bar{\eta}_o \bar{m}^o$ using the present naive averaging procedure, whereas $m_t = \bar{\eta}_o \bar{m}^o$ using the appropriate averaging procedure (D.10). These two procedures cannot coincide unless all agents are rational.

E Simple growth-optimal portfolios

In contrast to the machine-learning approach in Table 3, this appendix studies other growth-optimal proxies formed solely as combinations of the aggregate stock market with short-term bonds.

Assuming a portfolio that solely mixes the stock market and risk-free rate, the growth-optimal problem is

$$\max_{\theta_t} \mathbb{E}_t \left[\log(\theta_t R^m_{t+\Delta} + (1-\theta_t) R^f_t) \right] \tag{E.1}$$

for some re-investment time interval Δ. (Note also that R^f_t here denotes the risk-free rate between t and $t+\Delta$.) Using the log-normal approximation (or second-order approximation to the log), problem (E.1) becomes

$$\max_{\theta_t} \mathbb{E}_t \left[\theta_t R^m_{t+\Delta} + (1-\theta_t) R^f_t \right] - \frac{1}{2} \theta_t^2 \mathrm{Var}_t \left[R^m_{t+\Delta} \right], \tag{E.2}$$

which has the standard mean-variance optimal solution

$$\theta^*_t = \frac{\mathbb{E}_t \left[R^m_{t+\Delta} - R^f_t \right]}{\mathrm{Var}_t \left[R^m_{t+\Delta} \right]} \tag{E.3}$$

Implementation of (E.3) requires proxies for the objective conditional expected excess return and objective conditional variance.

I explore one of three sets of assumptions, detailed in the next three subsections. The assumptions are, roughly speaking, as follows:

1. Both expected returns and variances depend similarly on implied volatilities. Under this assumption, we recover the baseline with the market as the growth-optimal portfolio.

2. Both expected returns and variances are constant. Under this assumption, we recover a constant-leverage portfolio as growth-optimal.

3. Expected returns are constant, while volatility is predictable. Under this assumption, we recover "volatility-managed portfolios" as growth-optimal.

E.1 Benchmark assumptions: market as growth-optimal

First, one can use the theory from Martin (2017) to construct expected stock returns with option-implied variance (SVIX). In particular, assume (c.f., equation 15 of Martin, 2017)

$$\mathbb{E}_t[R^m_{t+\Delta} - R^f_t] = R^f_t \mathrm{SVIX}^2_t \times \Delta, \tag{E.4}$$

where SVIX^2_t represents the squared SVIX with the same horizon as the period length.

Assume further that risk-neutral and objective conditional variances coincide, which would arise in conditionally log-normal environments, so that (c.f., equation 13 of Martin, 2017)

$$\mathrm{Var}_t[R^m_{t+\Delta}] = (R^f_t)^2 \mathrm{SVIX}^2_t \times \Delta. \tag{E.5}$$

Under (E.4)-(E.5), we have $\theta^*_t = 1/R^f_t \approx 1$, so that the market is approximately growth-optimal, which would be sufficient to justify the estimates in Table 1 as valid lower bounds for the size of risk in the SDF.

equity share θ	Objective Premium $E[\log R^{\theta}/R^{f}]$	**A. Nagel-Xu.** Size of Risks in SDF $L(R^f \Delta\tilde{S})/L(R^f \Delta S)$	**B. Livingston.** Size of Risks in SDF $L(R^f \Delta\tilde{S})/L(R^f \Delta S)$	**C. CFO.** Size of Risks in SDF $L(R^f \Delta\tilde{S})/L(R^f \Delta S)$
$\theta = 0.5$	0.0313	0.8408	0.5620	0.5501
	(0.0079)	(0.2122)	(0.1372)	(0.1346)
$\theta = 1$ (baseline)	0.0555	0.8567	0.5412	0.5349
	(0.0161)	(0.2458)	(0.1506)	(0.1484)
$\theta = 1.5$	0.0722	0.8821	0.5179	0.5178
	(0.0250)	(0.3006)	(0.1712)	(0.1699)
$\theta = 2$	0.0786	0.9513	0.5055	0.5131
	(0.0361)	(0.4280)	(0.2185)	(0.2216)

Table E.1. Portfolios with constant leverage. This table displays estimates of the importance of risks versus belief volatility, using portfolios with constant leverage. The various rows indicate different equity shares θ in equation (E.7). Note that $\theta > 2$ is not possible, because it induces bankruptcy, i.e., $\theta R^m_{t+1} + (1-\theta)R^f_t \leq 0$ in some periods for $\theta > 2$. Panels A, B, and C use the Nagel-Xu, Livingston, and CFO survey expectations for the market, respectively. Newey-West asymptotic standard errors using 3 years of lags (i.e., 12 periods for Nagel-Xu, 6 periods for Livingston, 12 periods for CFO) are shown in parentheses.

E.2 Constant-leverage portfolios

Second, I assume that neither returns nor variances are predictable. This leads to

$$\theta^*_t = \frac{\widehat{\mathbb{E}}[R^m - R^f]}{\widehat{\mathrm{Var}}[R^m]} = \text{constant}, \tag{E.6}$$

which is an optimal portfolio that has no "market timing" (i.e., no time-variation in the weights) but does embed a leverage adjustment.

I implement this empirically as follows. For any constant equity share θ, I construct the following return:

$$R^{\theta}_{t+1} := \theta R^m_{t+1} + (1-\theta)R^f_t, \quad \theta \text{ constant}. \tag{E.7}$$

Starting from (E.7), I estimate the objective expected excess log return $\mathbb{E}[\log(R^{\theta}_{t+1})] - \mathbb{E}[\log(R^f_t)]$ for this strategy. Similarly, I can also use the surveys to compute proxies for the subjective version $\mathbb{E}[\tilde{\mathbb{E}}_t[\log(R^{\theta}_{t+1}) - \log(R^f_t)]]$ using the same log-normal approximation as in (25).[15]

Table E.1 presents the results of these leveraged returns for various values of θ. I continue to find that risk accounts for at least half the volatility of the SDF. Implementing leveraged versions of the market makes almost no difference to such results. The basic reason is that, while leverage does improve the objective return (the first column in Table E.1), it also improves the subjective return series by a similar amount.

[15] Specifically, for any portfolio θ_t, we have the approximation

$$\tilde{\mathbb{E}}_t[\log(R^{\theta}_{t+1})] \approx \log \tilde{\mathbb{E}}_t[R^{\theta}_{t+1}] - \frac{1}{2}\tilde{\mathrm{Var}}_t[\log R^{\theta}_{t+1}] \approx \log\left(\theta_t \tilde{\mathbb{E}}_t[R^m_{t+1}] + (1-\theta_t)R^f_t\right) - \frac{1}{2}\theta_t^2 \tilde{\mathrm{Var}}_t[\log R^m_{t+1}],$$

where the second "≈" uses the first-order approximation of the log function around 1. Then, to estimate $\mathbb{E}[\tilde{\mathbb{E}}_t[\log(R^{\theta}_{t+1}) - \log(R^f_t)]]$, we subtract $\log R^f_t$ and take the time series average.

E.3 Volatility-managed portfolios

Third, I refrain from assuming excess returns are predictable (as this is a notoriously difficult exercise), but I do allow for predictable volatility. In that case, the optimal portfolio is

$$\theta_t^* = \frac{\widehat{\mathbb{E}}[R^m - R^f]}{\widehat{\mathrm{Var}}_t[R^m_{t+1}]}, \tag{E.8}$$

where $\widehat{\mathrm{Var}}_t[R^m_{t+1}]$ is a conditional variance forecast. Here, the idea is closer to the "volatility-managed portfolios" model of Moreira and Muir (2017), where higher variance forecasts signal an optimal deleveraging from the stock market.

I empirically implement volatility-managed portfolios as follows. I construct

$$R^\theta_{t+1} := \theta_t R^m_{t+1} + (1-\theta_t)R^f_t, \quad \theta_t \propto \frac{1}{\text{variance measure at time } t}. \tag{E.9}$$

I take one of three measures for my market variance measure: (i) $\hat{v}_{t+1}$, the month-t forecast of next-month variance from a GARCH(1,1) model; (ii) last month's realized variance v_t; (iii) the time-t level of implied variance, VIX_t^2. To isolate the role of market timing and not introduce any average amount of leverage, I rescale portfolio weights so that the time-series average of θ_t is always one in each case. I also rescale θ_t so that it never exceeds 3 (i.e., this is the maximal allowable leverage).[16] One minor difference from the previous results is that these portfolios are rebalanced monthly, and so R^f_t is a one-month riskless rate here. I convert these monthly return series into annual returns by compounding. (Implementing this monthly is important to achieve non-trivial performance gains, as I show below.)

The results, displayed in Table E.2, are somewhat weaker than those so far. Rows 2-4 show that the size of risks in the SDF is lower-bounded by between 27% and 60%, depending on the survey and market-timing strategy. This is because the volatility-managed portfolios perform very well, with average log excess returns between 7.9-10.9% per annum, depending on which strategy is adopted.[17] Of course, the reader should keep in mind that Table E.2's comparison of objective versus subjective returns is a bit unfair: for each of the volatility-managed portfolios, the subjective counterpart is simply the baseline with $\theta_t = 1$ and so is not optimized in any way.

An additional concern with market-timing strategies is that they may be sensitive to the sample period, predictors used, and precise construction of the market-timing strategy. To illustrate this in a simple way here, the final three rows of Table E.2 compute very similar market-timing strategies, except that the variance signal is lagged by one month. Although variances are generally quite persistent, this lagging procedure dramatically changes the results, with the estimates for the size of risk in the SDF now much closer to the baseline levels. (The portfolio using VIX as its variance signal is more resilient to the lagging of its signal than the others, but this further emphasizes that the choice of predictor variables matters delicately.)

[16] This leverage constraint only affects the portfolio that is based on realized variance, which is highly volatile. The two portfolios based on the forecasted variance and the VIX-squared are unaffected.

[17] The three volatility-managed portfolios have Sharpe ratios of 0.59, 0.58, and 0.69, respectively. For comparison, the benchmark without volatility-management (i.e., $\theta_t = 1$) has a Sharpe ratio of 0.44.

equity share θ_t	Objective Premium $E[\log R^\theta / R^f]$	**A. Nagel-Xu.** Size of Risks in SDF $L(R^f \Delta\tilde{S})/L(R^f \Delta S)$	**B. Livingston.** Size of Risks in SDF $L(R^f \Delta\tilde{S})/L(R^f \Delta S)$	**C. CFO.** Size of Risks in SDF $L(R^f \Delta\tilde{S})/L(R^f \Delta S)$
$\theta_t = 1$ (baseline)	0.0588	0.8093	0.5112	0.5053
	(0.0158)	(0.2127)	(0.1336)	(0.1300)
$\theta_t \propto 1/\hat{v}_{t+1}$	0.0807	0.5898	0.3725	0.3682
	(0.0163)	(0.1079)	(0.0761)	(0.0659)
$\theta_t \propto 1/v_t$	0.0794	0.5990	0.3784	0.3740
	(0.0163)	(0.1122)	(0.0785)	(0.0685)
$\theta_t \propto 1/\text{VIX}_t^2$	0.1084	0.4386	0.2771	0.2739
	(0.0209)	(0.0683)	(0.0486)	(0.0440)
$\theta_t \propto 1/\hat{v}_t$ (lag)	0.0596	0.7986	0.5044	0.4986
	(0.0152)	(0.1973)	(0.1262)	(0.1208)
$\theta_t \propto 1/v_{t-1}$ (lag)	0.0594	0.8002	0.5054	0.4996
	(0.0152)	(0.1982)	(0.1267)	(0.1213)
$\theta_t \propto 1/\text{VIX}_{t-1}^2$ (lag)	0.0929	0.5122	0.3235	0.3198
	(0.0219)	(0.0994)	(0.0661)	(0.0631)

Table E.2. Portfolios with volatility market timing. This table displays estimates of the importance of risks versus belief volatility, using portfolios with time-varying equity shares. The various rows indicate different market-timing strategies θ_t in equation (E.9). In each strategy, I rescale portfolio weights so that the time-series average of θ_t is always one and so that $\max(\theta_t) = 3$. Panels A, B, and C use the Nagel-Xu, Livingston, and CFO survey expectations for the market, respectively. The subjective return series based on the surveys use $\theta_t = 1$ (i.e., no market timing). Newey-West asymptotic standard errors using 3 years of lags (i.e., 12 periods for Nagel-Xu, 6 periods for Livingston, 12 periods for CFO) are shown in parentheses.

F More details on data

F.1 Data samples and sources

	Frequency	Sample Period
A. Stocks		
Excess return	Monthly	1926:12–2020:12
Nagel-Xu	Quarterly	1972:06–1977:06 and 1987:03–2021:03
Livingston	Semi-annually	1952:05–2021:05
CFO	Quarterly	2000:08–2021:05
B. Treasuries		
Par yield, 1-year	Monthly	1953:04–2024:07
Par yield, 10-year	Monthly	1953:04–2024:07
Par yield, 30-year	Monthly	1961:06–2024:07
ZCB yield, 10-year	Monthly	1971:08–2024:07
ZCB yield, 30-year	Monthly	1985:11–2024:07
Return, 10-year	Monthly	1962:06–2024:07
Return, 30-year	Monthly	1962:06–2024:07
Return, index (> 10-year)	Monthly	1954:04–2020:12
BCFF	Monthly	2001:01–2018:06

Table F.1. Samples from key data sources. Note that the Nagel-Xu data from 1972 to 1977 is annual only.

F.2 Yield data

Figure F.1 displays the very tight relationship between the three data sources I use for par yields. The actual yield series I use takes the FRED yields, then fills any missing values with GSW yields, then fills the remaining missing values with the CRSP yields.

Figure F.2 displays par and zero-coupon bond (ZCB) yields. Some results (Table G.1) use separately par and ZCB yields as proxies for long-term bond expected returns, because par yields are available for a longer history (and importantly contain more of the period before 1980 when yields were not secularly declining).

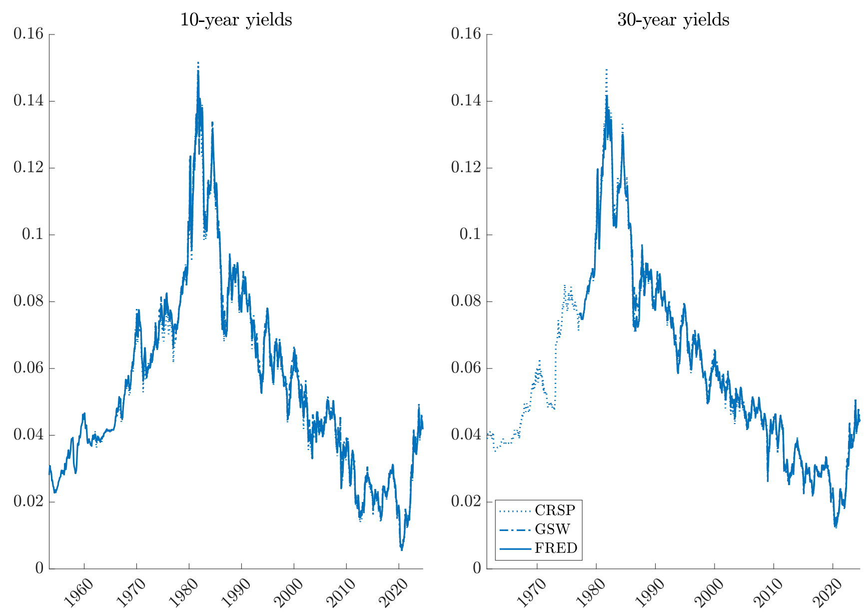

Figure F.1. Actual par yields from the three data sources used in the paper. Yields are annualized and in continuously compounded units (the par yields are converted to cc from coupon-equivalent units).

F.3 Bond survey interpolation and bootstrapping

To transform the BCFF survey data (which contains one-year forecasts of 6-month, 1-year, 2-year, 5-year, 10-year, and 30-year par yields) into a full zero-coupon yield curve forecast, I (i) interpolate the par forecasts and then (ii) "bootstrap" a zero-coupon forecast curve.

The interpolation step is a combination of linear interpolation (for maturities ≤ 5 years) and a fitted Nelson-Siegel model (for maturities > 5 years). The Nelson-Siegel model says that the cc par yields of maturity n should take the form

$$y(n) = \beta_0 + \beta_1 \frac{1-\exp(-n\theta)}{n\theta} + \beta_2 \Big[\frac{1-\exp(-n\theta)}{n\theta} - \exp(-n\theta)\Big]. \tag{F.1}$$

The parameters $(\beta_0, \beta_1, \beta_2, \theta)$ in equation (F.1) are estimated at every date (thus time-varying) to fit the 2-year, 5-year, 10-year, and 30-year par yield forecasts. I estimate this model by computing the fitting three model-implied slopes: between $n = 5$ and $n = 2$; between $n = 10$ and $n = 5$; and between $n = 30$ and $n = 10$. I compute these three slopes both model and survey, after first converting the survey yield forecasts to cc units. Since the slopes difference out β_0, this procedure estimates $(\beta_1, \beta_2, \theta)$ to fit these three slopes. I weight the three slopes by the number of maturity

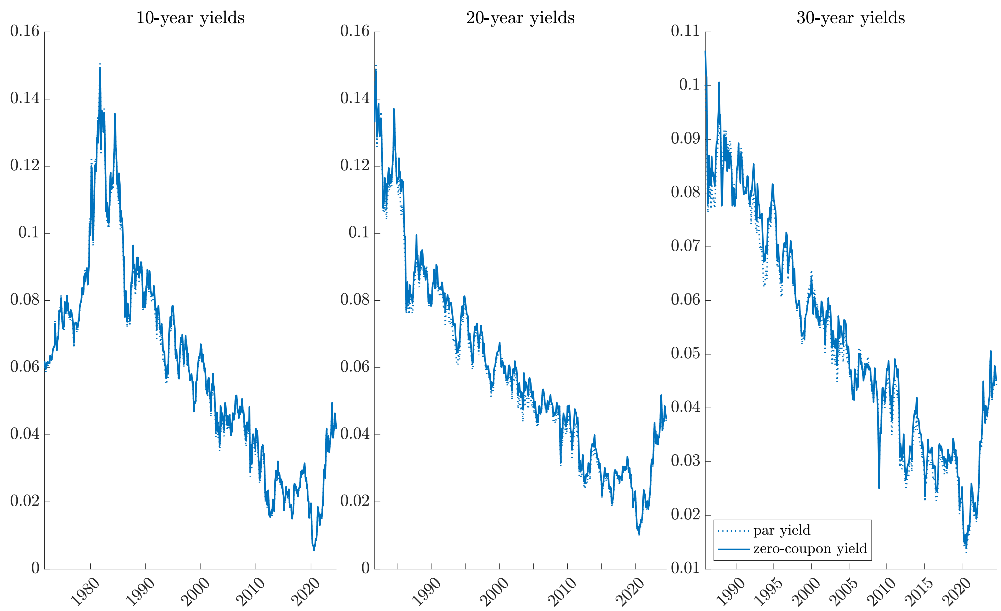

Figure F.2. Actual par and zero-coupon yields from GSW. Yields are annualized and in continuously compounded units (the par yields are converted to cc from coupon-equivalent units).

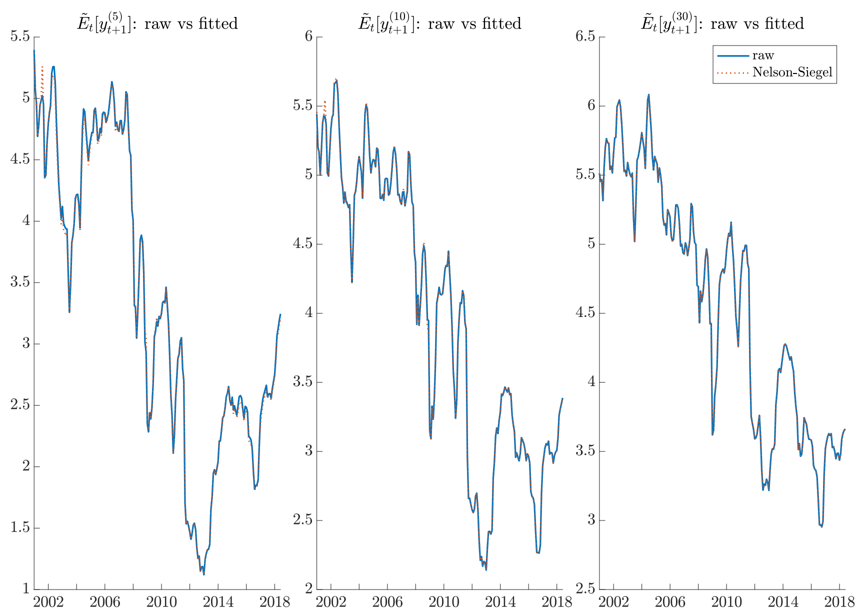

Figure F.3. BCFF consensus expectations of one-year ahead par yields ("raw", solid lines) as well as values from a Nelson-Siegel model fitted to the one-year ahead forecasts (dotted lines). All yield expectations are converted from coupon-equivalent to continuously compounded units.

years enveloped in the two maturities (i.e., weights of 3, 5, and 20). I then finally pick β_0 to match exactly the 30-year forecast level, since the long-maturity bonds are the most important in my analysis. Figure F.3 displays the resulting Nelson-Siegel fit to the key par yield forecasts. Figure F.4 contrasts the Nelson-Siegel fit to a linear interpolation in January of each year: the key distinction is the introduction of concavity at the long end, which is typically a feature of the yield curve.

Next, I bootstrap the zero-coupon forecasts as follows. A par yield Y_N^p (in coupon-equivalent

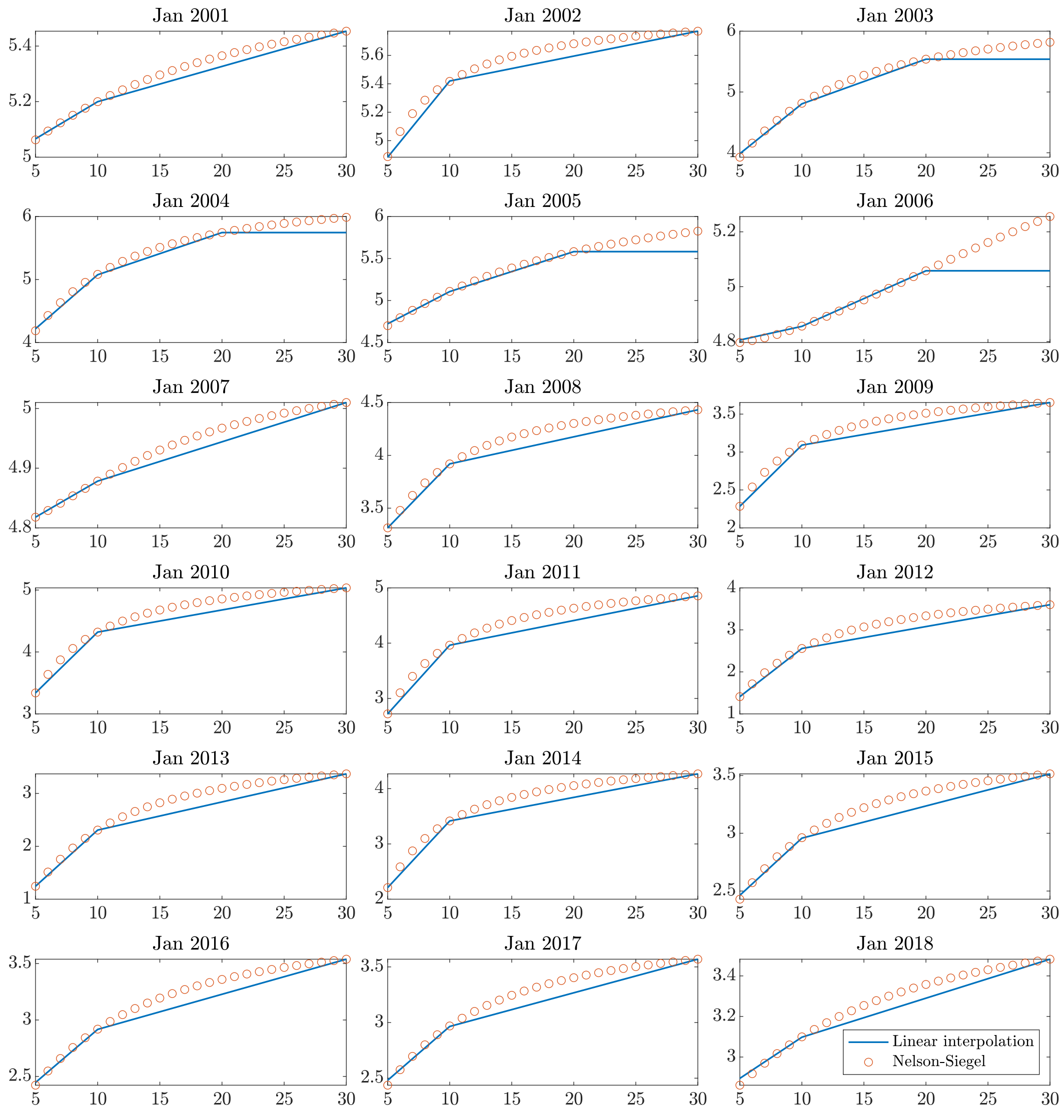

Figure F.4. BCFF consensus expectations of the one-year par yield curve (constructed via linear interpolation of different maturities versus interpolation via fitting a Nelson-Siegel model). All yield expectations are converted from coupon-equivalent to continuously compounded units.

units) is the yield such that an N-year bond paying that yield as its semi-annual coupon would trade at par, i.e.,

$$1 = \frac{Y_N^p}{2}\sum_{n=1}^{2N-1}\exp\Big(-\frac{n}{2}y_{n/2}\Big) + \Big(1+\frac{Y_N^p}{2}\Big)\exp\Big(-Ny_N\Big), \tag{F.2}$$

where y_j is the cc zero-coupon yield for maturity j. Thus, one can think of these par yields as the relevant discount rate for a coupon bond. One can solve for the implied N-period zero-coupon yield as

$$y_N = -\frac{1}{N}\log\left[\frac{1-\frac{Y_N^p}{2}\sum_{n=1}^{2N-1}\exp\big(-\frac{n}{2}y_{n/2}\big)}{1+\frac{Y_N^p}{2}}\right] \tag{F.3}$$

I recursively obtain $(y_{n/2})_{n=1}^{60}$ in this way. The only nuance is that I treat Y_N^p as the par yield *forecast* and obtain the zero-coupon yield *forecast* y_N, which effectively ignores the nonlinearities in this transformation.

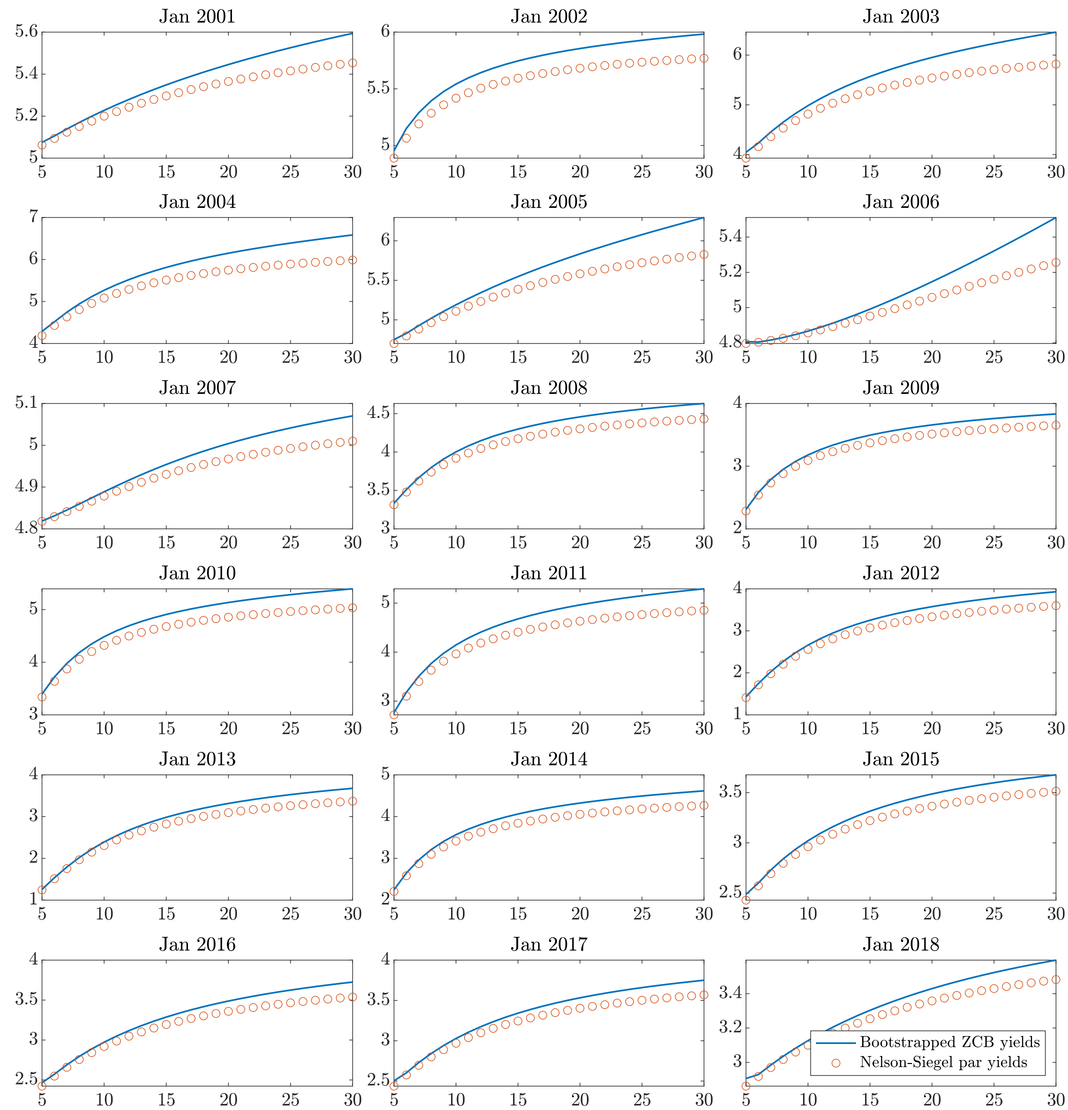

Figure F.5. BCFF consensus expectations of the one-year ahead par yield curve (constructed via fitting a Nelson-Siegel model) and the one-year ahead zero-coupon yield curve (constructed via boostrapping from the Nelson-Siegel par yield curve expectation). All yield expectations are converted from coupon-equivalent to continuously compounded units.

Figure F.5 shows yield curve forecasts (par vs zero-coupon) at the same January dates as above. Notice that the zero-coupon yield forecasts are higher than the par forecasts, mainly because zero-coupon bonds have higher durations due to their absence of coupons. Finally, Figure F.6 displays yield forecast time series and shows how the various transformations affect these forecasts. The left panel shows linearly interpolated par forecasts. The middle panel shows the Nelson-Siegel fitted par forecasts. The right panel shows the bootstrapped zero-coupon yield forecasts.

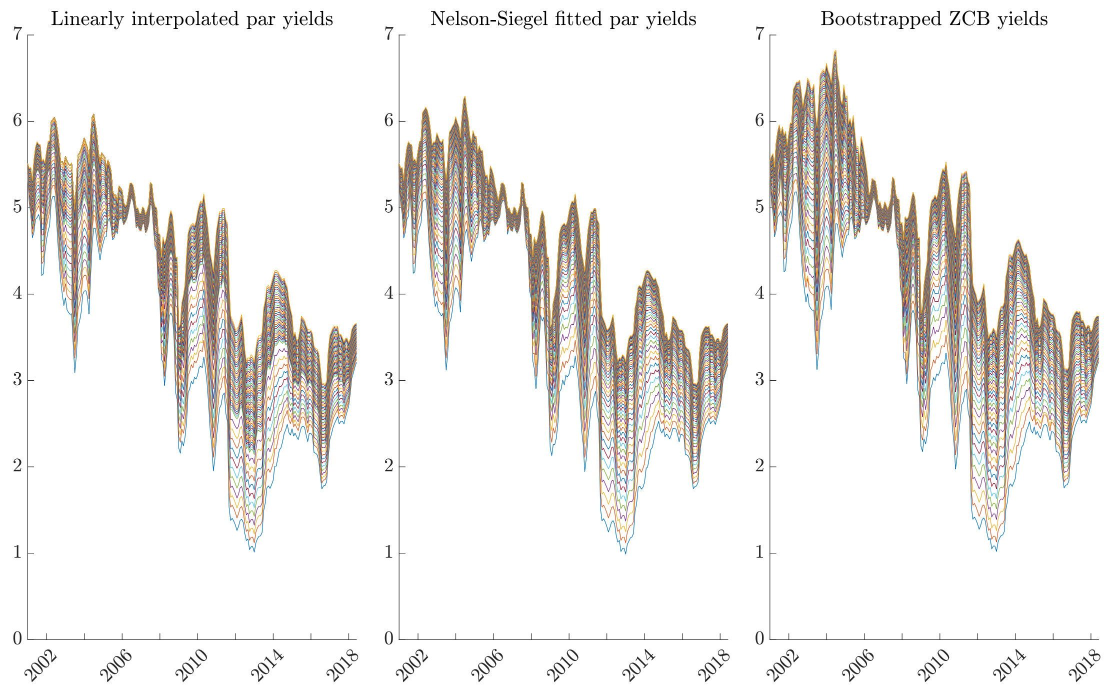

Figure F.6. BCFF consensus expectations of the one-year ahead yield curve. The first panel displays expectations of the par yield curve, constructed by linearly interpolating between maturities. The second panel displays expectations of the par yield curve, constructed by fitting Nelson-Siegel model to the expectations. The third panel displays expectations of the zero-coupon yield curve, constructed by bootstrapping from the Nelson-Siegel par curve. All yield expectations are converted from coupon-equivalent to continuously compounded units.

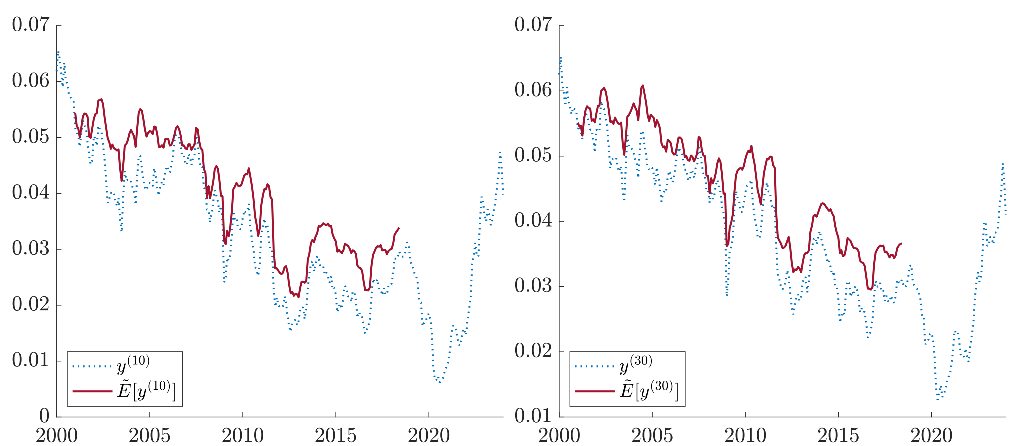

Figure F.7. Yields and yield forecasts. Par yields and the BCFF consensus one-year ahead forecasts of those par yields. All yields and yield forecasts are converted from coupon-equivalent to continuously compounded units.

G Updated Alvarez-Jermann bounds

This appendix refreshes the original Alvarez and Jermann (2005) results on the size of the permanent component in the SDF. First, I recall their main theoretical result:

Corollary G.1.

$$\frac{L(\Delta H_{t+1})}{L(\Delta S_{t+1})} \geq \min\left\{1, \frac{\mathbb{E}\left[\log R_{t+1} - \log R_{t+1}^{\infty}\right]}{\mathbb{E}\left[\log R_{t+1} - \log R_t^f\right] + L(1/R_t^f)}\right\} \tag{G.1}$$

for any return R such that $\mathbb{E}[\log R_{t+1} - \log R_t^f] + L(1/R_t^f) > 0$.

Table G.1 calculates the equity premium $\mathbb{E}[\log R^m - \log R^f]$ and the term premium $\mathbb{E}[\log R^{(k)} - \log R^f]$, and computes the lower bound on $L(\Delta H)/L(\Delta S)$ as the ratio of $\mathbb{E}[\log R^m - \log R^{(k)}]$ to $\mathbb{E}[\log R^m - \log R^f]$, along with the variance adjustment $L(1/R^f)$.

Standard errors, shown in parentheses, are computed using a Newey and West (1987) window with 36 months of lags to account for the overlap in returns and persistence in spreads, as in the main text. The standard errors for the size of the permanent component are computed using the Delta Method. Based on these standard errors, the estimates appear to be reasonably precise.

As in the original paper, I use several methods for the long bond component. I use both 10-year and 30-year bonds, as well as an index of long-term bonds, with the results similar for all proxies. I also use both holding period returns (Panel A) and yields (Panel B) as alternatives to measure expected returns. In a stationary and ergodic environment, the time-series average yield also measures a bond expected return; yields are much less volatile (though more persistent) than holding returns, which explains why Panel B features much tighter standard errors.

	Equity Premium $E[\log R^m/R^f]$	Term Premium $E[\log R^{(k)}/R^f]$	Adjustment for Volatility of Short Rate $L(1/R^f)$	Size of Permanent Component $L(\Delta H)/L(\Delta S)$
A. Holding returns.				
$k = 10$ years	0.0555	0.0100	0.0005	0.8134
	(0.0161)	(0.0099)	(0.0001)	(0.1641)
$k = 30$ years		0.0091		0.8292
		(0.0143)		(0.2308)
$k > 10$ (index)		0.0125		0.7686
		(0.0087)		(0.1564)
B. Yields.				
$k = 10$ years (par)	0.0555	0.0097	0.0005	0.8185
	(0.0161)	(0.0016)	(0.0001)	(0.0566)
$k = 10$ years (zcb)		0.0129		0.7615
		(0.0022)		(0.0718)
$k = 30$ years (par)		0.0119		0.7791
		(0.0027)		(0.0692)
$k = 30$ years (zcb)		0.0195		0.6428
		(0.0031)		(0.1096)

Table G.1. Size of permanent shocks in the SDF. This table displays estimates of the objective volatility bound from Corollary G.1, originally due to Alvarez and Jermann (2005). Panel A uses holding period returns to measure the term premium; each row uses a different proxy for R^{∞}. Panel B uses yields (par or zero-coupon bond yields) to measure the term premium. Newey-West asymptotic standard errors using 36 months of lags are shown in parentheses.